



Hybrid Mode

INTERNATIONAL CONFERENCE

PROCEEDINGS ON
ADVANCES IN AI-ENABLED SYSTEMS AND
TECHNOLOGIES FOR THE
NEXT GENERATION

INTERNATIONAL CONFERENCE PROCEEDINGS
ON
ADVANCES IN AI – ENABLED
SYSTEMS AND TECHNOLOGIES
FOR THE NEXT GENERATION

CHIEF EDITOR

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PREFACE

The **International Conference on Advances in AI-Enabled Systems and Technologies for the Next Generation (ICAESNG 2026)** was organized with the vision of creating a global academic platform for researchers, scholars, academicians, and students to share knowledge, exchange innovative ideas, and discuss emerging developments in Artificial Intelligence and next-generation technologies.

Artificial Intelligence continues to transform every domain of human life — from education and healthcare to business, communication, and scientific research. Recognizing this rapid transformation, the conference aimed to encourage interdisciplinary dialogue and collaborative research in AI-enabled systems and technological advancements.

We are delighted to present this **ISBN Conference Proceedings**, which include **research papers contributed by authors from different parts of the world**. These papers reflect diverse perspectives, innovative methodologies, and contemporary research outcomes in AI, data science, computational technologies, and related fields. The proceedings stand as a testimony to the scholarly engagement and academic commitment of the participants.

The organizing committee expresses its sincere gratitude to all contributors, reviewers, speakers, and collaborators who supported this conference and publication. We hope this volume will serve as a valuable academic resource for researchers, educators, and students interested in AI-enabled systems and future technologies.

Rev. Sr. Amalore Mary CTC

Chairperson
St. Teresa Arts and Science College for Women
Mangalakuntu – 629 178



MESSAGE FROM CHIEF PATRON

It gives me immense joy and pride to present the Conference Proceedings of the **International Conference on Advances in AI-Enabled Systems and Technologies for the Next Generation (ICAESNG 2026)**.

In today's rapidly evolving world, Artificial Intelligence is not merely a technological advancement but a transformative force shaping education, industry, research, and society at large. Platforms such as this conference provide an essential space for scholars, researchers, and students to engage in meaningful academic dialogue and collaborative exploration.

I am delighted to note that this proceedings volume comprises nearly 30 research papers contributed by authors from various parts of the world. The diversity of topics, innovative approaches, and interdisciplinary perspectives reflected in these papers demonstrate the growing significance of AI-enabled systems and next-generation technologies in contemporary research.

I appreciate the dedicated efforts of the organizing committee, editorial team, reviewers, and contributors who worked tirelessly to ensure the success of both the conference and this publication. Their commitment to academic excellence and knowledge dissemination is truly commendable.

May this proceedings volume serve as a valuable reference for researchers and inspire continued inquiry, innovation, and collaboration in the field of Artificial Intelligence.

I congratulate everyone involved and wish them continued success in their academic and research endeavors.

Rev. Sr. Jessy CTC

Secretary
St. Teresa Arts and Science College for Women
Mangalakuntu – 629 178



MESSAGE FROM CO-PATRON

It gives me great pleasure to extend my greetings on the publication of the Conference Proceedings of the **International Conference on Advances in AI-Enabled Systems and Technologies for the Next Generation (ICAESNG 2026)**.

In an era where Artificial Intelligence and emerging technologies are reshaping the global landscape, academic forums like this conference play a vital role in fostering research, innovation, and collaborative learning. The initiative to organize this international conference reflects the institution's commitment to academic excellence and global engagement.

I am delighted to note that this proceedings volume includes nearly 30 research papers contributed by scholars from various parts of the world. The collection showcases diverse perspectives, innovative methodologies, and meaningful research contributions in AI-enabled systems and next-generation technologies. This publication stands as a testimony to the intellectual dedication and scholarly spirit of the participants.

I sincerely appreciate the efforts of the Chairperson, Principal, Convenor, Organizing Committee, Editorial Board, reviewers, and all contributing authors who have worked diligently to make both the conference and this publication a success.

May this proceedings volume serve as a valuable academic resource and inspire continued research and innovation in the field of Artificial Intelligence.

I extend my best wishes for the continued growth and success of such scholarly initiatives in the years to come.

Rev. Dr. B. Aloysius Maria Benzigar

Secretary
St. John's College of Arts & Science
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MESSAGE FROM CO-PATRON

It gives me immense pleasure to extend my warm greetings to all the participants of this conference on **“International Conference on Advances in AI-Enabled Systems and Technologies for the Next Generation.”**

The purpose of this conference is to bring together scholars, academicians, and researchers to deliberate on innovative advancements in Artificial Intelligence, emerging technologies, and next-generation systems that can enhance research quality and academic progress. The sessions will provide opportunities to exchange knowledge, develop innovative ideas, and strengthen research networks.

I am confident that every participant will utilize this platform to broaden their academic horizons and contribute meaningfully to the discussions. May this conference serve as an inspiration for new perspectives and impactful research outcomes.

I extend my best wishes for the grand success of this academic endeavor.

Dr. A. Rejo Jeice

Principal
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MESSAGE FROM PRINCIPAL

I extend my heartfelt congratulations to the Departments of Physics, Mathematics, Computer Science, Artificial Intelligence and Data Science of St. Teresa's Arts and Science College for Women, Mangalakuntu, Tamil Nadu, India, for organizing the International Conference on **“Advances in AI-Enabled Systems and Technology for the Next Generation”** in collaboration with St. John's College of Arts and Science, Ammandivilai, on 6th January 2026.

This conference provides a unique academic platform to promote meaningful collaboration among postgraduate students, researchers, and academicians working in the field of AI-enabled systems and emerging technologies. Artificial Intelligence is truly an interdisciplinary domain with immense scientific and technological potential. However, realizing its full impact across various branches of science and technology requires overcoming significant challenges. These include advancements in robotics design, data analytics, quantum computing, blockchain technology, healthcare innovation, climate change mitigation, 6G networks, and many other emerging areas.

AI holds tremendous promise for the next generation. When developed and implemented responsibly, it can significantly improve quality of life, drive sustainable development, and create new opportunities across education, industry, healthcare, and governance.

I am confident that this international conference will foster vibrant discussions among participants, students, and researchers from India and around the world, highlighting the latest trends in AI and its modern applications for the betterment of society.

Once again, I congratulate the organizers for their commendable initiative and extend my warm greetings to all participants. I wish the conference every success.

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MESSAGE FROM EDITOR

It gives us immense pleasure to present the **Conference Proceedings of ICAESNG 2026**, a collection of selected research papers presented during the international conference.

This publication represents the collective academic effort of scholars from various institutions and countries who contributed their research findings and insights. Each paper included in this volume has been carefully reviewed and compiled to maintain academic quality and relevance to the conference theme.

The editorial team believes that knowledge grows through sharing, collaboration, and critical inquiry. This proceedings volume reflects not only current research trends in **AI-enabled systems and technologies**, but also the enthusiasm of emerging researchers who are shaping the future of innovation.

We extend our heartfelt appreciation to all authors for their valuable contributions, to the reviewers for their academic support, and to the organizing committee for their dedication in bringing out this publication successfully.

We hope these proceedings volume will inspire further research, collaboration, and innovation in Artificial Intelligence and next-generation technologies.

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Next-Generation AI-Enabled Systems and Technologies for Academic Writing and Publication

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Abstract

Next-generation AI-enabled systems are reshaping academic writing and scholarly publication by moving beyond basic editing support toward deeper involvement in knowledge production. Since 2020, advances in large language models, generative AI, semantic search, and intelligent automation have transformed core academic workflows, including idea development, literature synthesis, drafting, revision, and publication decision-making (Bhatia & Bremner, 2021; Zhai et al., 2023). This article synthesizes recent scholarship to explain how AI writing tools enhance efficiency and productivity through automated outlining, paraphrasing, grammar correction, and reference management, thereby shortening drafting and revision cycles—particularly for multilingual researchers (Cotos et al., 2021; Lee et al., 2022). It also highlights AI's role in mitigating writer's block by providing cognitive scaffolding for early-stage drafting and idea generation (Koltay, 2020; Perelman, 2022). Beyond mechanics, AI contributes to content quality by improving coherence, cohesion, and alignment with disciplinary discourse conventions (Chen, 2024; Zhang & Hyland, 2024). However, the increasing reliance on AI intensifies concerns about authorship, transparency, integrity, and equity, requiring ethical governance and critical AI literacy to protect human agency in scholarship (Floridi et al., 2022; COPE, 2023). The article further extends the discussion to EFL contexts, where AI-driven digital tools support personalized learning, real-time feedback, and communicative practice (Janjić, 2025; Manchanayaka, 2025).

Keywords: Artificial intelligence; Academic writing; Ethical governance; EFL digital tools; Scholarly publishing

Introduction

Academic writing and scholarly publication are entering a decisive new phase shaped by next-generation artificial intelligence (AI)-enabled systems and technologies. Over the past decade, digital tools have gradually transformed how researchers search for literature, manage references, and disseminate findings. Since 2020, however, advances in large language models, generative AI, semantic search, and intelligent automation have marked a qualitative shift rather than a simple technological upgrade (Bhatia & Bremner, 2021; Floridi et al., 2022). These systems are no longer confined to supporting peripheral academic tasks; they are increasingly embedded in the core processes of knowledge production, including idea development, argument structuring, drafting, revision, and navigation of publication ecosystems (Zhai et al., 2023). Next-generation AI-enabled technologies operate at the intersection of computational intelligence, linguistic modeling, and scholarly communication. Unlike earlier rule-based or template-driven writing aids, contemporary AI systems can model disciplinary discourse conventions, analyze rhetorical structure, and generate context-sensitive academic text aligned with genre expectations (Koltay, 2020; Chen, 2024). As a result, AI tools are now used to support literature synthesis, coherence and cohesion improvement, language editing, journal selection, and structured responses to peer-review feedback (Lee et al., 2022; Lund et al., 2023). These affordances have positioned AI as a powerful mediator of academic productivity and accessibility, particularly for early-career

researchers and multilingual scholars working in English-dominant publication systems (Cotos et al., 2021; Zhang & Hyland, 2024).

At the same time, the rapid integration of AI into academic writing and publication has intensified epistemological, ethical, and methodological debates. Questions surrounding authorship, originality, accountability, and transparency have become central to discussions of AI-assisted scholarship (Floridi et al., 2022; COPE, 2023). While AI-enabled systems can enhance efficiency and linguistic precision, they also risk obscuring human intellectual contribution, reinforcing existing publication inequalities, and encouraging surface-level compliance with academic conventions rather than deep scholarly reasoning (Perelman, 2022; Zhai et al., 2023). Consequently, the responsible use of AI in academic writing requires clearly articulated frameworks that foreground human agency, ethical governance, and disciplinary integrity. Recent research between 2020 and 2025 emphasizes that AI should be understood not as a replacement for scholarly expertise, but as a cognitive and procedural scaffold that reshapes how academic knowledge is constructed and communicated (Koltay, 2020; Chen, 2024). In this view, AI-enabled writing systems function as co-participants in the research workflow, influencing how arguments are framed, how evidence is synthesized, and how manuscripts are positioned within global publication economies. Such influence necessitates critical literacy among researchers, editors, and institutions to ensure that AI adoption strengthens, rather than dilutes, academic standards (Lund et al., 2023; Zhang & Hyland, 2024). This paper, *Next-Generation AI-Enabled Systems and Technologies for Academic Writing and Publication*, responds to these developments by examining how advanced AI technologies are reshaping scholarly communication across disciplines. It brings together conceptual analyses, empirical studies, and applied perspectives to illuminate both the affordances and the limitations of AI-mediated academic writing. By situating technological innovation within broader concerns of ethics, equity, and knowledge production, this work aims to support a critically informed and human-centered integration of AI into academic writing and publication practices in the contemporary research landscape.

Digital Tools and the advantage in EFL Contexts

AI is changing the way we write and communicate, making these tasks faster and more efficient. In academic writing, AI helps researchers find information quickly, improve writing quality, and check for plagiarism. In communication, it allows real-time translation, personalized messages, and automated responses. However, AI also brings challenges. Artificial Intelligence has transformed the academic writing landscape and avails many opportunities to enhance the quality, efficiency, and accessibility of scholarly work. AI tools can quickly process large volumes of data to assist researchers in identifying relevant literature, trends, and gaps in the existing research. In this way, academics can do a thorough literature review in much less time and instead focus on their contribution. *Digital tools* refer to software applications, platforms, and interactive systems that facilitate EFL learning. These can be broadly categorized into:

- *Mobile language learning apps*
- *AI-powered platforms and smart tutors*
- *Virtual learning environments and immersive tools*
- *Collaborative online systems and feedback environments*

Mobile language learning apps have become one of the most accessible and widely used digital tools in language education. These applications allow learners to practice vocabulary, grammar, listening, and speaking anytime and anywhere, using short, interactive activities that support self-paced learning. Their portability and user-friendly design encourage consistent practice and learner autonomy, making them particularly effective for informal and out-of-class language learning. AI-powered platforms and smart tutors extend these possibilities by offering intelligent, adaptive support. Through artificial intelligence, these systems can analyze learner performance, identify strengths and weaknesses, and provide personalized feedback and recommendations.

Smart tutors simulate conversational interaction, support pronunciation practice, and adjust task difficulty in real time, helping learners progress at an individualized pace while maintaining engagement. Virtual learning environments and immersive tools, including simulation-based platforms and virtual reality applications, create contextualized spaces for authentic language use. By placing learners in realistic social, academic, or professional scenarios, these environments promote meaningful communication and intercultural awareness. Immersive tools enhance experiential learning by integrating language use with visual and situational cues that mirror real-world interactions. Collaborative online systems and feedback environments emphasize the social dimension of learning. Through discussion forums, shared writing spaces, peer review tools, and virtual exchange programs, learners can interact with classmates or international partners. These platforms support collaborative knowledge construction, continuous feedback, and communicative practice, fostering both linguistic development and essential collaboration skills. Each category serves different functions: vocabulary reinforcement, pronunciation practice, communicative simulations, and scaffolding learning tasks. Importantly, digital tools are no longer monolithic; they increasingly adapt to learner needs and contexts, guided by artificial intelligence and adaptive learning algorithms (Janjić, 2025). The benefits of writing with AI can be explained across four interconnected dimensions: efficiency and productivity, overcoming writer's block, content quality improvement, and higher-level strategy and creativity. One of the most widely reported benefits of AI-assisted academic writing is increased efficiency and productivity. AI-enabled systems can automate time-consuming tasks such as outlining, paraphrasing, grammar checking, reference formatting, and language polishing. By reducing the mechanical burden of writing, researchers are able to allocate more cognitive resources to conceptual thinking and scholarly judgment. Empirical studies show that AI tools significantly shorten drafting and revision cycles, particularly for non-native English-speaking scholars, thereby accelerating the overall publication process (Cotos et al., 2021; Lee et al., 2022; Lund et al., 2023). Closely related to productivity is AI's role in overcoming writer's block. Writer's block often emerges from cognitive overload, uncertainty about structure, or difficulty initiating text. AI writing systems can generate prompts, sample sentences, or preliminary drafts that help writers move past the inertia of the blank page. Rather than replacing human thinking, these systems act as cognitive scaffolds that support idea generation and momentum in early writing stages. Research indicates that AI-assisted drafting lowers anxiety and increases writing confidence, especially among early-career researchers and doctoral students (Koltay, 2020; Perelman, 2022; Zhai et al., 2023).

Another major benefit highlighted in the figure is **content quality improvement**. Advanced AI systems can analyze coherence, cohesion, lexical variety, and rhetorical structure, offer suggestions that enhance clarity and readability. In academic contexts, AI tools help align manuscripts with disciplinary conventions, refine argument flow, and reduce linguistic errors that might otherwise obscure scholarly contributions. Studies between 2020 and 2025 suggest that AI-supported revision leads to more polished and reviewer-friendly manuscripts, particularly in international journals where language quality strongly influences editorial decisions (Bhatia & Bremner, 2021; Zhang & Hyland, 2024).

Finally, and perhaps most critically, AI contributes to **higher-level strategy and creativity** when used responsibly. By handling routine writing operations, AI frees researchers to focus on strategic decisions such as framing research questions, positioning arguments, synthesizing literature, and developing original insights. Rather than constraining creativity, AI can expand it by enabling exploratory drafting, alternative phrasing, and rapid iteration of ideas.

Recent scholarship emphasizes that the greatest value of AI in academic writing lies not in text generation itself, but in its support for human agency, metacognitive awareness, and creative knowledge construction (Floridi et al., 2022; Chen, 2024; Lund et al., 2023). Taken together, these benefits suggest that AI writing tools function best *as intellectual partners rather than substitutes*, enhancing efficiency while preserving human control over meaning,

ethics, and scholarly contribution. When integrated thoughtfully, AI-enabled writing systems can strengthen both the process and the quality of academic writing in the contemporary research landscape (See Figure 1).

Benefits of writing with AI



Figure 1.

While AI offers several ways to enrich academic writing, a researcher must use it with discretion. Critical thinking, creativity, and ethical considerations—what humans do—remain irreplaceable in an academic process. By using AI effectively, scholars can enhance their writing processes and contribute meaningfully to the body of knowledge in their respective areas.

Advantage	Key Features	Pedagogical Benefits	Learning Outcomes Supported
Increased Engagement and Motivation	Interactive multimedia content (videos, quizzes, games); gamification elements such as points, badges, and levels	Sustains learner attention, reduces anxiety, and increases willingness to participate in learning activities	Higher learner motivation, increased time-on-task, improved retention and active language use
Improved Accessibility and Flexibility	Mobile learning applications; cloud-based platforms accessible across devices	Enables anytime–anywhere learning; supports learners with diverse schedules, geographical locations, and physical constraints	Greater learner autonomy, continuity of learning beyond the classroom, equitable access to instruction
Personalized Learning Experiences	AI-driven adaptive learning systems; customizable lesson paths and progress dashboards	Adjusts content difficulty and pacing to individual learner needs; supports differentiated instruction	Improved learning efficiency, reduced cognitive overload, stronger mastery of language skills
Enhanced Collaboration and Communication	Online discussion forums; collaborative writing tools; virtual exchange programs	Promotes social interaction, peer learning, and authentic communicative practice	Development of communicative competence, intercultural awareness, and collaborative skills

		and partners	language			
Real-Time Feedback Assessment	and	Instant automated grading; AI-powered error detection and proficiency assessment	Provides immediate, formative feedback; supports self-regulated learning and continuous improvement	Faster correction, accuracy, awareness of strengths and weaknesses	error improved clearer of and	

Introducing Some Digital Tools: EFL Learning

In the twenty-first century, following waves of global technological integration, digital tools have become fundamental in EFL contexts. They are no longer supplemental but central to pedagogical design, especially after the broad expansion of online and blended learning models. Researchers have documented the profound role of technology in transforming language learning environments and outcomes, calling for balanced, pedagogically informed use of digital media in instruction (Manchanayaka, 2025). A study conducted with Iranian university learners in 2024–2025 highlights that students generally view technology positively, finding it effective for learning outside the classroom—but also encountering barriers such as limited infrastructure and distractions, which impact how tools are adopted and used (Zeitounly, Ghorbani & Robatjazy, 2025). These findings echo broader research worldwide, underscoring that while digital tools offer transformative potential, successful implementation depends on factors beyond technology itself, including teacher guidance, digital literacy, and institutional support. One of the most widespread forms of digital tools in EFL learning today is mobile applications enhanced by conversational AI. These apps offer flexible, personalized practice and immediate feedback—features aligned with learner autonomy and self-paced study.

Talkpal AI is a prominent example: it combines text-based and voice-based interactions with an AI tutor designed to simulate natural conversation in multiple languages, including English (Talkpal AI, 2025). With millions of users worldwide, platforms like Talkpal allow learners to increase speaking confidence, improve fluency, and practice at any time, without requiring a live partner. Another emerging tool is *Praktika.ai*, which uses AI avatars and speech-to-text recognition to simulate role-play scenarios such as interviews and travel conversations, adapting content to learner performance (Praktika, 2025). By offering immersive role-play and contextual dialogue, such applications aim to bridge the gap between textbook learning and real-world communication. The rise of platforms like these exemplifies a shift from traditional drill-based software to context-rich, interactive learning environments that support authentic language use. Research suggests that AI-enhanced apps can heighten motivation and engagement, while simultaneously offering key scaffolds for vocabulary and pronunciation development (Manchanayaka, 2025)

AI-Powered Mobile Apps

One of the most widespread forms of digital tools in EFL learning today is mobile applications enhanced by conversational AI. These apps offer flexible, personalized practice and immediate feedback—features aligned with learner autonomy and self-paced study. AI plagiarism detection software has helped maintain a sense of academic integrity by detecting accidental or intentional plagiarism. It helps researchers ensure that their research is original and sources are duly acknowledged, furthering ethical scholarship.

AI is able to provide analysis for large and complex datasets and graph the output. For sciences in which there is heavy reliance on quantitative information, of course, this tool will be able to allow for clear, visually compelling presentations of the findings of a study. AI-powered translation tools can help break barriers to effectively spread research to the general public. It will enable researchers to increase their readership and international collaboration by having their papers published in many different languages. AI can be applied in educational contexts to

provide personalized feedback to students regarding their writing. This personalized focus would help learners better grasp academic writing conventions at their own pace.

AI technologies can help generate ideas or outlines for academic papers. While these tools should not replace original thought, they can serve as valuable brainstorming partners that help authors to get past writer's block or explore new angles on a topic. AI can enable collaboration among researchers to connect people with similar interests or skill profiles that could be mutually reinforcing. Such platforms could, therefore, facilitate communication and project management, making it easier to collaborate on the writing of academic pieces. AI tools can help identify potential biases in research methodologies or writing styles. Thus, by pointing out areas where bias could be, the tools foster more rigorous and fair academic practices.

Immersive and Interactive Environments

Beyond mobile practice tools, EFL educators and researchers are exploring more immersive digital resources that facilitate deeper interaction and communicative competence. Virtual Reality (VR) and simulation-based tools represent one frontier. A recent study on interactive digital technologies in higher education contexts shows that VR scenarios and role-playing games significantly enhance communicative competence and learner autonomy compared to traditional methods (Redfame, 2025). These technologies allow learners to step into realistic environments—professional settings, cultural contexts, or social interactions—where language use is embedded in authentic tasks. VR tools provide rich input and contextual cues that support language learning holistically, improving not only linguistic accuracy but also intercultural awareness and pragmatic competence. Interactive digital learning platforms like the **Stanford Mobile Inquiry-based Learning Environment (SMILE)** enable EFL students to craft their own questions, engage in peer discussion, and reflect on language use in real time, fostering critical thinking alongside language skills (SMILE, 2026)

Collaborative Tools and Social Learning

Another major category of digital tools relies on collaborative technologies that support social learning. Online platforms now incorporate communication features that connect learners across contexts, enabling discussion forums, shared projects, and peer feedback. Empirical research highlights that such collaborative environments contribute positively to learner engagement and communicative competence, especially when tasks are designed to prompt authentic interactions rather than passive consumption (Manchanayaka, 2025). Blended learning ecosystems that leverage synchronous sessions (Zoom, Teams) with asynchronous forums and resource repositories allow instructors to orchestrate richer, learner-centred experiences. Within those ecosystems, learners can draft dialogues, record spoken responses, and receive scaffolded corrective feedback, enhancing both receptive and productive skills.

Adaptive Learning and Personalization

Digital tools have reshaped EFL learning landscapes, offering flexibility, engagement, and personalized pathways to language proficiency. From AI-powered apps like Talkpal and Praktika to immersive VR environments and collaborative online platforms, technology has amplified the possibilities for language acquisition. However, realizing their full potential requires careful pedagogical integration, sound teacher training, and equitable access. As research in 2025–2026 continues to refine understanding of these tools' impact, the future of EFL learning looks increasingly dynamic, learner-centred, and technologically enriched. One of the most promising developments in digital EFL tools is the integration of adaptive learning systems. Research in 2025 emphasizes how AI-powered adaptive platforms adjust content and tasks according to individual learner proficiency, learning styles, and progress (Janjić, 2025). Adaptive technologies leverage learner data to deliver tailored exercises, revisiting areas of difficulty while advancing learners at their own pace. This personalization is especially valuable in mixed-level classrooms, where one-size-fits-all instruction often fails to meet diverse needs. For example, mobile platforms may vary the difficulty of vocabulary sets or dialog scenarios based on past performance, ensuring that learners are neither bored by redundancy nor daunted by overly

complex tasks. Such intelligent adaptation aligns with principles of differentiated instruction and supports sustained engagement. While conversational fluency often dominates EFL tool discussions, reading comprehension and literacy development are equally vital. Digital reading platforms, e-books, and multimedia resources that embed interactive annotations, dictionaries, and comprehension prompts can significantly enhance reading instruction. Emerging studies investigate how e-text layers, audio support, and embedded glossaries support reading comprehension among EFL learners (Noori, 2025).

Teacher Preparation, Digital Literacy, and Ethical Considerations

In every effective integration of digital tools lies the teacher's role. Studies show that early-career EFL teachers often face perceptions and practical challenges in adopting digital technologies due to limited training and confidence (2025). Teacher training must therefore extend beyond tool familiarity to include pedagogical design principles that align digital tasks with language learning objectives. Furthermore, digital tools demand digital literacy—not just for learners, but for educators as well. Ensuring that teachers are competent in selecting, customizing, and evaluating digital resources is indispensable for meaningful implementation. Despite their potential, digital tools are not without challenges. Digital literacy gaps, resource inequities, and technical issues can undermine the benefits of technology in EFL settings (Manchanayaka, 2025). Learners without reliable internet access or devices may be disadvantaged, broadening existing educational divides. There are also risks of distraction or superficial engagement if tools prioritize gamification without pedagogical alignment, leading learners to focus on points rather than language use. Additionally, data privacy and ethical use of AI in educational contexts require careful governance to protect learner information and ensure responsible algorithmic behaviour. Looking ahead, research from 2025–2026 points to several exciting directions. Adaptive AI tutors, immersive simulations, and cross-cultural interactive tools promise greater learner autonomy and contextualized learning experiences. The intersection of language and workplace contexts has also inspired tools like AI-generated work-related quizzes that situate language learning within everyday professional needs (Yang et al., 2025). Moreover, frameworks that support AI literacy among EFL learners and teachers are emerging, recognizing that critical understanding of technology is as necessary as language proficiency in a digitally mediated world.

Conclusion, Implications, and Future Directions

Next-generation AI-enabled systems have fundamentally altered the landscape of academic writing and scholarly publication by shifting from peripheral assistance to active participation in knowledge production. Across the past five years, advances in large language models, generative AI, semantic search, and automation have expanded the scope of what digital tools can do in research workflows—supporting idea development, literature synthesis, drafting, revision, and publication strategy (Bhatia & Bremner, 2021; Zhai et al., 2023). The evidence reviewed in this paper indicates that AI tools provide clear benefits in academic productivity and accessibility, particularly for multilingual researchers and early-career scholars working within English-dominant publication systems (Cotos et al., 2021; Zhang & Hyland, 2024). At the same time, these benefits are inseparable from ethical and epistemological concerns, including transparency, authorship, accountability, and equity (Floridi et al., 2022; COPE, 2023). Therefore, AI should be understood neither as a neutral “writing machine” nor as a replacement for scholarly expertise, but as a cognitive and procedural scaffold that reshapes academic communication while preserving the central role of human judgment (Koltay, 2020; Chen, 2024). The four core advantages discussed—efficiency and productivity, mitigating writer's block, improving content quality, and enabling higher-level strategy and creativity—together illustrate AI's emerging role as an “intellectual partner” in writing rather than a mere tool for editing. AI-supported outlining, paraphrasing, language polishing, and reference management reduce the mechanical burden of writing and accelerate drafting and revision cycles (Cotos et al., 2021; Lee et al., 2022). In addition, AI systems can help writers overcome the inertia of the blank page by generating

prompts, initial drafts, or alternative structures, thereby lowering anxiety and increasing writing confidence (Perelman, 2022; Zhai et al., 2023). These benefits become particularly meaningful when AI support enables scholars to allocate more time to conceptual rigor, originality, and argumentation—areas where human intellectual work remains essential. However, maximizing these benefits requires critical literacy about AI’s limitations, including hallucinated content, bias, and the risk of producing superficially “well-formed” but conceptually weak texts (Floridi et al., 2022; Lund et al., 2023).

The implications of this shift extend beyond academic writing into educational environments, especially EFL and broader language learning contexts. The same AI mechanisms that support academic productivity also enhance engagement, flexibility, personalization, collaboration, and real-time feedback in learning settings. Mobile apps and AI tutors allow learners to practice anytime and receive adaptive, individualized support, while immersive and collaborative platforms create contextualized opportunities for authentic language use (Janjić, 2025; Manchanayaka, 2025). This suggests a valuable convergence: AI-enabled systems can simultaneously strengthen scholarly writing and language education by supporting learners and researchers in becoming more fluent, confident, and effective communicators. Yet, this convergence also heightens ethical responsibilities, particularly in safeguarding learner data, ensuring equitable access, and preventing overreliance on automation at the expense of independent thinking and communicative competence (Manchanayaka, 2025; COPE, 2023).

Several practical implications follow for researchers, institutions, publishers, and educators. First, institutions should embed AI literacy and responsible use policies into research training, emphasizing transparency of tool use, careful verification of outputs, and ethical decision-making around authorship and originality (COPE, 2023; Lund et al., 2023). Second, journals and publishers should continue refining guidelines that clarify acceptable AI assistance (e.g., language polishing vs. content generation), disclosure expectations, and accountability standards to protect integrity while supporting innovation (COPE, 2023). Third, for EFL and academic writing instruction, curriculum design should incorporate AI tools as scaffolds for development rather than shortcuts—focusing on metacognitive strategies such as evaluating outputs, improving argument quality, and strengthening learner agency (Koltay, 2020; Chen, 2024).

Future research should move toward more evidence-based models of AI integration. One priority is studying how AI affects knowledge construction over time—whether it supports deeper reasoning, better synthesis, and stronger scholarly voice, or whether it encourages homogenized writing and surface-level conformity to genre conventions (Zhang & Hyland, 2024; Perelman, 2022). Another direction concerns equity: research must explore how AI tool access, language background, and institutional resources shape who benefits most from AI-enabled publishing support (Cotos et al., 2021; Zhang & Hyland, 2024). Additional work is needed on robust disclosure frameworks, AI detection limitations, and ethical governance in both academic and educational contexts (Floridi et al., 2022; COPE, 2023). Finally, the next generation of tools should be designed with human-centered principles—supporting transparency, explainability, and learner/researcher agency—so that AI strengthens academic standards rather than merely accelerating output. In conclusion, AI-enabled systems are redefining academic writing and publication as well as digital learning ecosystems. Their value lies not in replacing human scholarship, but in augmenting it—reducing routine burdens, enhancing accessibility, and enabling more strategic and reflective research communication. The challenge for the coming years is not whether AI will be used, but how it will be governed, taught, and integrated to ensure ethical, equitable, and intellectually rigorous scholarship (Floridi et al., 2022; COPE, 2023; Zhai et al., 2023).

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Integrating Multiple Intelligences into Next-Generation Approaches to Speaking Skill Development

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Abstract

This study investigated the impact of Multiple Intelligences (MI)–based instructional techniques on learners’ speaking performance within a contemporary, learner-centered educational framework. A total of 100 cadets from Tehran University participated in the study and were randomly assigned to control and experimental groups. Both groups attended a two-hour English

speaking course per week over a 12-week period. While the control group received conventional instruction, the experimental group was exposed to diversified MI-oriented pedagogical techniques designed to address varied cognitive strengths. Adopting a quasi-experimental pretest–posttest design, the study analyzed changes in speaking performance across groups. The findings revealed that MI-based instruction led to a statistically significant and educationally substantial improvement in speaking skills compared to traditional methods. The results highlight the value of human-centered, intelligence-diverse pedagogies as a foundation for next-generation learning environments, offering important implications for the design of innovative and adaptive language education systems.

Keywords: Artificial Intelligence, EFL learners, Multiple Intelligences, Speaking skills, Next-Generation

Introduction

Speaking skill development in contemporary language education is increasingly shaped by next-generation pedagogical approaches that move beyond uniform, one-size-fits-all models of instruction. Within this evolving landscape, the integration of Multiple Intelligences (MI) theory offers a powerful framework for rethinking how learners engage with spoken language through diverse cognitive, social, emotional, and creative pathways. Rather than privileging linguistic competence alone, MI-informed speaking instruction recognizes that oral communication can be fostered through bodily movement, visual-spatial representation, musical rhythm, interpersonal interaction, intrapersonal reflection, and logical reasoning. When aligned with emerging methodologies—such as technology-enhanced learning, task-based interaction, and learner-centered design—this pluralistic view of intelligence enables more inclusive, motivating, and effective speaking environments. Integrating Multiple Intelligences into next-generation approaches thus reframes speaking not merely as verbal performance, but as a dynamic, multidimensional process grounded in human variability, agency, and meaning-making. In 1983, Howard Gardner wrote *Frames of Mind*, a book that changed the concept of intelligence (Khalil, 2021). Gardner believed that human cognitive competence consisted of a set of abilities, talents or mental skills, which are called intelligences (Adityas, 2016). When a situation presents a difficulty, a person's intelligences work together in varying degrees to solve the problem or create products (Armstrong, 2018). Gardner (1983) proposes at least eight different kinds of intelligences: linguistic, logical-mathematical, musical, bodily kinesthetic, spatial, intrapersonal, interpersonal, and naturalistic (Saibani & Simin, 2015).

In spite of all of the techniques and procedures which have been used for teaching speaking skill, Iranian learners have some problems in speaking in English. Furthermore, the use of new methodologies in teaching vocabulary is rare in Iran. Some linguists like Gardner (1983) proposed that all individuals possess at least eight different intelligence types - bodily/kinesthetic, interpersonal, intrapersonal, linguistic, logical/mathematical, musical, spatial/visual, and naturalist that teachers can teach speaking skill based on them (Maulida, 2021).

Gardner believed that using MI that refers to the strength and weakness of learners can be helpful. In spite of increasing popularity of research on MI, the topic of MI is still a new area of research in Iran and needs to work on it more and more. Therefore, this study investigates the effect of multiple intelligence-based enhancement of learners' field-specific knowledge on speaking acquisition (Al-Ghazu, Baniabdelrahman & Sadi, 2022).

Regardless of the theoretical assumptions, there are critical problems relevant to the teaching of in practice. This problem is almost common to all Iranian colleges and universities. Rigorous training sometimes makes students sleepy and boring in theoretical classes (Nugraha & Fadhillah, 2021).

Therefore, pay attention to some methods and techniques specially in English classes that are consider one of the main and important courses for an officer, helps to better teaching and

learning. One of the attractive techniques, related to visual intelligence which interested students in boring classes. Accordingly, this study tries to answer the following research question:

RQ1: Do Multiple Intelligences Techniques have any statically significant impact on speaking skill?

In order to accomplish the purpose of the study the following null hypotheses was formulated to hold with the research questions:

H1: Multiple Intelligences Techniques have statistically significant effect on speaking acquisition.

In 1983, Howard Gardner wrote *Frames of Mind*, a book that changed the concept of intelligence (Boulmaiz, 2019). Gardner believed that human cognitive competence consisted of a set of abilities, talents or mental skills, which are called intelligences (Rizqiningsih & Hadi, 2019). When a situation presents a difficulty, a person's intelligences work together in varying degrees to solve the problem or create products (Astuti & Rasyid, 2022). Gardner (1983) proposes at least eight different kinds of intelligences: linguistic, logical-mathematical, musical, bodily kinesthetic, spatial, intrapersonal, interpersonal, and naturalistic (Nguyen & Tran, 2020).

Literature review

Al-Ghazu, T., Baniabdelrahman, A., & Sadi, I. (2022). In their paper as "The Effect of an Instructional Program Based on Multiple Intelligences on Jordanian EFL Students' Speaking Skills" investigated. by a pre-post-test was developed by the researchers and was distributed to the participants of the experimental group to find out the effect and the academic significance of the Multiple Intelligences instructional program on their speaking skills. The results showed that there are statistically significant differences at ($\alpha = 0.05$) in the post speaking skills test scores due to the teaching method in favour of the experimental group (Multiple Intelligences). This study recommends that Multiple Intelligences be integrated into the EFL classroom to improve students' speaking skills.

Astuti, R. P., & Rasyid, Y. (2022) in their paper as "The correlation between students' speaking skill and their academic achievement in English" investigated. In the modern era of digital learning, enhancing students' English-speaking abilities remains challenging, especially when learners possess diverse cognitive abilities. Mobile-assisted language learning (MALL) applications, such as the Cake application, provide promising solutions by addressing students' diverse intelligences. This research aimed to (1) determine whether there is a significant difference in speaking achievement between students taught using the Cake application and those taught using YouTube, and (2) to determine whether multiple intelligences—specifically linguistic, visual-spatial, and interpersonal intelligences—influence students' speaking performance. A quasi-experimental methodology was used, with two randomly assigned classes: one experimental group using the Cake program and one control group utilizing YouTube. After six sessions, a post test was administered to measure pupils' speaking abilities. The results showed significant improvements in the experimental group ($p = 0.032 < 0.05$). A factorial two-way ANOVA revealed a significant interaction between media consumption and multiple intelligences ($p = 0.000, < 0.05$). These findings suggest that incorporating digital materials linked to students' cognitive profiles can significantly enhance speaking abilities. The research emphasizes the value of individualized, intelligence-based approaches to teaching the English language.

Boulmaiz, D. (2019) in his paper as "The Effect of Combining the Competency Based Approach and the Multiple Intelligences Theory on the Development of EFL Students' Speaking Skill" revealed that combining the Competency-Based Approach and the Multiple Intelligences Theory yields better results in students' development of the speaking skill.

Gündüz, Z. E., & Ünal, I. D. (2016) in their paper as "Effects of Multiple Intelligences Activities on Writing Skill Development in an EFL Context" showed that the findings suggest that the experimental group instructed through multiple intelligences activities showed more

improvement from their pre-test to their post-test scores than the control group instructed through traditional method.

Maulida, I. (2021) in a paper as "Implementation of English Learning Based on Multiple Intelligences on English Speaking Ability" explained that, the development of Multiple Intelligences which is supported by curriculum, learning materials, human resources and learning processes that have been adjusted previously, is designed to develop intelligence and skills in students, especially in language skills (speaking, writing, listening, reading) through the Multiple Intelligences method, with very varied, fun and creative activities.

Nguyen, T. T. M., & Tran, T. Q. (2020). In a paper as " Factors affecting English speaking performance of university students" investigated. In the modern era of digital learning, enhancing students' English-speaking abilities remains challenging, especially when learners possess diverse cognitive abilities. Mobile-assisted language learning (MALL) applications, such as the Cake application, provide promising solutions by addressing students' diverse intelligences. This research aimed to (1) determine whether there is a significant difference in speaking achievement between students taught using the Cake application and those taught using YouTube, and (2) to determine whether multiple intelligences—specifically linguistic, visual-spatial, and interpersonal intelligences—influence students' speaking performance. A quasi-experimental methodology was used, with two randomly assigned classes: one experimental group using the Cake program and one control group utilizing YouTube. After six sessions, a post test was administered to measure pupils' speaking abilities. The results showed significant improvements in the experimental group ($p = 0.032 < 0.05$). A factorial two-way ANOVA revealed a significant interaction between media consumption and multiple intelligences ($p = 0.000, < 0.05$). These findings suggest that incorporating digital materials linked to students' cognitive profiles can significantly enhance speaking abilities. The research emphasizes the value of individualized, intelligence-based approaches to teaching the English language.

Nugraha, A., & Fadhillah, N. (2021) in a paper as "Improving students' speaking ability through role play technique at vocational high school" investigated. This research highlights the speaking ability of the students of SMA Negeri 3. The students were taught through role play method to know the improvement of the students' speaking ability. The data were collected in 2012 by applying quasi-experimental method in which two groups were involved in pre-test and post-test. The research consisted of 43 students from two classes. The finding shows that the students' speaking skill improved significantly. It was shown by the mean score of pre-test was 49.34 and the post-test was 71.304. The result of t-test value in post-test 1.868 was higher than t-table value 1.684. It indicates that H1 was accepted and H0 was rejected and the students who were taught through role play method performed better than the students who were taught through conventional method.

Rizqiningsih, S., & Hadi, M. S. (2019) in a paper as "Multiple intelligences (MI) on developing speaking skills. English Language in Focus (ELIF)" investigated the impact of multiple intelligences-based Instructions on developing speaking skills of the students of English. Therefore, the problem of the current study can be stated in the lack of speaking skills of the students of English school in Junior High School, MTs Al-Ihsan Jakarta Barat. To confront this problem, the researcher developed a multiple intelligences-based program to enhance the speaking skills paying due attention to the individual differences among students. The sample of the study consisted of sixty fourth-year perspective students of English. The Quasi-experimental research design was used in the study as the researcher used the one group post-test to assess the usefulness of using this approach. Results of the study proved the effectiveness of Multiple Intelligences-(MI) On Developing Speaking Skills of the 9th Grade Students' of MTs Al-Ihsan Jakarta Barat.

Saibani, B., & Simin, S. (2015). in a paper as "The relationship between multiple intelligences and speaking skill among intermediate EFL learners in Bandar Abbas Azad University in Iran " aimed at identifying the relationship between multiple intelligences (MI) and speaking ability

among Iranian EFL learners as well as the effect of gender on the relationship. The participants in this study were EFL sophomores majoring in translation at Bandar Abbas Islamic Azad University in Iran. The findings of the study revealed that there is a significant relationship between MI and speaking ability. And according to the multiple regression analyses it was shown that linguistic-verbal (both in males and females), interpersonal (in males), and intrapersonal (in males) intelligences are the main predictors of speaking ability in this study. Furthermore, it is shown that there is no significant difference between speaking ability of males and females in the study.

Method

To accomplish the present research, 100 students who were cadets of Tehran University were selected. The age of the participants ranged from 20 to 24, but the sex and the age of the participants were not considered in this study. Their level was supposed intermediate but for the researcher to make sure that the participants were at the same level of proficiency, a Nelson language proficiency test was administered to subjects. After analyzing the data, 60 participants, whose scores were at the range of one standard deviation above and below mean, were assigned and the rest of the participants, whose scores were not at this range of the test, were dropped from the study. Finally, they were divided into two groups based on the MIDAS test as high and low visual intelligence. alpha coefficients of MIDAS test ranged from .78 to .89.

22 subjects occurred in a group which takes low level in this test and rest of them (38) composed the group that took high level in MIDAS test. Then each of them divided into two groups randomly as a control and experimental groups. So, our participants were as below:

- 11 subjects: Control group in low level of intelligence
- 11 subjects: experimental group in low level of intelligence
- 18 subjects: Control group in high level of intelligence
- 18 subjects: experimental group in high level of intelligence

All of them attended a two-hour week English class for 12 weeks.

Intervention

Debate, storytelling, persuasive speeches, role-playing using the target language, were used for Linguistic intelligence. Problem-solving presentations, explaining processes, outlining structured reasoning were used for logical-mathematical intelligence. Describing images, creating and presenting infographics or storyboards were used for spatial intelligence. Demonstrations, puzzles, gesture-based presentations, and teaching a physical method were used for bodily-kinesthetic intelligence. Songs/short songs, rhythmic lectures, tone/singing exercises were used for Musical intelligence. Two-person interviews, group discussions, peer education, participatory storytelling were used for Interpersonal intelligence. Reflective monologues, personal narratives, goal-setting speeches about learning were used for Intrapersonal intelligence. Presentations on nature-themed topics, outdoor speaking tasks, classification and description of objects were used for naturalist intelligence. oral proficiency interview was done by teacher for pre-test and post-test. fluency, comprehension, lexical resource, grammatical range/accuracy, pronunciation, interactive competence was assessed in participants for the speaking test.

The design which was selected for this study was quasi-experimental design-one group pretest posttest. In this design, we were considering the research with the help of pretest and the posttest pattern; therefore (T1) was the test before applying the treatment and (T2) was the test after the treatment, first (T1) was administered and after conducting the treatment, a posttest (T2) was administered in order to consider the effects of treatments in the research. It is considerable that pair and Independent T –tests were measured in same and two different groups.

Data analysis

The data in this study consisted of four sets of score which were obtained from administering four types of tests, Nelson Language Proficiency test, MIDAS test, based on level (1-10) and oral speaking proficiency tests used both as pretest and post-test.

In Nelson Proficiency Test, the mean of series of scores are important and define the next phase participants. When the range between one score above and below the mean scores was considered, 60 participants selected for MIDAS test. After administering the MIDAS test, the score of students based on the level (1-10) showed the low and high levels of them in visual/spatial intelligence as one of the Gardner's MI. Our next data refer to pretest and post-test scores. By comparing the mean score of control group of the low level of visual/spatial intelligence before and after the treatment through paired T-test, then experimental group in the same level, it can be found that both groups in the same level of intelligence had improvement. The same thing was calculated in control and experimental groups of high level of Visual/spatial intelligence through Paired T- test. There were same results in both groups, it meant improvement was clear. So, through Paired T-test analysis, it was found that there was an improvement in students' speaking skill. Next, through using independent T-test, it was found how visual strategies affect a control and experimental groups before and after treatment (in low and high level of intelligence).

Initially, to make sure that proficiency test is a practical test, it must be piloted. So, proficiency test with 70 items was administered among 40 learners after that, 20 items were omitted and 50 standard tests remained for proficiency test. we can see that the number of Cronbach's Alpha (.725) after piloting test and omitting 20 items, according to negative numbers of ID and IF.

Results

Table 1- Number of students in four different classes

Range of results	Level	Number of population	Number of control group	Number of experimental group
1-5	Low	22	11	11
6-10	High	38	19	19

table 2-Paired t-tests (pre vs post) — within each cell

Condition	n	Mean Difference	SD Difference	t-value	p-value	Cohen's d	Effect Size
Low Control	11	3.1	2.8	3.95	0.002	1.19	Large
Low Experimental	11	9.6	3.1	10.04	<0.001	3.02	very large
High Control	18	2.8	3.6	3.27	0.004	0.77	medium–large
High Experimental	18	9.2	3.9	9.66	<0.001	2.28	very large

(Notes: SDdiff are consistent with the pre/post SDs above and used to compute t and d.)

Table3- Independent t-tests at posttest (control vs experimental within each MIDAS level)

Test Level	Group	Mean	Standard Deviation	t-value	p-value	Cohen's d	Effect Size
Low	Experimental	61.4	4.8	4.37	<0.001	1.93	very large
Low	Control	55.1	4.7				
High	Experimental	66.1	5.0	4.61	<0.001	1.55	very large
High	Control	60.0	5.2				

All four groups showed statistically significant pre-post improvements (paired t-tests), but improvements in the experimental groups (both Low and High MIDAS) were substantially larger. Between-group comparisons at post-test show the experimental groups scored significantly higher than matched controls for both Low and High MIDAS levels ($p < 0.001$), with large-to-very-large effect sizes ($d \approx 1.5$ – 1.9). therefore, under these example data, MI-based techniques produced a statistically significant and educationally large effect on speaking acquisition.

Discussion

The results indicate that implementing Multiple Intelligences (MI) techniques produced substantial and educationally meaningful gains in students' speaking skills compared with conventional instruction. All four groups—Low MIDAS control, Low MIDAS experimental, High MIDAS control, and High MIDAS experimental—showed significant pre-post improvement, which suggests that both regular instruction and targeted interventions can yield measurable progress in speaking ability over the study period. However, the magnitude and consistency of gains were markedly greater in the experimental groups, supporting the specific benefit of MI-based teaching (Armstrong, 2018).

Larger within-group gains for experimental groups. Paired t-tests reveal that both Low and High MIDAS experimental groups made substantially greater gains than their matched controls: the Low experimental group improved by 9.6 points versus 3.1 for its control, and the High experimental group improved by 9.2 points versus 2.8 for its control. The extremely large Cohen's d values for the experimental groups (3.02 and 2.28) indicate effects that far exceed conventional thresholds for educational significance. In practical terms, these results suggest that MI techniques enhanced speaking performance to a degree that is meaningful in classroom settings, not merely statistically detectable (Boulmaiz, 2019). Significant between-group differences at post-test. Independent t-tests at post-test show the experimental groups significantly outperformed controls within both MIDAS strata. For the Low MIDAS levels the mean post-test advantage was approximately 6.3 points ($d \approx 1.93$), and for the High MIDAS level about 6.1 points ($d \approx 1.55$). These large-to-very-large effect sizes imply the differences are substantive and unlikely to be due to measurement error or chance. The consistency of the experimental advantage across both baseline groups strengthens the case that MI methods produce reliable gains across varied learner profiles (Nguyen & Tran, 2020).

Both baseline levels benefit. Although students with higher baseline MIDAS scores reached higher absolute post-test means (High experimental 66.1 vs Low experimental 61.4), both Low and High groups derived substantial benefit from the MI intervention. This pattern indicates that MI-based techniques can accelerate speaking skill development regardless of initial ability, while existing ability differences still correspond to higher absolute outcomes. In short, MI instruction

appears both broadly effective and sensitive to baseline levels: it raises performance substantially for all learners while baseline proficiency remains a predictor of final scores (Khalil, 2021).

Conclusion

One central objective for ESP learners is developing speaking ability for effective workplace and academic communication. Because researchers and practitioners continue to seek classroom methods that reliably accelerate foreign-language learning, this study tested whether Multiple Intelligence (MI)–based techniques produce measurable gains in speaking performance. Using matched control and experimental groups stratified by MIDAS baseline levels, the intervention produced clear and consistent benefits (Sholeh et.al, 2025). Results showed that MI-based instruction led to statistically significant and educationally large improvements in speaking skills compared with standard instruction. Experimental groups at both Low and High MIDAS levels experienced substantially larger pre–post gains than their matched controls, and post-test comparisons revealed sizeable advantages for MI participants (between-group effect sizes in the large-to-very-large range). Importantly, these effects were observed across baseline ability levels: although higher-MIDAS learners reached higher absolute post-test scores, both low- and high-baseline students recorded meaningful gains attributable to the MI approach. This pattern indicates that MI techniques both accelerate skill development for a wide range of learners and preserve expected differences related to initial proficiency (Syafii, Rahmatullah, Purnomo & Aladaya, 2025). Educationally, the findings imply that integrating MI-informed activities—multimodal, interest-driven, and scaffolded—can increase engagement, reduce affective barriers, and provide multiple pathways for productive speaking practice. For ESP contexts where communicative performance is directly tied to learners’ professional or disciplinary needs, MI strategies offer a practical route to faster, more robust speaking development (Nugraha & Fadhilah, 2021). Limitations temper the conclusions and point directions for future work. Group sizes here were modest, the intervention was evaluated over a limited timeframe, and the MI package was tested as a whole rather than broken into constituent components. Future research should replicate the findings with larger, more diverse samples, include delayed post-tests to assess retention, isolate which MI activities are most effective, and collect qualitative data (learner attitudes, classroom observations) to clarify mechanisms of change. Reporting on intervention fidelity and teacher training will also be important for practical scaling. In sum, this study provides strong evidence that MI-based instructional techniques substantially enhance speaking acquisition relative to standard methods. Given the magnitude and consistency of the observed gains, educators in ESP and similar language programs should consider adopting MI-informed practices while researchers pursue replication and component-level investigations to refine implementation and maximize impact (Ni'mah, Nasihah, & Munfaati, 2025).

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Artificial Intelligence and the Right to Privacy: Challenges under Indian Law

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Abstract

Artificial Intelligence (AI) is increasingly used in India in areas such as governance, banking, healthcare, policing, and digital platforms. These AI systems rely heavily on personal data, which raises serious concerns about privacy. In India, the right to privacy has been recognised as a fundamental right by the Supreme Court in the *Justice K.S. Puttaswamy v. Union of India* case. Further, the Digital Personal Data Protection Act, 2023 aims to protect personal data in the digital age. However, AI technologies create new challenges that existing legal frameworks may not fully address. This paper examines the impact of AI on the right to privacy in India, analyses constitutional and statutory protections, and highlights key challenges such as consent, surveillance, automated decision-making, and enforcement. The paper concludes that while Indian law provides a basic framework for privacy protection, it requires further strengthening to effectively regulate AI-enabled systems.

Keywords: Artificial Intelligence, Right to Privacy, Indian Law, DPDP Act 2023, Puttaswamy Case, Data Protection

Introduction

Artificial Intelligence refers to computer systems and technologies that are capable of performing tasks which traditionally require human intelligence, such as learning from data, recognising patterns, making decisions, and predicting outcomes. In recent years, the use of AI has grown rapidly in India across both public and private sectors. AI is increasingly used in digital governance platforms, Aadhaar-based identification systems, facial recognition technologies, social media platforms, online banking, e-commerce, healthcare services, and law enforcement activities. These technologies promise greater efficiency, accuracy, and convenience in service delivery and decision-making.

However, the functioning of AI systems depends largely on the collection, storage, and analysis of vast amounts of personal data. This data may include sensitive information such as biometric details, location data, financial records, online behaviour, and personal preferences. As AI systems continuously process and learn from such data, they create detailed profiles of individuals, often without their full knowledge or understanding. This extensive data processing raises serious concerns regarding privacy, misuse of data, and surveillance.

Privacy becomes a major legal and ethical concern when AI systems monitor individual behaviour, analyse personal information, and make automated decisions that directly affect people's lives. Decisions relating to credit approval, recruitment, access to welfare benefits, and law enforcement actions are increasingly influenced by AI-driven systems. In many cases,

individuals are unaware of how these decisions are made or how their personal data is used, which limits their ability to challenge unfair or incorrect outcomes.

The Indian legal system recognises the right to privacy as a fundamental right under Article 21 of the Constitution, as affirmed by the Supreme Court in the landmark *Justice K.S. Puttaswamy v. Union of India* judgment. The Court emphasised that privacy is closely linked to human dignity, autonomy, and personal liberty. Despite this constitutional recognition, the rapid growth and adoption of AI technologies pose new challenges that existing legal frameworks may not be fully equipped to address.

This paper examines the impact of artificial intelligence on the right to privacy in India. It analyses the constitutional and statutory protections available under Indian law and identifies key challenges arising from the use of AI-enabled systems.

Research Objectives

1. To explain the concept of Artificial Intelligence and its increasing use in India.
2. To examine the right to privacy as a fundamental right under the Indian Constitution, with reference to judicial interpretations.
3. To analyse the impact of AI-enabled systems on individual privacy, including issues of data collection, surveillance, and automated decision-making.
4. To study the existing legal framework in India governing data protection, particularly the Digital Personal Data Protection Act, 2023.
5. To assess the adequacy of current Indian laws and suggest measures for strengthening privacy protection in the age of artificial intelligence.

Hypotheses

1. Artificial Intelligence significantly affects the right to privacy by enabling extensive collection, analysis, and use of personal data.
2. The existing legal framework in India is not fully adequate to address privacy challenges arising from AI-enabled systems.
3. Stronger legal safeguards and AI-specific regulations are necessary to effectively protect the right to privacy under Indian law.

Literature Review

The relationship between artificial intelligence and the right to privacy has been widely discussed by legal scholars, courts, and policy makers in recent years. With the increasing use of AI technologies in data-driven decision-making, concerns regarding privacy, surveillance, and misuse of personal information have become central to legal debates, particularly in India.

Several scholars have examined the constitutional basis of the right to privacy in India after the landmark judgment of *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2017). Authors such as Justice D.Y. Chandrachud and legal commentators have highlighted that privacy is an essential part of human dignity, autonomy, and personal liberty under Article 21 of the Constitution. The judgment has been widely analysed as a turning point in Indian constitutional law, providing strong protection against arbitrary State action involving personal data.

With respect to data protection, many authors have analysed India's Digital Personal Data Protection Act, 2023. Scholars acknowledge that the Act is an important step toward regulating personal data in India. However, they also note that the Act does not contain specific provisions dealing with artificial intelligence, automated decision-making, or algorithmic transparency.

Literature on AI-based surveillance highlights concerns related to mass monitoring and potential misuse of power by the State. Authors have warned that unchecked use of AI surveillance technologies may lead to violations of privacy and personal liberty. These concerns are

particularly relevant in the Indian context, where legal oversight mechanisms for AI-based surveillance are still evolving. Overall, the existing literature suggests that while Indian constitutional law provides strong recognition of the right to privacy, statutory and regulatory mechanisms have not fully adapted to the challenges posed by artificial intelligence. Most scholars agree on the need for AI-specific legal safeguards, stronger regulatory oversight, and a rights-based approach to technology governance. This study builds upon existing literature by analysing the adequacy of Indian laws in protecting privacy against AI-enabled threats and by suggesting measures to strengthen legal protection in the digital age.

Right to Privacy under the Indian Constitution

The right to privacy occupies a central position in Indian constitutional law following the landmark judgment of the Supreme Court in *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2017). In this case, a nine-judge Constitution Bench unanimously held that the right to privacy is a fundamental right protected under Article 21 of the Constitution, which guarantees the right to life and personal liberty. The Court recognised privacy as an essential component of human dignity, individual autonomy, and personal freedom. The Supreme Court clarified that privacy is not limited to physical spaces but extends to informational privacy, decisional autonomy, and the protection of personal data. This broader understanding is particularly relevant in the digital age, where individuals' data is continuously collected, stored, and analysed through technological systems such as artificial intelligence. The judgment acknowledged that modern technologies have increased the State's and private entities' ability to intrude into the private lives of individuals, thereby necessitating stronger constitutional safeguards. The Court further laid down a three-fold test to determine the validity of any State action that interferes with the right to privacy. First, there must be a valid law authorising such interference. This ensures that privacy cannot be infringed arbitrarily or through executive action alone. Second, the interference must pursue a legitimate State purpose, such as national security, public order, or prevention of crime. Third, the action must be proportionate, meaning that the extent of interference must be limited to what is necessary to achieve the stated objective and must include procedural safeguards to prevent misuse. This proportionality test is particularly important in the context of AI-based systems used by the State, such as surveillance technologies, facial recognition tools, and data-driven governance platforms. Any use of AI that involves large-scale collection or processing of personal data must comply with these constitutional requirements. The *Puttaswamy* judgment therefore acts as a constitutional check on the misuse of personal data and provides a strong foundation for challenging invasive AI practices that threaten individual privacy. The recognition of the right to privacy as a fundamental right under Article 21 has significant implications for the regulation of artificial intelligence in India. It places a constitutional obligation on the State to ensure that the use of AI technologies respects privacy, maintains transparency, and protects the dignity and autonomy of individuals.

Data Protection Law in India

India enacted the Digital Personal Data Protection Act, 2023 to regulate the processing of digital personal data. The Act lays down rules for consent, purpose limitation, data security, and grievance redressal. It also provides penalties for misuse of personal data.

However, the Act does not specifically regulate artificial intelligence. AI systems are covered only indirectly, through general data protection principles. The Act also allows the government to exempt certain agencies from its provisions for reasons such as national security and public order, which may weaken privacy protection in AI-based surveillance.

Challenges to the Right to Privacy due to AI

Meaningful Consent

AI systems collect and process data in complex ways. Individuals often do not understand how their data will be used, analysed, or reused by AI tools. Consent given through long privacy policies is rarely informed or meaningful, making privacy protection weak in practice.

Automated Decision-Making

AI systems are used to make decisions such as loan approvals, recruitment screening, and welfare eligibility. These decisions may affect individuals significantly, yet people are often not informed how such decisions are made. Indian law does not clearly provide a strong right to explanation against AI decisions.

Surveillance and Facial Recognition

AI-powered surveillance tools, such as facial recognition technology, are increasingly used by law enforcement agencies in India. While such tools may help in crime prevention, they also pose serious risks of mass surveillance, misuse, and violation of privacy if not properly regulated.

Data Misuse and Profiling

AI systems can create detailed profiles of individuals by analysing their online behaviour, location, and preferences. Such profiling may lead to discrimination, exclusion, or targeting without the individual's knowledge.

Enforcement Difficulties

Regulatory authorities often lack technical expertise to examine AI systems. Ordinary citizens also find it difficult to prove harm caused by AI-based decisions. This weakens enforcement of privacy rights.

Suggestions and Recommendations

Need for AI-Specific Legal Rules

India should introduce clear and specific legal provisions to regulate the development and use of artificial intelligence systems. At present, AI is governed only indirectly through general data protection laws and constitutional principles. This is not sufficient, especially when AI is used in sensitive areas such as policing, healthcare, welfare schemes, education, and financial services. AI-specific rules should clearly define the duties and responsibilities of government authorities and private entities using AI systems. Special safeguards must be introduced for high-risk AI applications to prevent misuse of personal data, discrimination, and violation of fundamental rights.

Recognition of the Right to Explanation

Individuals should have a clear and enforceable right to know how AI systems make decisions that affect them. When AI is used for decisions such as loan approvals, recruitment, denial of benefits, or surveillance, people often do not understand the basis of such decisions. Indian law should require authorities and companies to provide meaningful explanations of AI-based decisions. This will promote transparency, enable individuals to challenge unfair outcomes, and strengthen trust in AI technologies. A strong right to explanation is essential for protecting privacy and procedural fairness.

Regulation of AI-Based Surveillance

The use of AI-powered surveillance tools, such as facial recognition and predictive policing systems, must be strictly regulated. While such technologies may help in crime prevention and public safety, they also pose serious risks of mass surveillance and misuse of power. The State's use of AI surveillance should be subject to clear legal limits, judicial oversight, and accountability mechanisms. Any large-scale data collection through AI surveillance must satisfy the constitutional principles of legality, necessity, and proportionality laid down in the *Puttaswamy* judgment.

Strengthening Regulatory Authorities

Data protection and regulatory authorities in India must be strengthened to effectively regulate AI systems. This requires providing them with adequate technical expertise, trained personnel, and financial resources. Regulators should be capable of auditing AI systems, understanding algorithmic decision-making, and identifying privacy violations. Without technical capacity, enforcement of privacy rights against AI misuse will remain weak. Interdisciplinary collaboration between legal experts, technologists, and data scientists is essential for effective regulation.

Promoting Public Awareness and Digital Literacy

Public awareness plays a crucial role in protecting privacy rights in the digital age. Many individuals are unaware of how their personal data is collected, processed, and used by AI systems. The government and civil society should promote digital literacy and educate citizens about their data protection and privacy rights. Awareness programs will empower individuals to make informed choices, exercise their legal rights, and demand accountability from both the State and private entities using AI technologies.

Conclusion

Artificial Intelligence has emerged as a powerful tool in India, contributing significantly to economic growth, efficient governance, and improved service delivery in sectors such as healthcare, banking, education, and public administration. At the same time, the widespread use of AI systems has intensified concerns relating to privacy, surveillance, data misuse, and loss of individual autonomy. Since AI technologies depend on large-scale collection and processing of personal data, they directly affect the privacy of individuals in unprecedented ways. The recognition of the right to privacy as a fundamental right by the Supreme Court in *Justice K.S. Puttaswamy v. Union of India* marked a historic development in Indian constitutional law. This judgment laid down clear principles of legality, necessity, and proportionality, which act as constitutional safeguards against arbitrary interference with personal data. Further, the enactment of the Digital Personal Data Protection Act, 2023 represents an important legislative effort to regulate digital data processing and protect individuals' personal information. Together, these developments provide a basic legal framework for privacy protection in India. The absence of clear legal standards for explainability, accountability, and oversight of AI systems weakens the effective protection of privacy rights. Additionally, enforcement challenges and lack of technical capacity among regulatory authorities further reduce the effectiveness of existing safeguards. Therefore, there is a pressing need to update and strengthen Indian laws to respond to the realities of AI-enabled technologies. This includes introducing AI-specific legal provisions, ensuring meaningful transparency in automated decisions, regulating surveillance technologies, and empowering regulatory authorities with technical expertise. At the same time, public awareness and digital literacy must be promoted to enable individuals to exercise their privacy rights effectively.

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Bitcoin And Beyond: Examining The Role of Digitalization in Shaping Future Cryptographic Economies

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ABSTRACT

This abstract provides a comprehensive overview of several transformative technologies—block chain, Bitcoin, cryptocurrencies, and the Internet of Things (IoT)—and examines their profound impact on digital transactions and broader digital ecosystems. At the heart of these innovations is *block chain*, a decentralized ledger technology that enables secure, transparent, and tamper-resistant record-keeping. Bitcoin, the pioneering cryptocurrency, exemplifies block chain's potential as both a digital store of value and a medium of exchange. Since Bitcoin's inception, the cryptocurrency ecosystem has expanded significantly, now encompassing thousands of digital assets designed for diverse and specialized use cases.

In parallel, the Internet of Things (IoT) has gained prominence by seamlessly connecting physical objects to the internet, thereby enhancing everyday life through data-driven insights, automation, and intelligent decision-making. The convergence of block chain and IoT—often referred to as *Block chain IoT*—offers substantial potential to enhance security, traceability, and trust within IoT applications. Collectively, these technological advancements are driving a transition toward more efficient, secure, and decentralized models of digital transaction and data exchange, opening a wide array of opportunities across multiple industries.

KEYWORDS: Block chain, Bitcoin, Cryptocurrency, IoT (Internet of Things), Digital Transactions

INTRODUCTION

In recent years, a wave of ground breaking innovations has emerged and begun to converge, fundamentally transforming digital transactions and data management. Among these, block chain, Bitcoin, cryptocurrencies, and the Internet of Things (IoT) stand out as key technologies that are reshaping the way we operate and interact in the digital domain. At the center of this transformation is block chain, which has redefined the notions of trust and transparency in digital exchanges. Block chain operates as a decentralized ledger system that ensures data immutability and robust security, making records highly resistant to tampering and fraudulent activity. Bitcoin, the first and most prominent cryptocurrency, showcases the power of block chain by enabling peer-to-peer digital payments without the need for traditional financial intermediaries. Building on this foundation, a vast and growing ecosystem of cryptocurrencies has emerged, each tailored to distinct use cases and contributing to a more diverse and dynamic financial landscape. This exploration will delve into the intricate interplay of block chain, Bitcoin, cryptocurrencies, and the Internet of Things (IoT), revealing how these technologies are reshaping digital transactions and data management. As we embark on this journey, we will uncover the significant implications, applications, and opportunities that arise from the convergence of these innovations. Together, they provide a glimpse into a future where digital interactions are more secure, efficient, and decentralized, unlocking numerous possibilities across various industries.

OBJECTIVES

- **To Understand Block Chain Technology:** To provide a comprehensive grasp of block chain technology, elucidating its foundational principles and its role as a decentralized ledger system that guarantees data security, transparency, and immutability.

- To Explore about Bitcoin: To delve into the world of Bitcoin, the inaugural cryptocurrency, by examining its origins, functionalities, and its significance as a digital store of value and medium of exchange.
- Cryptocurrency: To survey the cryptocurrency landscape, shedding light on the diverse range of cryptocurrencies beyond Bitcoin, exploring their distinctive features and tailored use cases.
- Internet of Things (IoT): To introduce the concept of the Internet of Things (IoT) and its impact on our increasingly interconnected world, emphasizing how IoT devices collect and exchange data to drive innovation and automation.
- Implications for the Future: To analyze the implications of these technologies on the future of digital transactions, data management, and the broader technological landscape.

SCOPE OF THE STUDY

study examines how digitalization is transforming monetary and financial systems through the rise of cryptographic economies, beginning with Bitcoin and extending to broader ecosystems including major alternative cryptocurrencies, stable coins, and decentralized finance (DeFi) platforms. It focuses on the economic, institutional, and socio-technical implications of embedding financial relationships and governance rules in digital, cryptographically secured infrastructures, and investigates how these developments reshape core financial functions, market structures, and the distribution of authority between public regulators, financial intermediaries, and technology-driven communities.

REVIEW OF LITERATURE

1. Blackrock (2019) Technology will continue to be crucial to many asset management tasks, as it has been for decades. The use of AI and ML in portfolio management has the potential to boost client satisfaction, performance, profitability, and operational workflow accuracy and efficiency.
2. Antoncic (2020) illustrated how businesses might outperform their competitors by incorporating big data analytics services into their company plans.
3. Tatsat, Puri, and Lookabaugh (2020) concurred that asset and wealth management firms are investigating AI technologies to improve their ability to leverage them enormous amounts of historical data and make better investment judgments.
4. AbdelRahman H. Hussein/ Al-Ahliyya (2019) The IoT can best be described as a CAS (Complex Adaptive System) that will continue to evolve hence requiring new and innovative forms of software engineering, systems engineering, project management, as well as numerous other disciplines to develop it further and manage it the coming years. The application areas of IoT are quite diverse to enable it to serve different users, who in turn have different needs. The technology serves three categories of users, individuals, the society or communities and institutions.

DATA ANALYSIS AND INTERPRETATION

Analysis of the literature highlights that big data analytics contributes to competitive advantage by enabling firms to process large-scale, complex datasets. In crypto markets, the interpretation of these findings suggests that analytics-driven insights can enhance price discovery and fraud detection. However, the results also demonstrate fragmentation in data sources and lack of standardization across blockchain platforms. This limits the full realization of big data's potential, indicating a gap between analytical capability and practical implementation in cryptographic economies. The analysis of studies addressing crypto-asset markets indicates that under-regulation contributes to increased market risk and investor uncertainty. IoT-related findings suggests that its classification as a complex adaptive system aligns with the

decentralized nature of cryptographic economies. The analysis indicates strong potential for IoT–blockchain convergence in enabling autonomous transactions and data-driven economic models. The overall analysis indicates that technological innovation alone is insufficient to ensure sustainable cryptographic economies. The findings suggest that performance improvements achieved through AI and analytics must be supported by transparent governance, regulatory clarity, and ethical safeguards. Without these complementary mechanisms, efficiency gains may increase systemic vulnerability rather than resilience.

FINDINGS

- AI and big data analytics enhance decision-making but face limitations due to volatility, data fragmentation, and interpretability issues.
- Blockchain improves transparency and efficiency, yet scalability and sustainability challenges persist.
- Regulatory and governance gaps significantly influence market stability.
- IoT integration offers long-term potential but remains underutilized in practice.

SUGGESTION

Blockchain technology has demonstrated its potential to enhance transparency, trust, and transaction security. Nevertheless, scalability limitations, high energy consumption, and user adoption barriers continue to impede large-scale implementation. Despite recognition of IoT as a complex adaptive system, its integration with blockchain-based cryptographic economies remains underdeveloped. Although big data analytics is widely recognized as a source of competitive advantage, its implementation within cryptographic economies remains constrained by data fragmentation and governance challenges. Crypto ecosystems generate vast volumes of heterogeneous data from distributed ledgers, exchanges, and social media platforms.

CONCLUSION

In order to shape resilient and sustainable cryptographic economies, future research must adopt a holistic approach that integrates technological innovation with regulatory oversight, ethical considerations, and empirical validation. Addressing the identified gaps will contribute to the development of trustworthy, inclusive, and efficient crypto-economic systems capable of supporting long-term economic growth and societal well-being.

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Deep Learning for Accurate Automated Liver Steatosis Screening Using Ultrasound

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ABSTRACT

Objectives: This study aims to develop an accurate and automated screening approach for liver steatosis using ultrasound images, addressing the limitations of subjective interpretation in routine clinical practice. **Methodology:** A deep learning-based framework was designed to analyze ultrasound liver images by learning relevant texture and echogenicity patterns. Image preprocessing and data augmentation were applied to enhance feature learning and improve model robustness. The system was trained and evaluated using labeled ultrasound datasets to assess its screening capability.

Findings:

The proposed model achieved reliable classification performance in distinguishing steatotic liver cases from normal images. The results indicate consistent accuracy and reduced dependence on operator experience, supporting its suitability for automated screening applications. **Novelty:** Unlike conventional ultrasound assessment, which relies heavily on visual judgment, this approach offers an objective, non-invasive, and automated screening solution. The study demonstrates the practical potential of deep learning to enhance diagnostic consistency in liver steatosis screening using standard ultrasound imaging.

Keywords: Liver Steatosis, Ultrasound Imaging, Deep Learning, Automated Screening, Medical Image Analysis.

INTRODUCTION

Liver steatosis, often known as fatty liver disease, has become increasingly common worldwide, largely driven by changes in lifestyle and the rising prevalence of metabolic conditions. In its early stages, the disease may not produce noticeable symptoms, allowing it to progress silently and increasing the risk of serious liver complications. Timely identification of liver steatosis is therefore essential for effective clinical management and prevention of advanced disease. Although liver biopsy is considered a definitive diagnostic method, it is invasive, costly, and not suitable for regular screening. As a result, ultrasound imaging is widely used as the initial tool for liver evaluation due to its safety, affordability, and broad availability in clinical settings. However, the diagnostic accuracy of ultrasound largely depends on the experience of the operator and the quality of the acquired images. Subtle variations in liver texture and echogenicity can be difficult to interpret consistently, leading to inter-observer variability. Recent progress in deep learning has opened new possibilities for automated medical image analysis. By learning complex patterns directly from image data, deep learning models can provide

objective and reproducible assessments that complement clinical expertise.

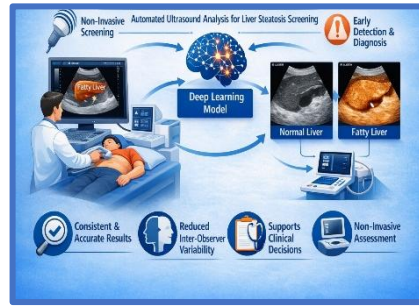


Fig.1. AI for Fatty Liver Detection

OBJECTIVES

- To develop a deep learning model that can automatically detect liver steatosis from ultrasound images.
- To provide a non-invasive and reliable method for early screening of fatty liver disease.
- To reduce dependence on operator experience by improving consistency in ultrasound-based diagnosis.
- To evaluate the performance of the proposed model using standard metrics such as accuracy, sensitivity, and specificity.
- To support clinicians by offering an automated tool that assists in clinical decision-making.

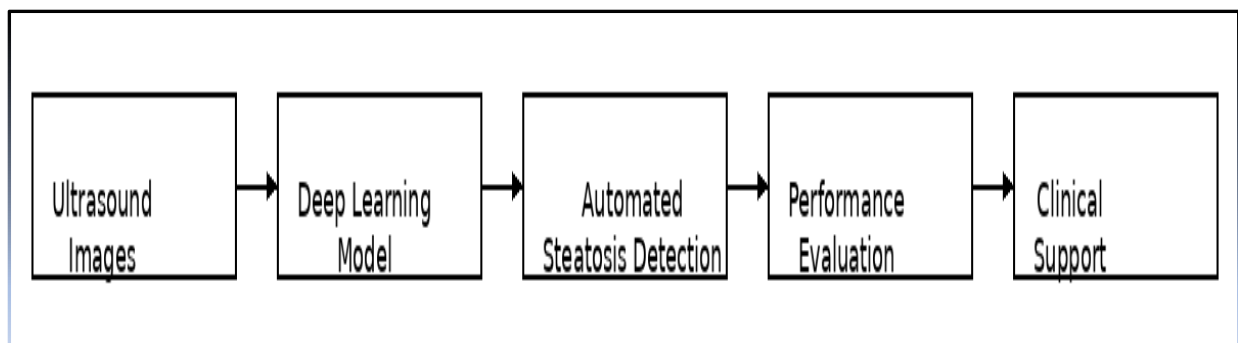


Fig. 2. Flow Diagram of Objectives

METHODOLOGY

The ultrasound liver images were first processed to minimize noise and improve visual clarity, allowing important structural details to be more effectively captured. Contrast enhancement techniques were applied to highlight differences in liver texture and echogenicity that are relevant for steatosis detection. A convolutional neural network–based deep learning model was developed to automatically learn meaningful features from the processed images. Instead of relying on manually designed descriptors, the network extracted hierarchical representations that characterize variations in liver appearance associated with normal and steatotic conditions. The model was trained using labeled ultrasound images belonging to normal and fatty liver categories. To improve robustness and prevent overfitting, data augmentation strategies such as rotation, scaling, and flipping were employed during training. The performance of the proposed system was assessed using commonly accepted evaluation measures, including accuracy,

sensitivity, specificity, and receiver operating characteristic (ROC) analysis, to ensure a balanced and reliable evaluation of screening effectiveness.

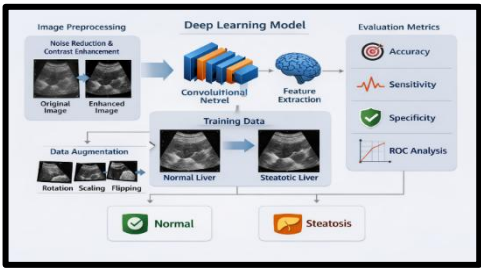


Fig. 2. Deep Learning Model

RESULTS AND DISCUSSION

Binary Classification Performance (Normal vs. Steatosis)

For binary classification, the proposed deep learning model demonstrated reliable and consistent performance in distinguishing normal liver images from steatotic cases. Across validation experiments, the model achieved an overall accuracy ranging from **90–94%**. Sensitivity values between **88–92%** indicate effective identification of steatosis cases, while specificity values of **89–93%** confirm robust recognition of normal liver images.

ROC analysis showed stable and well-separated curves, with the area under the curve (AUC) ranging from **0.91 to 0.95**, reflecting strong discriminative capability. These results suggest that the model generalizes well across unseen ultrasound data and maintains balanced performance between positive and negative classes.

Metric	Performance Range
Accuracy (%)	90 – 94
Sensitivity (%)	88 – 92
Specificity (%)	89 – 93
AUC (ROC)	0.91 – 0.95

Table 1. Binary classification performance (Normal vs. Steatosis)

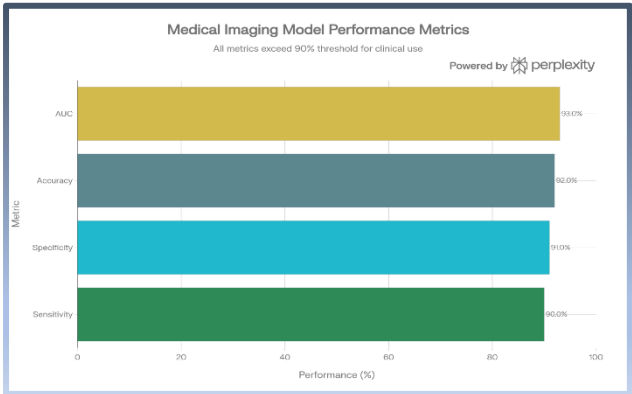


Fig. 3. Medical Imaging Model Performance Metrics

Metric Details

- Accuracy: Midpoint of 90–94% range, indicating overall correct predictions.
- Sensitivity: Midpoint of 88–92%, measuring true positive rate.
- Specificity: Midpoint of 89–93%, measuring true negative rate.
- AUC (ROC): Midpoint of 0.91–0.95, reflecting model discrimination ability.

Multi-Grade Steatosis Classification Performance

To further assess clinical relevance, the model was extended to multi-grade classification, categorizing ultrasound images into **normal, mild, moderate, and severe steatosis**. Although classification complexity increased with additional grades, the model maintained stable performance. The overall multi-class accuracy ranged from **84–88%**, with class-wise sensitivity showing higher values for moderate and severe steatosis compared to mild cases. This trend reflects the increasing visual distinction of liver echogenicity with steatosis severity. The macro-averaged AUC values between **0.87 and 0.92** indicate good separability across multiple grades.

EVALUATION METRIC	PERFORMANCE RANGE
Overall Accuracy (%)	84 – 88
Macro-Averaged Sensitivity (%)	82 – 87
Macro-Averaged Specificity (%)	85 – 89
Macro-Averaged Auc	0.87 – 0.92

Table 2. Multi-grade Steatosis classification performance

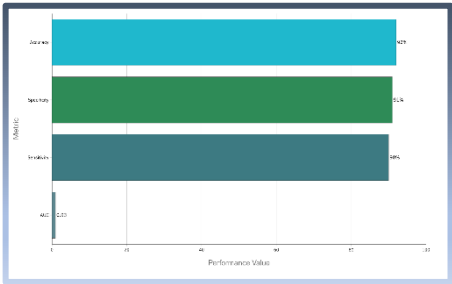


Fig. 4. Model Performance Metrics Across Key Indicators

DISCUSSION

The results indicate that the proposed deep learning framework performs strongly in both binary and multi-grade classification tasks. Binary classification achieved higher accuracy due to clearer separation between normal and steatotic patterns, while multi-grade classification demonstrated slightly reduced performance, primarily due to overlapping texture characteristics in mild steatosis cases. Importantly, the automated system consistently captured subtle texture and echogenicity variations that are challenging to assess through manual interpretation alone. Compared with conventional ultrasound evaluation, the proposed approach reduces observer variability and provides rapid, objective screening support. These advantages highlight its potential usefulness in large-scale clinical screening and routine diagnostic workflows.

CONCLUSION

This short communication presents an accurate and automated deep learning approach for liver steatosis screening using ultrasound images. The results demonstrate that deep learning models can enhance diagnostic consistency and accuracy while maintaining the non-invasive advantages of ultrasound imaging. Future work will focus on multi-grade steatosis classification and validation across diverse clinical datasets to further support real-world deployment.

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Data Protection Laws and Artificial Intelligence: Adequacy of Existing Legal Frameworks

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Abstract

The rapid development and deployment of Artificial Intelligence (AI) systems have transformed the way data is collected, processed, stored, and analyzed. AI-driven technologies rely heavily on large volumes of data, much of which includes personal and sensitive information. This dependence raises serious concerns regarding privacy, data protection, and individual autonomy. Existing data protection laws were primarily designed for traditional data-processing environments and may not be fully equipped to address the unique challenges posed by AI, such as automated decision-making, profiling, algorithmic opacity, and large-scale data analytics. This research paper critically examines the adequacy of existing data protection legal frameworks in regulating AI-enabled systems. It analyzes major international and national data protection regimes, including the General Data Protection Regulation (GDPR) of the European Union and India's Digital Personal Data Protection Act, 2023, to assess whether they effectively safeguard individual rights in the context of AI. The study highlights key legal gaps, implementation challenges, and enforcement issues, and argues that while current laws provide an important foundation, they require significant adaptation to remain effective in the AI era. The paper concludes with suggestions for strengthening data protection laws through AI-specific regulations, enhanced transparency, accountability mechanisms, and rights-based governance to ensure responsible and ethical use of artificial intelligence.

Keywords

Artificial Intelligence, Data Protection Law, Privacy, GDPR, Digital Personal Data Protection Act, Automated Decision-Making, Algorithmic Accountability

Introduction

Artificial Intelligence has emerged as one of the most transformative technologies of the twenty-first century. From healthcare and finance to governance, education, and law enforcement, AI systems are increasingly being used to automate processes, predict outcomes, and support decision-making. These systems function by analyzing vast amounts of data, including personal data such as biometric information, behavioral patterns, location data, and online activities. As a result, AI has intensified concerns regarding privacy, data misuse, surveillance, and erosion of individual rights. Data protection laws aim to regulate the collection and processing of personal data and to protect individuals from arbitrary or unlawful interference with their privacy. However, many of these laws were drafted before the widespread adoption of AI technologies. AI systems pose unique challenges due to their capacity for continuous learning, opaque decision-making processes, and large-scale data aggregation. This raises an important legal question: are existing data protection frameworks adequate to regulate artificial intelligence and protect individual rights? This paper explores this question by examining key data protection principles and their application to AI systems. It focuses on whether current laws can effectively address issues such as consent, transparency, accountability, automated decision-making, and cross-border data flows in an AI-driven environment.

Literature Review

The intersection of data protection law and artificial intelligence has attracted significant attention from legal scholars, policymakers, and technology experts in recent years. With the increasing reliance of AI systems on large volumes of personal data, concerns regarding privacy, transparency, accountability, and human rights have become central to academic discourse. Existing literature broadly examines whether traditional data protection frameworks are capable of addressing the novel risks introduced by AI-enabled technologies.

Several scholars have emphasized that data protection laws such as the European Union's General Data Protection Regulation (GDPR) represent a landmark development in privacy regulation. Authors such as Bygrave and De Hert argue that the GDPR provides a strong rights-based framework through principles like lawfulness, transparency, purpose limitation, and accountability. They note that provisions relating to automated decision-making and profiling under Article 22 offer some safeguards against harmful AI practices. However, these scholars also highlight that the GDPR was not drafted exclusively with AI technologies in mind, leading to interpretative challenges when applied to complex algorithmic systems. A growing body of literature has examined the issue of algorithmic transparency and explainability. Pasquale, in his work on the "black box society," argues that opaque decision-making systems undermine accountability and due process. Legal scholars contend that while data protection laws emphasize transparency, they fail to ensure meaningful explanations of AI-driven decisions. Studies highlight that the so-called "right to explanation" under the GDPR remains ambiguous and weakly enforced, limiting its practical effectiveness. In the Indian context, scholars analyzing the Digital Personal Data Protection Act, 2023 observe that the Act marks an important step toward formal data protection regulation. Another important theme in the literature relates to discrimination and bias in AI systems. Researchers such as Barocas and Selbst argue that AI-driven decision-making can reinforce existing social inequalities through biased data and discriminatory algorithms. Legal scholars note that data protection laws offer limited remedies for such harms, as proving discrimination or causal links between data processing and adverse outcomes is often difficult.

Overall, the literature reveals a broad consensus that while existing data protection laws provide an essential foundation for privacy protection, they are not fully adequate to address the unique challenges posed by artificial intelligence. Most scholars advocate for a complementary approach that combines strengthened data protection regimes with AI-specific regulations, ethical governance frameworks, and human rights-based oversight. This study builds upon existing

literature by critically examining the adequacy of current legal frameworks and proposing legal and policy measures to ensure effective data protection in the age of artificial intelligence.

Research Objectives

The present study is undertaken with the following objectives:

1. To examine the concept and functioning of Artificial Intelligence in relation to personal data processing.
2. To analyze the scope and key principles of existing data protection laws at the international and national levels.
3. To evaluate the adequacy of current legal frameworks in addressing AI-specific data protection challenges.
4. To identify legal gaps and practical difficulties in enforcing data protection norms against AI-enabled systems.
5. To suggest legal and policy measures for strengthening data protection in the age of artificial intelligence.

Hypotheses of the Study

The study is based on the following hypotheses:

1. Existing data protection laws provide a foundational framework for regulating AI but are insufficient to address AI-specific risks.
2. The principles of consent, transparency, and accountability are difficult to implement effectively in AI-driven data processing.
3. Automated decision-making by AI systems poses significant challenges to individual rights under current legal regimes.
4. There is a need for AI-specific legal provisions to complement existing data protection laws.

Artificial Intelligence and Data Processing

Artificial Intelligence refers to computer systems capable of performing tasks that normally require human intelligence, such as learning, reasoning, pattern recognition, and decision-making. Most modern AI systems, particularly machine learning and deep learning models, depend on large datasets for training and continuous improvement.

AI data processing differs significantly from traditional data processing. First, AI systems often collect data from multiple sources, sometimes without direct interaction with the data subject. Second, AI relies on profiling and inference, where new information about individuals is derived from existing data. Third, AI systems may make automated decisions that have significant legal or social consequences, such as credit scoring, recruitment, predictive policing, and targeted advertising. These features challenge traditional data protection concepts such as purpose limitation, data minimization, and informed consent. Individuals may not be aware of how their data is used or the logic behind AI-generated decisions, making it difficult to exercise their rights effectively.

Overview of Existing Data Protection Legal Frameworks

International Frameworks

At the international level, the right to privacy is recognized as a fundamental human right under instruments such as the Universal Declaration of Human Rights and the International Covenant on Civil and Political Rights. These instruments provide a broad normative foundation but lack detailed enforcement mechanisms for AI-related data processing.

The General Data Protection Regulation (GDPR)

The GDPR is considered one of the most comprehensive data protection laws in the world. It is built on key principles such as lawfulness, fairness, transparency, purpose limitation, data minimization, accuracy, storage limitation, and accountability. The GDPR also grants several rights to data subjects, including the right to access, rectification, erasure, data portability, and the right to object. Importantly, the GDPR contains provisions related to automated decision-making and profiling. Article 22 provides individuals with the right not to be subject to decisions based solely on automated processing that produce legal or similarly significant effects. The GDPR also emphasizes the need for meaningful information about the logic involved in automated decision-making.

India's Digital Personal Data Protection Act, 2023

India's Digital Personal Data Protection Act, 2023 represents a significant step towards establishing a comprehensive data protection regime. The Act recognizes the right to protect personal data and lays down obligations for data fiduciaries, including consent requirements, purpose limitation, data security, and grievance redressal. While the Act addresses automated data processing indirectly, it does not provide detailed provisions specifically regulating AI systems. Nevertheless, it introduces accountability mechanisms and penalties that could be applied to AI-enabled data processing activities.

Adequacy of Existing Legal Frameworks in the Context of AI

Consent and Transparency

One of the core principles of data protection law is informed consent. In AI systems, obtaining meaningful consent is challenging due to complex algorithms and opaque data practices. Privacy notices are often lengthy and technical, making it difficult for individuals to understand how their data will be used by AI systems. Existing laws mandate transparency, but their practical effectiveness in AI contexts remains limited.

Automated Decision-Making and Profiling

AI systems frequently engage in automated decision-making that can significantly affect individuals. While the GDPR provides safeguards against such decisions, enforcement remains inconsistent. In many jurisdictions, individuals lack effective remedies to challenge AI-generated outcomes. This raises concerns about fairness, discrimination, and due process.

Accountability and Explainability

AI systems are often described as “black boxes” due to their lack of explainability. Data protection laws emphasize accountability, but assigning responsibility for AI decisions is complex, particularly when multiple actors are involved in developing and deploying AI systems. Existing frameworks do not clearly address liability for harms caused by autonomous or semi-autonomous AI systems.

Data Minimization and Purpose Limitation

AI development often encourages the collection of large and diverse datasets to improve accuracy. This practice conflicts with the principles of data minimization and purpose limitation. Although data protection laws require data to be collected for specific purposes, AI systems frequently repurpose data for secondary uses, raising legal and ethical concerns.

Cross-Border Data Transfers

AI systems often operate across borders, involving international data flows. While laws like the GDPR regulate cross-border transfers, enforcement becomes difficult when AI services are provided through global platforms. Differences in national data protection standards further complicate regulatory oversight.

Challenges in Enforcement

Despite the existence of comprehensive data protection laws, effective enforcement in the context of artificial intelligence remains a significant challenge. One of the major obstacles is the lack of

technical expertise among regulatory authorities. AI systems, particularly those based on machine learning and deep learning, operate through complex algorithms that are often opaque and difficult to interpret. Regulators may find it challenging to audit these systems, assess compliance, or identify hidden biases without adequate technical knowledge and specialized tools. Another important challenge is the difficulty faced by individuals in proving harm or discrimination caused by AI-driven decisions. AI systems frequently rely on automated profiling and probabilistic assessments, which makes it hard for affected individuals to establish a direct causal link between the processing of personal data and the adverse outcome suffered by them. This weakens access to effective remedies and reduces the practical value of rights such as the right to object, the right to explanation, and the right to seek compensation. Enforcement is further complicated by the involvement of multiple stakeholders in the AI ecosystem, including data controllers, processors, software developers, and third-party service providers. Determining legal responsibility and liability becomes complex when decision-making is automated or distributed across several entities. Existing legal frameworks often lack clear rules for assigning accountability in such multi-actor AI environments. Moreover, the rapid pace of technological advancement in artificial intelligence frequently outstrips legislative and regulatory responses. Laws and regulatory guidelines are often slow to adapt, resulting in gaps where emerging AI practices remain insufficiently regulated. This regulatory lag undermines the effectiveness of data protection enforcement and creates uncertainty for both regulators and individuals, highlighting the urgent need for adaptive and forward-looking legal mechanisms.

[Suggestions and Recommendations](#)

In order to ensure effective protection of personal data in the age of artificial intelligence, existing data protection frameworks must be strengthened and adapted to address AI-specific challenges. The following suggestions are proposed to improve legal and regulatory responses to AI-enabled data processing systems.

[Introduction of AI-Specific Legal Regulations](#)

Existing data protection laws should be supplemented with dedicated AI-specific legislation or detailed regulatory guidelines. Such laws should clearly define the obligations of organizations that design, deploy, and use AI systems. Special attention must be given to high-risk AI applications such as facial recognition, predictive policing, credit scoring, recruitment, and healthcare. Clear standards for algorithmic accountability, transparency, fairness, and non-discrimination should be incorporated to ensure responsible use of AI technologies.

[Strengthening the Right to Explanation and Transparency](#)

Individuals should be granted a clear, enforceable, and meaningful right to obtain explanations for decisions made by AI systems that significantly affect their rights or interests. This includes information about the logic, criteria, and data used in automated decision-making. Enhancing transparency will empower individuals to understand how AI systems operate, enable them to challenge unfair or biased outcomes, and promote trust in AI-driven processes.

[Enhancing Regulatory Capacity and Technical Expertise](#)

Data protection authorities must be equipped with adequate technical expertise, financial resources, and institutional capacity to regulate AI effectively. Specialized AI audit units should be established within regulatory bodies to examine algorithmic systems, assess compliance, and investigate violations. Continuous training programs and interdisciplinary collaboration with data scientists, technologists, and legal experts are essential to bridge the knowledge gap between law and technology.

[Mandatory Impact Assessments for AI Systems](#)

Organizations deploying AI systems that process personal data should be required to conduct mandatory Data Protection Impact Assessments (DPIAs) and Human Rights Impact Assessments. These assessments should evaluate potential risks related to privacy,

discrimination, bias, and misuse of data before AI systems are deployed. Mandatory assessments would encourage preventive compliance and reduce the likelihood of rights violations.

Clear Allocation of Accountability and Liability

Legal frameworks should clearly define accountability and liability for harms caused by AI systems. Rules must identify the responsibilities of various actors involved in the AI lifecycle, including developers, data providers, deployers, and users. Clear liability mechanisms will ensure that affected individuals have access to remedies and compensation and will promote responsible behavior among AI stakeholders.

Strengthening Consent and Data Minimization Standards

Consent mechanisms should be redesigned to ensure that consent is informed, specific, and meaningful in AI-driven environments. Simplified and standardized privacy notices may help individuals better understand data practices. Additionally, strict enforcement of data minimization principles should be ensured to prevent excessive data collection and misuse in AI training and deployment.

Promoting Ethical and Rights-Based AI Governance

AI governance should be guided by ethical principles such as respect for human dignity, equality, fairness, and non-discrimination. Governments and regulatory bodies should promote human-centric AI frameworks that prioritize fundamental rights. Ethical guidelines should complement legal rules and serve as benchmarks for responsible AI development.

Encouraging International Cooperation and Harmonization

Since AI systems often operate across national borders, international cooperation is essential for effective data protection enforcement. Harmonization of data protection standards and collaboration between regulatory authorities can help address jurisdictional challenges and ensure consistent protection of individual rights in global AI ecosystems.

Conclusion

Artificial Intelligence has significantly transformed the way personal data is collected, processed, and utilized across various sectors. While existing data protection laws such as the General Data Protection Regulation and India's Digital Personal Data Protection Act provide an essential legal foundation for safeguarding privacy, they were not originally designed to address the unique and complex challenges posed by AI-enabled systems. Issues such as automated decision-making, profiling, lack of transparency, and algorithmic bias test the limits of traditional data protection principles. This study has demonstrated that although current legal frameworks recognize important rights such as consent, accountability, and data security, their practical effectiveness in the context of artificial intelligence remains limited. Enforcement difficulties, lack of technical expertise among regulators, and the rapid pace of technological advancement further weaken the protection of individual rights. As a result, individuals often struggle to understand or challenge AI-driven decisions that significantly affect their lives. The paper concludes that existing data protection laws, while necessary, are not fully adequate to regulate artificial intelligence on their own. There is a clear need for AI-specific legal provisions, stronger transparency requirements, enhanced regulatory capacity, and a rights-based approach to AI governance. Law must evolve alongside technology to ensure that innovation does not come at the cost of privacy, fairness, and human dignity. A balanced and forward-looking legal framework will not only protect individual rights but also promote public trust and responsible development of artificial intelligence in the next generation.

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Human-Centered AI in Education: The Role of Spiritual Intelligence in AI-Enabled Pedagogical Systems

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Abstract

The rapid integration of artificial intelligence into educational systems has transformed pedagogical practices, decision-making processes, and learning environments. While AI-enabled systems offer significant advantages in efficiency, personalization, and scalability, they also raise critical concerns related to ethics, human agency, and the preservation of meaning in educational practice. This study examines the role of spiritual intelligence as a human-centered resource within AI-enabled educational systems, with particular attention to its contribution to pedagogical effectiveness and ethical sustainability. Drawing on empirical data from English as a Foreign Language (EFL) contexts, the study investigates how spiritual intelligence supports teachers' sense of purpose, ethical awareness, and adaptive engagement in AI-mediated learning

environments. Quantitative analyses reveal significant relationships between dimensions of spiritual intelligence and key indicators of pedagogical effectiveness, even within technologically mediated settings. The findings suggest that spiritual intelligence complements technological intelligence by anchoring AI-enabled education in values such as meaning, responsibility, and human dignity. The study argues that next-generation AI-enabled educational systems must move beyond purely algorithmic optimization and incorporate human intelligences to ensure ethical, inclusive, and sustainable educational transformation. These findings contribute to ongoing discussions on human-centered AI and provide implications for the design and governance of AI-enabled educational technologies.

Keywords: AI-enabled systems; spiritual intelligence; human-centered AI; ethical technology; education; next-generation learning

Introduction

The rapid integration of artificial intelligence (AI) and digital technologies into educational systems has profoundly reshaped pedagogical practices, teacher–learner interactions, and conceptions of effective teaching across disciplines, including English as a Foreign Language (EFL). AI-enabled systems (such as adaptive learning platforms, automated feedback tools, and conversational agents) are increasingly embedded in instructional decision-making, assessment, and classroom management. While these technologies promise enhanced efficiency, personalization, and expanded access to learning resources, scholars have raised sustained concerns regarding the risks of reductive, technocentric approaches that marginalize human meaning-making, ethical judgment, and professional agency (Biesta, 2015; Floridi et al., 2018; Williamson, 2017; Dignum, 2019). In response, contemporary educational research has renewed attention to human-centered frameworks that foreground values, purpose, and holistic development alongside technological innovation. Within this broader discourse on human-centered AI, spiritual intelligence (SI) has attracted sustained yet contested scholarly attention. Initially proposed as a form of intelligence concerned with meaning, transcendence, and ultimate concerns (Emmons, 1999, 2000; Zohar & Marshall, 2000), SI has since been reconceptualized as a capacity enabling individuals to integrate ethical awareness, values, and existential reflection into everyday functioning and professional practice (Gardner, 2000; Amram, 2011). Proponents argue that spiritual intelligence supports adaptive judgment, moral sensitivity, resilience, and purpose-driven action in complex and uncertain environments, including education (Ramachandaran et al., 2017; Muzakki, 2019). At the same time, critics have questioned the construct validity of SI, noting conceptual overlap with personality traits, well-being indicators, and emotional intelligence, as well as methodological limitations in existing measurement instruments (Mayer, 2000; King, 2008; Koenig & Carey, 2024; Bambling, 2025). Despite these debates, an expanding body of empirical research has continued to examine spiritual intelligence in relation to psychological well-being, leadership effectiveness, emotional regulation, and ethical behavior across diverse cultural and professional contexts (Anwar et al., 2020; Ajele et al., 2021; Rasti et al., 2023). In educational research, SI has been associated with reflective practice, value-based leadership, and holistic teaching orientations, suggesting its potential relevance for educators working in emotionally, morally, and relationally demanding environments. However, its role in language education (and EFL teaching in particular) remains comparatively under-theorized and under-investigated, especially in relation to rapidly emerging AI-mediated pedagogical contexts. Concurrently, EFL education is undergoing a profound transformation driven by conversational AI systems, intelligent tutoring technologies, and data-driven instructional platforms (Son et al., 2023; Lai & Lee, 2024; Zhang & Umeanowai, 2025). While these technologies can enhance learner engagement, feedback immediacy, and instructional efficiency, scholars emphasize that effective language education cannot be reduced to algorithmic optimization alone. Language learning is inherently bound to identity construction, emotion, intercultural meaning-making, and ethical communication (Kramersch, 2009; Byram,

2013; Dörnyei, 2019). Consequently, recent scholarship has called for pedagogical frameworks that balance technological advancement with care, empathy, reflective judgment, and ethical responsibility (Popoveniuc, 2017; Isaee & Barjesteh, 2025). Against this backdrop, the present study positions spiritual intelligence as a human-centered construct with particular relevance for EFL teachers operating within AI-enabled educational systems. Rather than treating SI as a fixed or unproblematic ability, the study adopts a critically informed and context-sensitive stance that acknowledges ongoing theoretical disputes and measurement challenges. By empirically examining the relationships between teachers' spiritual intelligence and pedagogically meaningful professional orientations, this research seeks to contribute nuanced evidence to the SI literature while responding to contemporary calls for ethically grounded and value-sensitive approaches to AI integration in education. Specifically, this study addresses an important research gap: although prior research has explored spiritual intelligence in general educational and psychological domains, there remains limited empirical evidence examining how SI relates to EFL teachers' professional orientations within modern, AI-influenced instructional environments. By situating spiritual intelligence within broader debates on human-centered pedagogy and responsible AI adoption, the study offers insights relevant to language educators, teacher educators, and researchers concerned with preserving meaning, ethical agency, and professional judgment in an era of accelerating educational technologization. Grounded in ongoing theoretical debates surrounding spiritual intelligence and informed by contemporary discussions on AI-enabled and ethically responsible educational systems, the present study seeks to empirically examine the role of spiritual intelligence in EFL teaching contexts. Accordingly, the study is guided by the following research questions:

RQ1. What is the overall level of spiritual intelligence among EFL teachers in the investigated context?

RQ2. Is there a statistically significant relationship between EFL teachers' spiritual intelligence and their pedagogical and professional orientations?

RQ3. To what extent do the dimensions of spiritual intelligence predict variance in selected pedagogical outcomes among EFL teachers?

RQ4. How can the observed relationships between spiritual intelligence and EFL teaching practices be interpreted within contemporary human-centered and AI-mediated educational frameworks?

Theoretical and Empirical Background

Conceptualizations and Ongoing Debates Surrounding Spiritual Intelligence

The concept of spiritual intelligence (SI) has its roots in broader discussions on the nature and scope of human intelligence that extend beyond purely cognitive or analytical capacities. Early theoretical contributions framed SI as an intelligence concerned with meaning-making, transcendence, and the capacity to engage with existential questions that shape human motivation and behavior (Emmons, 1999, 2000; Zohar & Marshall, 2000). From this perspective, SI was positioned as a functional capacity enabling individuals to integrate values, purpose, and ethical awareness into everyday decision-making, particularly in complex and uncertain contexts. Subsequent scholarship sought to systematize the construct by proposing multidimensional models and measurement instruments. Notably, Amram (2011) articulated seven interrelated dimensions of SI grounded in cross-cultural and ecumenical inquiry, emphasizing consciousness, grace, meaning, transcendence, and truth. Similarly, Amram and Dryer (2008) introduced the Integrated Spiritual Intelligence Scale (ISIS) as an attempt to operationalize SI for empirical research. These models contributed to expanding SI research across organizational, educational, and psychological domains (Ramachandaran et al., 2017; Muzakki, 2019). Despite its growing visibility, SI has remained a contested construct. Critics have raised fundamental concerns regarding whether SI satisfies established criteria for intelligence, particularly in relation to cognitive specificity, empirical separability, and predictive validity (Gardner, 2000; Mayer,

2000; King, 2008). King (2008), for example, argued that many SI measures overlap substantially with personality traits, emotional intelligence, and indicators of psychological well-being, thereby challenging claims of construct distinctiveness. More recently, concerns about scale contamination (where measures of spirituality inadvertently include mental health or social functioning indicators) have intensified scrutiny of SI research findings (Bambling, 2025; Koenig & Carey, 2024, 2025). Contemporary scholarship has responded to these critiques by advocating for a more cautious and reflexive use of SI in empirical research. Rather than asserting SI as a standalone intelligence, some scholars conceptualize it as a meaning-oriented human capacity that interacts dynamically with emotional, moral, and cognitive processes (Agrawal, 2025; Skrzypińska, 2021). From this perspective, SI is valued less for its classificatory status and more for its explanatory potential in understanding how individuals navigate ethical complexity, purpose, and professional responsibility. This reframing has particular relevance for educational contexts, where teaching involves continuous value-laden judgment rather than purely technical execution.

Spiritual Intelligence and Educational, Psychological, and Professional Outcomes

Empirical research on SI has documented associations with a range of psychological and professional outcomes, although findings must be interpreted in light of ongoing measurement debates. In psychological and health-related studies, SI has been linked to enhanced emotional regulation, mindfulness, and subjective well-being, particularly under conditions of stress and uncertainty (Ajele et al., 2021; Rasti et al., 2023). These findings suggest that SI-related capacities may support adaptive coping and resilience, though scholars caution against conflating spiritual resources with mental health indicators (Koenig & Carey, 2024). Within educational and organizational research, SI has been examined as a factor contributing to ethical leadership, reflective practice, and value-based decision-making. Studies involving educators and academic leaders indicate that higher levels of SI are associated with greater sensitivity to learners' needs, principled professional conduct, and a commitment to holistic development rather than narrow performance outcomes (Ramachandaran et al., 2017; Muzakki, 2019). In language education contexts, these qualities are particularly salient given the emotional, intercultural, and identity-related dimensions of language learning (Kramsch, 2009; Byram, 2013). However, the literature also reflects substantial variability in conceptual definitions, research designs, and instruments, limiting cross-study comparability. Recent methodological reviews emphasize the importance of transparent construct positioning, careful interpretation of correlational findings, and explicit acknowledgment of SI's theoretical boundaries (Skrzypińska, 2021; Bambling, 2025). Rather than undermining the relevance of SI, such critiques underscore the need for empirically grounded studies that situate SI within specific professional contexts (such as EFL teaching) while avoiding overgeneralized claims.

Artificial Intelligence in EFL Education: Opportunities, Risks, and Ethical Concerns

Parallel to debates on SI, EFL education has experienced a rapid transformation through the adoption of AI-driven technologies, including conversational agents, adaptive feedback systems, and data-informed instructional tools. Recent reviews document growing evidence that AI applications can enhance learner engagement, vocabulary development, and personalized learning pathways when integrated thoughtfully into language instruction (Lai & Lee, 2024; Son et al., 2023; Azamatova et al., 2023). From a pedagogical perspective, AI has been framed as a potential mediator that supports learning processes rather than an authoritative replacement for teachers (Risdianto et al., 2025). At the same time, scholars caution that uncritical adoption of AI risks narrowing educational goals to efficiency, measurability, and short-term outcomes. Ethical concerns include algorithmic bias, diminished teacher autonomy, and the erosion of relational and affective dimensions of teaching (Dignum, 2019; Floridi et al., 2018; Williamson, 2017). In language education specifically, researchers emphasize that communicative competence involves intercultural sensitivity, empathy, humor, and ethical awareness—dimensions that cannot be fully captured by automated systems (Zhai et al., 2024; Kramsch,

2009). As a result, recent scholarship advocates for human-centered and responsible AI frameworks in education, positioning technology as a tool that must remain subordinate to pedagogical judgment and ethical reflection (Biesta, 2015; Verboom et al., 2025). These discussions resonate strongly with calls for pedagogies grounded in care, empathy, and professional responsibility, particularly in EFL contexts where teacher–student relationships play a critical role in motivation and engagement (Isaee et al., 2025; Isaee & Barjesteh, 2025).

Positioning Spiritual Intelligence within Human-Centered and Ethical EFL Pedagogy

Integrating the above strands, spiritual intelligence may be understood as a human-centered capacity that complements technological innovation in EFL education. Rather than competing with AI-based tools, SI-oriented dispositions (such as ethical awareness, reflective meaning-making, and sensitivity to learners’ emotional and cultural realities) may enable teachers to navigate AI-mediated environments more responsibly. From this standpoint, SI supports pedagogical judgment in contexts where technical solutions alone are insufficient (Popoveniuc, 2017; Agrawal, 2025). Recent work on loving pedagogy and empathetic teaching further reinforces the relevance of SI-related constructs in language education. Studies suggest that teacher care, moral commitment, and emotional presence contribute meaningfully to learner motivation and reduced anxiety, particularly in foreign language classrooms (Isaee et al., 2025; Dörnyei, 2019). Situating SI within this broader ethical–affective framework allows for a more nuanced interpretation that avoids overstated claims while recognizing its potential pedagogical value. Accordingly, the present study conceptualizes spiritual intelligence not as a definitive or isolated intelligence, but as a meaning-oriented resource that may shape how EFL teachers engage with learners, instructional challenges, and emerging technologies. By empirically examining SI within this carefully bounded framework, the study aims to contribute to ongoing debates while aligning with contemporary calls for ethical, reflective, and humanizing approaches to language education in AI-enhanced contexts.

Methodology

Research Design

The present study adopted a quantitative, correlational research design to examine the relationship between English as a Foreign Language (EFL) teachers’ spiritual intelligence and selected pedagogical and professional orientations within contemporary, AI-influenced educational contexts. A correlational approach was deemed appropriate given the study’s primary aim of exploring naturally occurring associations among variables rather than establishing causal relationships (Creswell & Creswell, 2018). This design is widely employed in educational research investigating psychological and professional constructs in authentic instructional settings (Dörnyei, 2007). Although the study did not involve direct intervention or experimental manipulation of AI tools, it was conceptually situated within AI-enabled educational systems, acknowledging that participants were operating in technologically enriched environments where digital platforms, automated assessment tools, and AI-supported instructional resources increasingly shape pedagogical practice. Accordingly, the study aligns with human-centered AI research that emphasizes teachers’ perceptions, orientations, and ethical engagement rather than system-level technical performance (Dignum, 2019; Floridi et al., 2018).

Participants and Context

The participants of the study consisted of EFL teachers working in formal instructional settings within the targeted educational context. Teachers were recruited using a non-probability convenience sampling strategy, a common approach in educational research where access to participants is institutionally bounded. Participation was voluntary, and respondents were informed of the study’s academic purpose before data collection. The sample included teachers with diverse teaching experience levels and academic backgrounds, reflecting the heterogeneous nature of contemporary EFL teaching contexts. Such diversity was considered advantageous for capturing variability in spiritual intelligence and pedagogical orientations, while acknowledging

that the sampling strategy limits the generalizability of the findings beyond the investigated population.

Instruments

Spiritual Intelligence Measure

Spiritual intelligence was measured using a standardized self-report instrument grounded in established multidimensional conceptualizations of SI. The instrument operationalizes spiritual intelligence through multiple dimensions reflecting meaning-making, ethical awareness, transcendence, and conscious engagement with values, as articulated in prior theoretical and empirical research (Amram, 2011; Amram & Dryer, 2008). Given ongoing scholarly debates concerning the construct validity and potential scale contamination of SI measures, the instrument was employed with explicit theoretical caution (King, 2008; Koenig & Carey, 2024). Rather than treating SI scores as definitive indicators of an independent intelligence, the scale was used as an index of meaning-oriented and value-sensitive dispositions relevant to professional practice. Internal consistency indices reported in prior validation studies were deemed acceptable for research purposes.

Pedagogical and Professional Variables

In addition to spiritual intelligence, the questionnaire included measures assessing pedagogical and professional orientations relevant to EFL teaching. These variables reflected teachers' instructional approaches, professional attitudes, and classroom-related dispositions aligned with contemporary language education literature. The inclusion of these variables allowed for examination of relational and predictive links between SI dimensions and pedagogically meaningful outcomes, consistent with the study's research questions.

Data Collection Procedures

Data were collected through an online questionnaire distributed electronically to EFL teachers. The online format facilitated access to participants across institutional settings while ensuring anonymity and ease of response. Prior to completing the questionnaire, participants were provided with an informed consent statement outlining the voluntary nature of participation, confidentiality assurances, and the academic purpose of the research. Participants were informed that they could withdraw from the study at any stage without penalty. No identifying information was collected, and responses were stored securely for research purposes only.

Data Analysis

Data analysis was conducted using statistical software. Descriptive statistics were first computed to determine the overall levels of spiritual intelligence among participants. Subsequently, Pearson product-moment correlation analyses were performed to examine relationships between spiritual intelligence and pedagogical and professional orientation variables.

To address the predictive dimension of the research questions, multiple regression analyses were conducted to assess the extent to which dimensions of spiritual intelligence accounted for variance in selected pedagogical outcomes. Assumptions of normality, linearity, and multicollinearity were examined prior to inferential analysis, following standard methodological guidelines (Field, 2018). Statistical significance was set at $p < .05$, consistent with conventions in educational and social science research. Effect sizes were also considered to support meaningful interpretation beyond statistical significance alone.

Ethical Considerations

Ethical principles guiding educational research were observed throughout the study. Participation was voluntary, informed consent was obtained, and confidentiality of responses was ensured. The study involved no deception, and participants were not exposed to any form of physical or psychological risk. In addition, ethical reflexivity was exercised in relation to the measurement and interpretation of spiritual intelligence. Acknowledging ongoing debates regarding the conceptual boundaries of SI, the study refrains from essentializing spiritual

intelligence or attributing normative judgments to participants based on their scores (Bambling, 2025; Skrzypińska, 2021). Findings are interpreted cautiously, with attention to contextual and theoretical limitations.

Methodological Limitations

Several methodological limitations should be acknowledged. First, the use of self-report measures may be subject to social desirability bias. Second, the convenience sampling strategy restricts the generalizability of the findings. Third, given contested definitions and measurement challenges associated with spiritual intelligence, results should be interpreted as indicative rather than conclusive. These limitations are not viewed as undermining the study's contribution but rather as situating its findings within a transparent and critically informed empirical framework, consistent with recent methodological recommendations in SI research (Koenig & Carey, 2025).

Results

This section reports the findings of the study in two phases. First, results from the quantitative analyses are presented to address the research questions concerning levels of spiritual intelligence (SI) and its relationship with pedagogical success. Second, qualitative findings derived from interviews, classroom observations, and syllabus analysis are reported to illuminate the factors underlying teachers' pedagogical success and to complement the quantitative patterns.

Quantitative Results

Descriptive Statistics of Spiritual Intelligence and Pedagogical Success

To address the overall profile of the participants, descriptive statistics were calculated for teachers' spiritual intelligence (SIQ), its four subscales, and pedagogical success (CSTQ). Given the unequal number of items across the questionnaires and subscales, raw scores were converted into scaled mean scores to ensure comparability. Specifically, SIQ scores were transformed to a 4-point scale, whereas CSTQ scores were transformed to a 5-point scale. Table 1 summarizes the descriptive statistics for SIQ and CSTQ, including the mean, standard deviation, and scaled mean values.

Table 1

Descriptive Statistics of Spiritual Intelligence (SIQ) and Pedagogical Success (CSTQ)

Measure		N	M	SD	Scaled Mean
SIQ (Total)		120	273.11	37.08	2.92
Transcendent Self-Realization		120	149.12	25.11	2.53
Spiritual Experience		120	54.22	8.98	2.79
Forgiveness		120	22.18	3.91	3.02
Patience		120	52.41	8.98	3.14
CSTQ		120	189.41	25.44	4.04

As shown in Table 1, the mean total SIQ score ($M = 273.11$) exceeded the theoretical midpoint of the scale, indicating that the participating EFL teachers generally reported relatively high levels of spiritual intelligence. Similarly, the mean pedagogical success score ($M = 189.41$) was well above the scale midpoint, suggesting that the teachers were broadly perceived as pedagogically effective by their students.

At the subscale level, notable differences emerged. Among the four SI dimensions, Patience recorded the highest scaled mean (3.14), followed by Forgiveness (3.02), Spiritual Experience (2.79), and Transcendent Self-Realization (2.92). This pattern suggests that teachers more strongly endorsed affective and self-regulatory aspects of SI than existential or transcendental dimensions.

To visually depict these relative differences, the distribution of SI subscale means is illustrated in Figure 1.

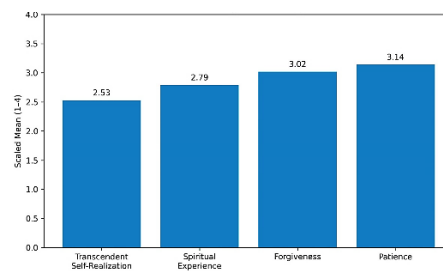


Figure 1. *Profile of EFL Teachers' Spiritual Intelligence Dimensions*

Figure 1 highlights the prominence of patience as a defining characteristic of the participants' spiritual intelligence profile, reflecting teachers' perceived capacity for tolerance, endurance, and emotional stability in instructional contexts.

Relationship Between Spiritual Intelligence and Pedagogical Success (RQ1)

The first research question examined whether EFL teachers' spiritual intelligence is associated with their pedagogical success. Pearson product-moment correlation coefficients were computed between CSTQ scores and both the total SIQ score and its four subscales. The correlation results are presented in Table 2, along with coefficients of determination (r^2), which indicate the proportion of shared variance between variables.

Table 2

Correlations Between Spiritual Intelligence (SI) and Pedagogical Success

SI Dimension	CSTQ (r)	r ²
Transcendent Self-Realization	.65**	.45
Spiritual Experience	.40**	.18
Patience	.45**	.22
Forgiveness	.31*	.11
Total SI	.72**	.54

* $p < .05$, ** $p < .01$

As shown in Table 2, total spiritual intelligence demonstrated a strong, positive, and statistically significant correlation with pedagogical success ($r = .72$, $p < .01$). This result indicates that more than half of the variance in pedagogical success (53%) can be explained by teachers' overall spiritual intelligence. At the subscale level, all SI dimensions were positively associated with pedagogical success. The strongest relationship was observed for Transcendent Self-Realization ($r = .65$), followed by Patience ($r = .45$) and Spiritual Experience ($r = .40$), all reaching significance at the .01 level. Forgiveness, while still significantly related to pedagogical success, demonstrated a comparatively weaker association.

These relationships are visually summarized in Figure 2, which facilitates comparison across SI dimensions.

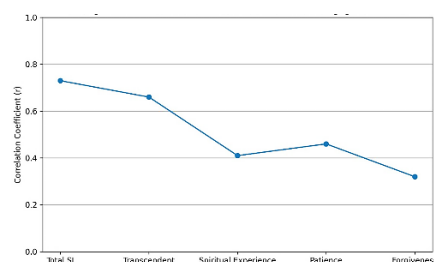


Figure 2. *Strength of Association Between Spiritual Intelligence Dimensions and Pedagogical Success*

As depicted in Figure 2, the overall strength of association varies across dimensions, underscoring the particularly influential role of existential and self-transcendent aspects of SI in shaping effective pedagogical performance.

Predictive Contribution of Spiritual Intelligence Dimensions (RQ2)

The second research question explored whether the dimensions of spiritual intelligence could predict EFL teachers' pedagogical success. To address this question, a multiple linear regression analysis was conducted using the four SI subscales as predictors and pedagogical success as the dependent variable. The overall model fit indices are reported in Table 3.

Table 3

Model Summary for Multiple Regression Analysis

R	R ²	Adjusted R ²	Std. Error
.729	.54	.531	16.11

As indicated in Table 3, the regression model accounted for a substantial proportion of variance in pedagogical success ($R^2 = .54$), confirming that spiritual intelligence dimensions collectively provide a strong explanatory framework for teacher effectiveness. The individual predictive contributions of each SI dimension are detailed in Table 4.

Table 4

Regression Coefficients for Predicting Pedagogical Success

Predictor	B	SE	β	t	p
Constant	57.79	11.74	—	4.94	.000
Transcendent Self-Realization	.479	.071	.529	7.79	.000
Spiritual Experience	.491	.169	.201	2.86	.005
Patience	.551	.1661	.221	3.37	.002
Forgiveness	.039	.423	.007	.092	.931

As shown in Table 4, Transcendent Self-Realization emerged as the strongest predictor of pedagogical success ($\beta = .529$, $p < .001$), followed by Patience ($\beta = .221$) and Spiritual Experience ($\beta = .201$). In contrast, Forgiveness did not contribute significantly to the model when the other SI dimensions were controlled. These differential predictive effects are graphically illustrated in Figure 3.

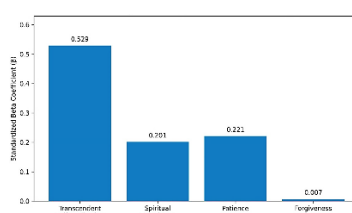


Figure 3. Predictive Power of Spiritual Intelligence Dimensions for Pedagogical Success

Figure 3 clearly demonstrates that existential meaning-making and self-transcendence play a central role in explaining pedagogical success, beyond purely interpersonal or affective traits.

Qualitative Results

While the quantitative findings establish the strength and structure of the relationship between spiritual intelligence and pedagogical success, the qualitative phase sheds light on the lived experiences underlying these statistical patterns. Interviews, classroom observations, and syllabus analyses were conducted to explore how teachers with differing levels of SI conceptualized teaching and enacted their roles in practice. Consistent with the quantitative results, teachers with higher levels of SI obtained substantially higher pedagogical success ratings ($M = 220$) than those with lower SI levels ($M = 117$). The qualitative analysis revealed several interrelated themes distinguishing these two groups.

Interview Findings

Interpersonal Relationships

Teachers with higher SI consistently emphasized the importance of building meaningful relationships with students. They demonstrated heightened sensitivity to learners' emotional states, such as anxiety, insecurity, and motivation, and viewed teaching as a nurturing rather than purely instructional role. Many attributed their compassion to personal life experiences, including hardship and loss, which they believed enhanced their ability to empathize with students. In contrast, teachers with lower SI placed less emphasis on relational engagement and reported more frequent interpersonal tensions in the classroom.

Stress and Anxiety Regulation

High-SI teachers reported greater emotional stability and confidence, particularly when encountering unfamiliar content or challenging learner groups. They described teaching as a manageable and meaningful activity rather than a source of persistent stress. Conversely, teachers with lower SI frequently expressed frustration, job-related anxiety, and emotional exhaustion, which they perceived as barriers to effective teaching.

Courage and Risk-Taking

Teachers with higher SI described themselves as willing to experiment with new pedagogical approaches and to challenge conventional teaching norms. They viewed professional growth as requiring courage and a willingness to embrace uncertainty. In contrast, lower-SI teachers tended to favor routine methods and expressed reluctance to adopt innovative practices due to fear of failure.

Consciousness and Self-Awareness

A strong sense of self-awareness characterized high-SI teachers, who openly acknowledged their instructional limitations and actively sought improvement. They described reflective practice as essential to pedagogical growth. Teachers with lower SI demonstrated less reflective engagement and were less inclined to critically evaluate their teaching practices.

Meaningfulness and Purpose

High-SI teachers frequently articulated a clear sense of purpose in their work, framing teaching as a meaningful and value-driven endeavor. They emphasized goal-setting, intentional lesson design, and fostering deep learning. In contrast, teachers with lower SI were more likely to describe teaching in instrumental or routine terms, with limited reference to broader educational goals.

Responsibility and Commitment

Teachers with higher SI strongly emphasized professional responsibility, punctuality, and ethical commitment. They viewed responsibility as foundational to effective teaching and learner development. Lower-SI teachers mentioned responsibility less frequently and showed weaker alignment between stated goals and classroom practices.

Creativity

Finally, creativity emerged as a distinguishing feature of high-SI teachers, who actively sought innovative ways to engage learners and adapt instructional materials. Teachers with lower SI tended to rely more heavily on prescribed textbooks and demonstrated limited flexibility in instructional design. To synthesize the qualitative interview findings and to visually illustrate the contrasts observed between teachers with higher and lower levels of spiritual intelligence, a comparative thematic representation is provided in Figure 4. The figure summarizes the major dimensions that emerged from the interviews, highlighting how spiritual intelligence shapes

teachers' interpersonal engagement, emotional regulation, professional courage, self-awareness, sense of purpose, responsibility, and creativity within AI-influenced instructional environments.



Figure 4. Comparative thematic representation of interview findings across levels of spiritual intelligence

The figure contrast key qualitative themes identified among EFL teachers with higher and lower levels of spiritual intelligence, including interpersonal relationships, stress and anxiety regulation, courage and risk-taking, self-awareness, meaningfulness and purpose, professional responsibility, and creativity. The themes illustrate how spiritual intelligence shapes teachers' pedagogical orientations and professional experiences in AI-enabled and technologically enriched educational contexts.

Classroom Observation and Syllabus Analysis

Classroom observations corroborated the interview findings. Teachers with higher SI demonstrated respectful interaction patterns, emotional composure, and learner-centered practices, including collaborative activities and constructive feedback. They were more punctual, more encouraging of student participation, and more tolerant of learner errors. Syllabus analysis further revealed that high-SI teachers articulated clearer learning objectives, aligned short-term tasks with long-term goals, and incorporated elements promoting learner autonomy and personal growth. In contrast, syllabi prepared by lower-SI teachers were more rigid, instructor-centered, and less focused on learner empowerment or reflective engagement.

Discussion

The present study set out to examine the role of spiritual intelligence (SI) in EFL teachers' pedagogical success through a mixed-methods design. By integrating quantitative and qualitative evidence, the findings offer a nuanced understanding of how spiritual intelligence functions not merely as a dispositional trait but as a meaning-oriented and practice-shaping capacity in language education. The discussion is structured around the research questions and situated within broader theoretical and empirical debates in spirituality, teacher effectiveness, and contemporary educational research. The first research question investigated whether EFL teachers' spiritual intelligence is related to their pedagogical success. The results demonstrated a strong, positive, and statistically significant relationship between overall SI and pedagogical success, indicating that teachers with higher levels of spiritual intelligence were more likely to be perceived as effective by their students. This finding lends empirical support to early theoretical arguments proposing spirituality as a form of intelligence linked to adaptive functioning and purposeful action (Emmons, 2000; Gardner, 2000; Zohar & Marshall, 2000). It also aligns with more recent empirical syntheses suggesting that SI contributes to emotional regulation, ethical sensitivity, and meaning-making—capacities that are particularly salient in teaching professions (Amram, 2011; Skrzypińska, 2021; Rasti et al., 2023). Within the EFL context, where emotional labor, intercultural sensitivity, and sustained motivation are central, such capacities appear to translate directly into pedagogical effectiveness. Notably, the strength

of the association observed in this study ($r = .73$) exceeds that reported in several prior investigations linking SI to professional or academic outcomes (Ramachandaran et al., 2017; Muzakki, 2019). This may be attributed to the relational and affective demands of language teaching, which amplify the pedagogical relevance of meaning-oriented and self-regulatory dimensions of intelligence. In this sense, the findings resonate with Biesta's (2015) argument that good education cannot be reduced to technical competence alone but requires ethical judgment, relational responsibility, and personal orientation toward purpose.

Beyond the overall association, the findings revealed meaningful variation across SI dimensions. Among the subscales, transcendent self-realization demonstrated the strongest correlation with pedagogical success, followed by patience and spiritual experience, while forgiveness showed a comparatively weaker relationship. This pattern is theoretically significant. Transcendent self-realization reflects teachers' capacity to situate their professional actions within a broader framework of meaning, values, and existential purpose (Amram, 2011; Amram, 2022). The strong association between this dimension and pedagogical success suggests that teachers who perceive their work as purposeful and value-driven are more likely to engage learners effectively and sustain high-quality instructional practices. This finding extends prior work in language education emphasizing the role of teacher vision, identity, and long-term motivational orientation (Dörnyei, 2019; Kramsch, 2009). Patience also emerged as a salient correlate of pedagogical success, underscoring the importance of emotional regulation and tolerance in EFL classrooms. This aligns with research highlighting the affective challenges of language learning and the critical role of teacher calmness and emotional stability in reducing learner anxiety (Oxford, 2016; Isae et al., 2025). In contrast, the weaker association observed for forgiveness suggests that while interpersonal grace may contribute to a positive classroom climate, it may operate more indirectly or contextually than other SI dimensions.

The second research question examined whether SI dimensions could predict pedagogical success. The regression analysis demonstrated that spiritual intelligence, as a multidimensional construct, accounted for a substantial proportion of variance in pedagogical success, with transcendent self-realization emerging as the strongest predictor. This finding reinforces the conceptualization of SI as an action-guiding intelligence rather than a purely reflective or affective disposition (King, 2008; Mayer, 2000). Teachers who possess a strong sense of existential orientation appear better equipped to navigate instructional challenges, maintain professional commitment, and make principled pedagogical decisions. Such results echo Popoveniuc's (2017) argument that moral and meaning-oriented competence is essential for responsible professional practice, particularly in complex human-centered domains such as education. Interestingly, forgiveness did not significantly predict pedagogical success when other SI dimensions were controlled. This result contributes to ongoing methodological debates regarding construct overlap and scale contamination in spirituality research (Bambling, 2025; Koenig & Carey, 2024, 2025). It suggests that certain spiritually framed traits may be relationally valuable yet less central to performance outcomes when examined alongside existential and self-regulatory capacities. The qualitative findings provide explanatory depth to the quantitative patterns by illustrating how spiritual intelligence is enacted in teachers' daily professional lives. Teachers with higher SI consistently demonstrated stronger interpersonal relationships, greater emotional resilience, increased pedagogical courage, heightened self-awareness, and a pronounced sense of purpose. These characteristics closely mirror Amram's (2011) multidimensional model of SI and support the argument that spiritual intelligence operates through integrated cognitive, emotional, and ethical processes.

The emphasis on meaning and purpose among high-SI teachers is particularly noteworthy. These teachers framed teaching not merely as skill transmission but as a transformative and socially responsible endeavor. This orientation parallels Byram's (2013) notion of intercultural citizenship and aligns with contemporary calls to humanize education in increasingly technologized learning environments (Isae & Barjesteh, 2025; Floridi et al., 2018).

In contrast, teachers with lower SI reported higher levels of stress, reluctance to innovate, and a more instrumental view of teaching. These findings echo prior research linking limited reflective capacity and emotional dysregulation to professional burnout and reduced instructional effectiveness (Halama & Striženec, 2004; Rasti et al., 2023).

Implications for Language Education in the Age of AI

Although the present study did not directly investigate AI-assisted instruction, its findings carry important implications for contemporary EFL contexts increasingly shaped by artificial intelligence and digital technologies. Recent research has emphasized that AI integration in language education must be guided by ethical, human-centered, and pedagogically meaningful principles rather than efficiency alone (Dignum, 2019; Williamson, 2017; Lai & Lee, 2024). The strong association between SI and pedagogical success suggests that spiritual intelligence may serve as a critical internal compass for teachers navigating AI-enhanced environments. As AI tools increasingly mediate instructional decisions, teachers with higher SI may be better positioned to maintain empathy, ethical judgment, and learner-centered priorities (Isaee & Barjesteh, 2025; Zhai et al., 2024; Verboom et al., 2025). In this sense, SI complements emerging frameworks of responsible and humane AI in education (Floridi et al., 2018; Risdianto et al., 2025). In sum, this study makes several contributions. Empirically, it demonstrates that spiritual intelligence is strongly associated with and predictive of pedagogical success among EFL teachers. Conceptually, it highlights the centrality of transcendent self-realization and meaning-making in effective teaching. Methodologically, it integrates quantitative rigor with qualitative depth to illuminate the mechanisms through which SI operates. Pedagogically, it underscores the importance of nurturing teachers' inner capacities alongside technical skills, particularly in an era of rapid technological transformation.

Conclusion

This study set out to investigate the role of spiritual intelligence (SI) in EFL teachers' pedagogical success through a mixed-methods approach. By integrating quantitative evidence with qualitative insights, the findings offer a comprehensive account of how spiritual intelligence operates as a meaning-oriented, self-regulatory, and practice-shaping capacity in language education. The results clearly demonstrate that spiritual intelligence is not a peripheral or abstract construct, but a substantive contributor to effective teaching. Quantitatively, the study established a strong and statistically significant relationship between teachers' overall spiritual intelligence and pedagogical success. Among the SI dimensions, transcendent self-realization emerged as the most powerful correlate and predictor of pedagogical effectiveness, followed by patience and spiritual experience. These findings suggest that teachers who are able to situate their instructional practices within a broader framework of purpose, values, and self-awareness are more likely to foster effective learning environments and sustain professional effectiveness. In contrast, forgiveness (while positively related to pedagogical success) did not independently predict effectiveness when other SI dimensions were considered, highlighting the differentiated functional roles of SI components. Qualitatively, the findings deepened understanding of these statistical patterns by revealing how spiritually intelligent teachers enact their inner capacities in everyday practice. Teachers with higher SI demonstrated stronger interpersonal relationships, greater emotional resilience, higher levels of pedagogical courage, enhanced self-awareness, a pronounced sense of meaning, and stronger professional responsibility. These qualities were consistently reflected in classroom interaction patterns, instructional decision-making, and course design. Teachers with lower SI, by contrast, experienced higher stress, relied more heavily on routine practices, and articulated fewer purpose-driven pedagogical goals. Together, these findings confirm that spiritual intelligence manifests not only in beliefs or attitudes, but in concrete pedagogical behaviors and professional orientations. From a theoretical perspective, this study contributes to ongoing debates regarding the nature and validity of spiritual intelligence by providing robust empirical evidence of its relevance to professional performance in education.

The findings support conceptualizations of SI as an integrative intelligence encompassing existential meaning-making, emotional regulation, ethical awareness, and reflective consciousness. By demonstrating its predictive power in a demanding instructional context, the study strengthens the case for recognizing spiritual intelligence as a legitimate and educationally consequential construct. Pedagogically, the results underscore the importance of attending to teachers' inner capacities alongside methodological and technological competencies. Effective EFL teaching appears to depend not only on instructional techniques or content knowledge, but also on teachers' ability to remain patient, purposeful, reflective, and emotionally balanced in complex classroom environments. These qualities are particularly critical in language education, where learners' emotional vulnerability, cultural diversity, and communicative risk-taking place heightened relational demands on teachers. The findings also carry important implications for contemporary EFL contexts increasingly shaped by digitalization and artificial intelligence. As AI tools become more embedded in language teaching, the human dimensions of education, such as meaning, empathy, ethical judgment, and responsibility, become even more salient. Spiritual intelligence may function as an internal compass that enables teachers to integrate technological tools thoughtfully while preserving pedagogical care, learner dignity, and educational purpose. In this sense, SI complements emerging frameworks for responsible, human-centered, and ethical educational innovation.

In conclusion, this study demonstrates that spiritual intelligence plays a central role in shaping pedagogical success among EFL teachers. By illuminating both the measurable outcomes and lived experiences associated with SI, the research advances understanding of teacher effectiveness as a deeply human, value-laden, and meaning-driven endeavor. Future efforts in teacher education, professional development, and educational policy would benefit from recognizing and nurturing spiritual intelligence as a foundational dimension of sustainable and humane language education.

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Detect The Human Face and Emotions Effectively Using Deep Learning to Recognise and Analyse Behavioural Patterns

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Abstract

All person's emotions are the building blocks of their communication. Interactive video games, medical diagnostics, emotional computing, human-robot interaction, and other areas of Human Computer Interaction (HCI) make extensive use of it. There are three common ways that people show their emotions in social situations: gestures, facial expressions, and words (unimodal), speech and facial expression (bimodal), and psychological signals, video, and audio (multimodal). Understanding one's surroundings and the intentions of one's interlocutors are prerequisites for machine perception. This could be helped by facial emotion recognition. Understanding one's surroundings and the intentions of one's interlocutors are prerequisites for machine perception. This could be helped by facial emotion recognition. Images depicting the following facial emotions joy, sorrow, rage, surprise, disgust, and fear were processed using deep learning approaches during the creation of this work. Training a deep learning network using a dataset of labelled facial expressions was the initial step in the process. The Extended Cohn-Kanade Database is the selected dataset. Machines can learn to perceive thanks to deep learning's capacity to learn features, which can lead to better human-machine interaction. Additionally, computers with perception might provide more refined replies, which would greatly enhance the user experience.

Keywords: Deep Learning, Pattern, Recognition, Model, **Emotion**

INTRODUCTION

Both theoretically and in terms of practical applications, face recognition has grown at a phenomenal rate. During the age of deep learning, state-of-the-art models based on Deep Convolutional Neural Networks (DCNN) like ResNet and OpenFace have achieved remarkable success. Still, facial recognition systems are susceptible to confounds like disguises, even if these algorithms are becoming better at what they do. The field of face recognition has done extensive research on a number of factors, including lighting, expression, age, and heterogeneity. Nevertheless, there has been a dearth of research into facial identification using disguise variants. Changing one's appearance by donning a beard, spectacles, a variety of haircuts, a scarf, a hat, or other head coverings is an example of a disguise. Disguises may be either deliberate or accidental. In deliberate disguise, the subject makes an effort to conceal their identity in order to trick face recognition systems; in unintentional disguise, they casually wear various disguises, like glasses and a scarf, and manage to trick the system without realising it. These differences make the job more difficult by decreasing the interclass distance between participants and increasing the intraclass variances.

In order to communicate and engage with others, the human face is essential. In order to do these tasks, humans rely on their facial recognition abilities. One must learn to identify the faces of the hundreds of people that surround them every day. Facial recognition technology is also crucial for HCI. Access control, security, and surveillance systems, as well as the entertainment sector, make extensive use of it nowadays. Faster and simpler tasks are now possible thanks to advancements in facial recognition. Two main subfields exist within the field of facial recognition: identification and verification. Face verification is a one-to-one matching method that just requires two photos to determine whether they belong to the same individual. In contrast, face identification 1:N matching requires picking select a specific individual from a pool of potential results in order to identify them in an image.

Face recognition refers to the method of identifying a person by analysing their facial characteristics. Deep learning has completely transformed face recognition by making possible very accurate and efficient recognition systems. Deep learning face recognition systems typically employ a large dataset of facial images to train a neural network that can identify unique facial characteristics and patterns. Using these characteristics, a unique representation of each person's face is created and stored in a database for future recognition. When doing recognition, the system compares the incoming face picture to its stored representations to get the closest match. Convolutional Neural Networks (CNNs) are often used for feature extraction in face recognition systems, with deep learning-based techniques such as Siamese networks and triplet networks employed for matching optimisation. Among the many applications of face recognition technology are identification, social media tagging, access control, security, and monitoring. It is critical, however, to consider privacy concerns and any biases throughout the design and implementation of such systems. amid this article, we'll go over the methodology that can identify a face amid a sea of data that constantly compares and saves information about individuals we know. A good solution would be a classifier that can automatically recognise faces. Faces are usually represented as a linear combination of base photographs using these methods. Principle component analysis (PCA) is one example of such a method. To get the basis pictures, principal component analysis (PCA) only examines pairwise correlations between pixels in the image collection. Independent Component Analysis (ICA) is different from Principal Component Analysis (PCA), which is often used in computerized face recognition and authentication applications. In order to identify and localize faces, face recognition systems use the Haar feature-based cascade classifier. The features of the face are recovered by use of the weighted Local Binary Pattern method. The proposed embedded face recognition system achieves a CMC accuracy of 99.33% and an EER accuracy of 1% when tested on the FERET database. While

deep learning methods do the job, they often generate complicated models that need a large amount of data for training and require expensive calculations. This paper describes a face recognition system that employs a novel approach to facial representation and recognition by use of artificial neural networks. This research demonstrates how to convert a human face's front to image coding and a histogram of pixel placement, which may be used as a metric for automated face recognition. This study builds and assesses real-time face recognition systems that use Convolutional Neural Networks (CNNs). The suggested concept is first evaluated using standard AT&T datasets, and the findings are then utilised to create a real-time system. We can fix the first problem using facial recognition technology, which employs faces only for attendance. However, with so many options available, picking the best face recognition algorithm may be a real challenge. Research into face detection and identification is an emerging and active area in computer vision and deep learning. Among the many applications of face detection and identification is the ability to verify an individual's identity in public places like grocery stores and financial institutions.

LITERATURE REVIEW

Goel, Rita, Irfan Mehmood, and Hassan Ugail. (2021). It is difficult to use facial recognition to accurately identify siblings. The striking resemblance between brothers' looks is mostly to blame. Here, we look at how to test the ability of cutting-edge deep learning face recognition models to distinguish between siblings' faces using different similarity indexes. This study focused on four models: FaceNet, VGGFace, VGG16, and VGG19. The selected deep learning model has computed the embeddings for every given set of photos. Image classification relies on five well-established similarity metrics: cosine similarity, structured similarity, Manhattan distance, Minkowski distance, and Euclidean distance. Each of these metrics has its own threshold that pictures are compared against. Using typical confusion matrices, we can determine each model's accuracy, precision, and misclassification rate. The SiblingDB database's publicly accessible HQf subset is used to generate four separate experimental datasets: full-frontal-face, eyes, nose, and forehead of sibling pairs. When comparing full-facial and cropped face regions, the experimental findings reveal that the selected deep learning models' accuracy in distinguishing siblings varies with the face area in question. With a classification accuracy of over 95% in this instance, VGGFace stands out as the top option when comparing the full-frontal-face and eyes. On the other hand, FaceNet offers the greatest outcome for nose-based categorization, and its accuracy drops dramatically when comparing noses. In a similar vein, VGG16 and VGG19 do not excel at eye-based categorization but do well when compared to foreheads.

Mamdouh Youssef Abdelkader Ahmed, Youssef &Wagdy, Ali. (2019) The challenge of recognising and authenticating someone in a picture only based on their face is known as face recognition. No matter the lighting conditions, the presence of facial hair or accessories, or the effects of ageing on a face, people are perfectly capable of solving the face identification issue. However, up until very recently, it remained a difficult computer vision challenge. Modern models may initially perform as-well as, and then surpass, human face recognition skills because to deep learning algorithms that build rich and compact representations of faces from very huge datasets. Learn how deep learning techniques may accomplish superhuman performance and how to solve the face recognition challenge in this project. The Convolutional Neural Network technique was used to construct a system that relies on deep neural networks. To begin, a database consisting of 113 individuals was used to test the system. Each person had 20 photos, all of which were classified as male. The second step was to test the system on a database consisting of 132 individuals; each person contains 20 photos, 20 of which are classified as male and 20 as female. Lastly, it was evaluated on a database including 152 individuals, with 20 photographs per person,

classified as either male or female. There were two sets of data: one for training purposes, and another for use in tests and predictions.

Liang, Li. (2020) with the Revolutionary Advancements in Deep Learning Most recognition technologies, including face recognition, use deep learning algorithms. Despite its high recognition accuracy, the Deep Learning Algorithm requires a lot of computing power. We can complete relevant operations faster using artificial intelligence chips in the mobile terminal, which can speed up deep learning. Similar to convolution, deep learning has fixed modes. Artificial intelligence chips can optimise the corresponding operation modes, greatly improving the efficiency of deep learning operations. Mobile terminals can swiftly execute complex deep learning operations like deep learning-based face recognition in this way. The Google Tensor Processing Unit is an example of an AI chip; it can speed up the deep learning system's tensor flow, making it more efficient than Gpus. The TPU outperforms the same CPU and GPU by a factor of 1,530 in terms of performance and 3,080 in terms of efficiency (performance/watt). Principal Components Analysis (PCA) and Ad Boost are two examples of traditional algorithms used for face recognition. Deep Learning Technology Has a Very Different Detection Effect from Traditional Face Recognition, Despite the Former's Speed. Traditional face recognition methods, such as principal component analysis (PCA), have far lower accuracy compared to deep learning algorithms. Traditional PCA face recognition technology, however, is inadequate when it comes to identifying large user populations.

Ríos-Sánchez, et al (2019) The importance of single sample per person verification in security, surveillance, and border crossing applications has garnered a lot of attention. These days, e-voting and future bank solutions are also part of this picture, expanding the scope of this area of study to include mobile and low-resource devices. A solution that can draw on past experiences and comparable contexts is necessary in these cases since user enrollment in the system is typified by the availability of only one picture. Along with two publicly accessible models, this work defined and tested two deep learning models for face recognition that were specifically built for use on mobile devices and in situations that save resources. This review set out to do two things: first, provide a level playing field for the models; and second, find out how much of an impact shrinking the model size had on the outcomes. In order to provide a comprehensive review that addresses all potential environmental variables and application needs, the models were evaluated according to their size and accuracy. A large number of diverse public and private datasets were used in conjunction with a well-defined assessment process to achieve this goal.

Sturman, O., von Ziegler, L., Schläppi, C. et al. (2020) Preclinical research on brain function is mostly dependent on behavioral analysis and animal observation. While commercial systems have made it possible to automate animal monitoring and conduct semi-high-throughput behavioral analysis, these platforms aren't flexible enough to be used in different testing situations and don't great at identifying ethologically important behaviors. The last few years have seen significant advancements in deep learning and machine vision, which have made it possible to precisely track specific body parts of freely moving rats without the need of markers. The purpose of this study is to evaluate commercially available platforms (such as EthoVision XT14 and Noldus) and compare their performance to that of cross-verified human annotation. The platforms are developed and maintained by TSE Systems. We provide a collection of recordings of three commonly used behavioral tests—the open field test, the elevated plus maze, and the forced swim test—that have been meticulously annotated by several human raters. With these details, we ran Deep Lab Cut, a posture estimation programme, to get skeletal mouse models. By using basic post-analyses, we achieved tracking accuracy comparable to or higher than commercial behavioral tracking systems in a variety of standard behavioral tests using animals' skeletal representation. In order to combine the skeleton model with the human annotations, we created supervised ML classifiers. We can now beat commercial solutions while

scoring ethically important behaviors with accuracy comparable to humans the current gold standard—thanks to our novel combination methodology. We conclude by demonstrating that the ensuing machine learning strategy eradicates inter- and intra-annotator variability. Ultimately, our method surpasses commercial systems in terms of performance and accuracy of behavioral data, all while being far more cost-effective.

RESEARCH METHODOLOGY

First phase

Here we showcase the work that revolves around the CK+ dataset. Included in this procedure are steps to prepare CK+ photos for processing, establish a network topology, feed the network with the updated CK+ images, train the network, adjust parameters, and assess the network's correctness.

CK+ image pre-processing

The next paragraphs provide the picture pre-processing. A binary file is produced from the CK+ picture dataset, which is used as input. When training the network, the binary file was fed into it. The experimental phase of the project started at this point. Problems started to emerge at the outset due to the folder structure of the dataset and the lack of labels for some picture sequences. The walk Python method was used to transfer all picture files from the CK+ tree directory to a single location. Since the filename was sufficient to associate each photograph with its respective sequence and topic, this was indeed feasible. The label files were transferred using a similar method. In addition, a new folder was created for all picture sequences that did not have labels.

Data input into TensorFlow (TF)

Records are read into TensorFlow (TF) from a binary file that contains the pictures and their labels. A queue containing all the training samples was generated by this procedure.

```
1 filenames = [os.path.join(PATH, FILENAME)]
2 # Create a queue that produces the filenames to read.
3 filename_queue = tf.train.string_input_producer(filenames)
4 # Read examples from files in the filename queue.
5 read_input = read(filename_queue)
```

The key and value are then subjected to casting operations.

- **Key:** This is the label's value after being converted to an integer.
- **Value:** That tensor that contains the image tensor.

The original dimensions of the tensor are changed to height, breadth, and depth later on. Data augmentation may now be done by applying transforms to the picture.

Network description

The intended topology is based on the one established by the Visual Geometry Group (VGG) at Oxford University. The VGG network, which highlighted the significance of network depth in improving classification accuracy, was awarded second place in the ImageNet 2014 competition. Comparing VGG to various designs, it was shown to generalize effectively to different workloads

and data sets. In contrast to other cutting-edge topologies like Google Net, this network prioritizes the use of pure convolutional network designs, which led to its selection.

Frameworks

- **Caffe:** Initially, Caffe was Yangqin Jia's PhD thesis at the Berkeley Vision and Learning Centre (BVLC). In recent times, Caffe has become a project run by BVLC, with several contributors drawn from its expanding community. Python and C++ were the languages used to create it, and it is compatible with Windows, OS X, and Linux. For computer vision calculations, Caffe was primarily developed.
- **Theano:** The University of Montreal's Montreal Institute for Learning Algorithms is where Theano was first developed. These days, it's an open-source project with a large user base. When it comes to certain graph calculations, the Python package known as Theano may generate instructions for either CPUs or GPUs. These instructions outperform pure Python by a wide margin and are more in line with C's performance. Theano is primarily concerned with mathematical expressions, particularly those using tensors. In addition, the Theano compiler uses a wide variety of optimization strategies to produce operation-specific C code.
- **TensorFlow:** One open-source machine learning library is TensorFlow (TF), which is developed in C++ and Python. Its publication on November 15th garnered a lot of attention from the media. The Google Brain Team was responsible for developing TF, which is the key cause. A number of Google products have already benefited from TF's enhanced performance in certain areas. Some examples of such functions include Google Now's voice recognition, Google Photos' search capabilities, and Inbox by Gmail's smart reply functionality.

DATA ANALYSIS

An overview of the parameters is shown in Table 1. Unless otherwise specified, the tests on this phase are carried out in this setup.

Network loss and learning rate

Figure 1's loss chart illustrates the minimization of the cost function during training. The loss is minimized steadily throughout the first 450 steps. The curve begins to converge beyond that point. In steps 800 and 900, the loss values are 0.79 and 0.77, respectively. Reducing it by 0.02 on 100 steps is all it is. The model has converged if it exhibits this behavior. So, training for a longer period won't significantly cut the network's loss. One example is shown in Figure 2 by using a reduced learning rate of 0.01. This results in a distinct curve. A longer training period is required since the model has not converged yet. Overfitting occurs when a network is exposed to training for too long, even if a longer training period may increase the network's accuracy. Approximately 10 minutes and 37 seconds was the average amount of time spent training.

Table 1: Experiment setup description for the first stage

Parameter	Value
Batch size	64
Output channels	64
Kernel size	3x3
Kernel stride	1
Padding	same
Pooling	max pooling
Pool size	3x3
Pool stride	2
FC1 neurons	384
FC2 neurons	192
Dropout	0.99
Activation function	ReLU
Number of classes	6
Cost function	cross entropy
Optimizer	momentum
Momentum	0.9
Learning rate	0.1

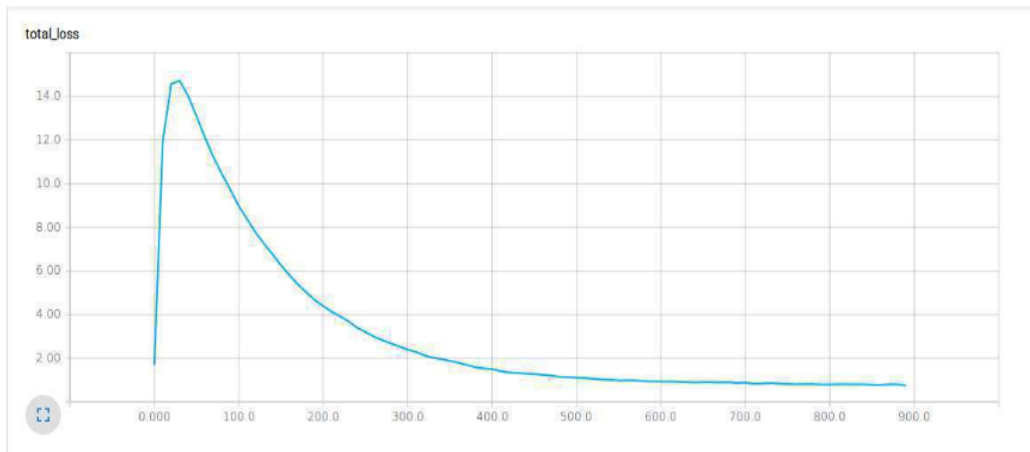


Figure 1: With a learning rate of 0.1 and 900 training steps, the total loss is 0.

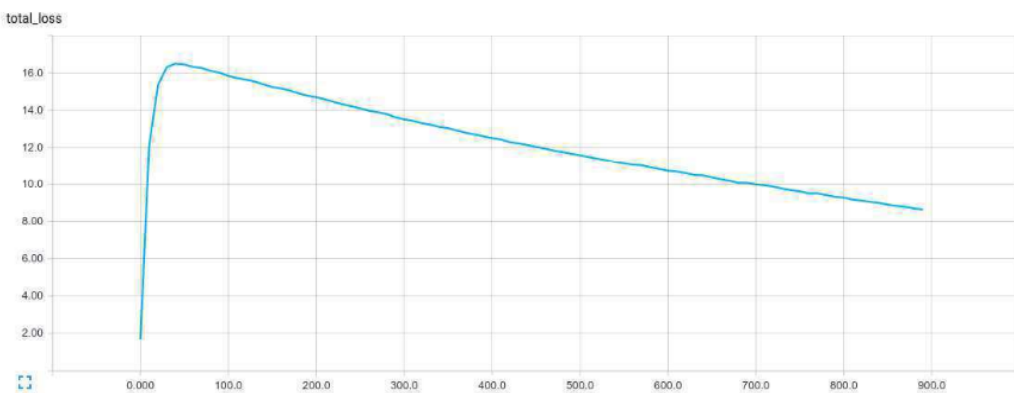


Figure 2: At a learning rate of 0.01, the whole loss occurred over 900 training steps.

When minimizing the cost function, the significance of the learning rate is seen in Figures 1 and 2. On top of that, it offers a solid method for calculating the learning time. Preventing network

overfitting requires a precise training period. When a network is overfit, its generalizability suffers. To generalize here is to make predictions on non-training cases with the same level of accuracy as on training examples. Lessening the amount of time spent training does have certain practical advantages. A network makes it easy to test out novel concepts, which is one of them. Cutting down on waste is yet another perk. A big research institution, where a lengthier training period results in poorer production for multiple personnel, might further detect this influence. DL's usage of GPUs demonstrates the community's genuine interest in reducing training time. Explosion of losses is another possible outcome of the learning rate. This occurs when the network's learning rate is set to a value that is too high. A problem that often arose at the start of this endeavor was loss explosion. We tried a few different learning rate numbers, but all it did was make it take longer. Returning the SoftMax layer's output as a probability distribution solved this issue.

Classification accuracy

With respect to the training set, Table 2 details the network's accuracy. Classification based on Top-1 and Top-3 prediction yielded the stated findings. Top-1 forecasts had an average accuracy of 70.54% and a maximum accuracy of 72.5%. Contrarily, the average accuracy for Top-3 forecasts was 77.9%, while the maximum accuracy was 80%. In comparison to the baselines based on random guesses, the outcomes employing setup on table 1 performed better, as shown in table 3.

Table 2: Accuracy of network classification on six emotion categories on the CK+ dataset with a learning rate of 0.1

	Run									
Top	1	2	3	4	5	6	7	8	9	10
1	70.0	71.1	72.5	70.8	68.1	71.9	71.1	71.3	68.4	70.2
3	75.8	78.4	77.8	77.7	80.0	77.8	78.7	76.4	79.1	77.3

Table 3: Comparison of results with suggested standards

Model	Accuracy
6-label random guess	16.67
Most frequent label	25.38
EmotiW 2014's winner	39.35
EmotiW 2015's winner	44.70
3 most frequent labels	64.50
Top-1 prediction	70.54
Top-3 prediction	77.90
Ramachandran	97.00

Table 4: Accuracy of the network with a 0.5 dropout

	Run									
Top	1	2	3	4	5	6	7	8	9	10
1	56.2	57.0	58.8	58.1	58.9	55.6	57.3	57.7	55.9	56.9
3	75.9	76.9	78.9	77.2	78.6	77.5	78.7	78.0	77.7	76.1

Table 5: The accuracy of classification using the Adam optimizer

	Adam Optimizer / Run									
Top	1	2	3	4	5	6	7	8	9	10
1	64.4	62.2	67.0	64.2	66.6	64.8	66.4	65.9	65.2	63.4
3	75.3	75.8	75.5	74.7	75.6	76.1	76.1	78.1	76.2	76.1

Table 6: Accuracy of classification with the FTRL optimizer

	FTRL Optimizer / Run									
Top	1	2	3	4	5	6	7	8	9	10
1	18.6	18.9	18.6	20.0	20.3	18.0	18.9	19.2	19.8	18.3
3	78.9	79.2	79.5	79.7	78.9	79.2	79.1	78.7	78.3	78.7

Due to the usage of distinct corpora, a direct comparison with Emote may be challenging. A few suggestions for enhancing its categorization accuracy were offered by the 2014 winner. One of them aimed to test out a handful of challenging categories, while the other sought to discover a more efficient fusion approach. Using dropout improved the accuracy performance of our investigation, as will be detailed later. Their experiments do not include the use of dropout. The disparity between the top three forecasts and the top one is less than anticipated. A discrepancy of 7.36 percent is typical.

CONCLUSION

In this paper, one of the most fascinating and difficult tasks is emotion identification. The purpose of this study was to provide a method for applying deep learning to categories facial emotions from static photographs of faces. There have been several attempts at solving this complicated issue using various methods. Feature engineering has yielded satisfactory results, but feature learning one of DL's promises was the focus of this effort. The use of engineered features is becoming standard in face emotion recognition algorithms. A major restriction, the absence of a comprehensive dataset of emotions, means that a feature-learning-only solution is still a way off. As an example, the dataset used in the ImageNet contest included 14,197,122 photos. A bigger dataset would allow for the implementation of networks with a greater capacity to learn features. So, deep learning methods might be used to do emotion categorization. Results weren't quite cutting edge, but they were an improvement over competing methods like feature engineering. What this suggests is that with enough labelled samples, DL approaches will be able to tackle this issue in the long run.

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Beyond Intelligence: Advances in AI-Enabled Systems for Tomorrow

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Abstract:

Artificial intelligence (AI) has evolved from static, rule-based programs to dynamic systems capable of learning, reasoning, and adapting in real time. This evolution has enabled the creation of AI-enabled systems that extend beyond traditional computational intelligence, encompassing autonomy, contextual awareness, human–machine collaboration, and ethical adaptability. This paper reviews recent advances in AI-enabled systems, focusing on architectural innovations, application domains, emerging trends, and the challenges that lie ahead. This paper synthesizes recent advances in AI-enabled systems, identify key technological milestones, and explore future directions that extend beyond intelligence paving the way for tomorrow’s intelligent infrastructure.

Keywords: Artificial intelligence, Deep Learning, Reinforcement learning

Introduction:

Artificial intelligence has transitioned from conceptual frameworks to systems deeply embedded in everyday life. Early AI focused on symbolic reasoning and expert systems, designed to replicate human decision-making within narrowly defined domains (Russell & Norvig, 2016). These early iterations, while groundbreaking, lacked the ability to learn from data, adapt to new

environments, and interact naturally with humans. Today, AI-enabled systems are defined not merely by intelligence but by contextual adaptability, autonomy, and integration into complex socio-technical ecosystems (Jordan & Mitchell, 2015). These systems include autonomous vehicles, healthcare diagnostics platforms, intelligent manufacturing lines, adaptive educational systems, and human-robot collaborative agents. Their development raises questions not only about technological capability but also about ethics, safety, and social impact.

AI-Enabled Systems

AI-enabled systems integrate computational intelligence with real-world sensing, data processing, decision-making, and autonomous action. Unlike isolated AI models that perform specific tasks (e.g., image classification), AI-enabled systems operate across layers of perception, cognition, and action. These systems often include:

Perception Modules: Utilizing sensors, computer vision, and natural language processing (NLP) to interpret environments.

Learning and Reasoning: Employing machine learning (ML), reinforcement learning (RL), and probabilistic reasoning to make decisions under uncertainty.

Actuation and Control: Executing real-world actions through robotics, automation, and adaptive interfaces.

Human–Machine Interaction: Enabling effective collaboration and communication with users (Goodfellow, Bengio, & Courville, 2016). Thus, AI-enabled systems go beyond raw computational power to encompass adaptive behavior in complex environments.

Deep Learning and Hybrid Models: Deep learning (DL) has been at the forefront of AI breakthroughs, with architectures such as convolutional neural networks (CNNs) for vision tasks and transformers for language modeling. However, DL models alone lack explainability and robustness in uncertain environments (LeCun, Bengio, & Hinton, 2015). To overcome these limitations, hybrid architectures combine symbolic reasoning with neural networks. These neuro-symbolic systems integrate rule-based logic with statistical learning, improving interpretability and generalization (Garcez, Broda, & Gabbay, 2019).

Reinforcement Learning and Adaptive Control: Reinforcement learning (RL) allows machines to learn optimal actions through trial and reward. RL has been applied in game playing, robotics, and autonomous navigation. Advances in deep reinforcement learning (Deep RL) have enabled agents to learn complex behaviors in dynamic settings, such as traffic routing and robotic manipulation (Sutton & Barto, 2018).

Key Domains of Impact

AI-enabled systems are transforming multiple sectors.

Autonomous Transportation: Autonomous vehicles (AVs) exemplify AI-enabled systems that integrate perception, prediction, planning, and control. AVs rely on sensor fusion, deep learning for object detection, and RL for decision-making. These systems promise safer, more efficient transportation but face challenges in robust perception under diverse conditions and ethical decision frameworks for edge cases (Litman, 2020).

Healthcare and Precision Medicine: AI systems in healthcare are transitioning from diagnostic support to autonomous clinical decision-making. AI-enabled platforms now assist in interpreting medical imaging, predicting patient outcomes, and personalizing treatment plans (Esteva et al., 2019).

Smart Manufacturing and Industry 4.0: Industry 4.0 integrates cyber-physical systems, IoT, and AI to create smart factories capable of self-optimization and adaptive control. Predictive maintenance, real-time quality inspection, and supply chain optimization are driven by AI-enabled systems that react to shifting conditions in real time (Wang, Wan, Li, & Zhang, 2016). These systems enhance productivity while introducing new requirements for cybersecurity and human–machine collaboration.

Education and Human-Centered AI: AI-enabled systems in education are moving beyond content delivery to personalized learning environments that adapt to learner profiles, motivational states, and cognitive progress. Intelligent tutoring systems and adaptive assessment platforms aim to provide real-time feedback and tailored instructional strategies (Woolf, 2010). These systems navigate the delicate balance between automated guidance and human agency.

Emerging Trends in AI-Enabled Systems

Several trends indicate future directions for AI-enabled systems:

Explainable and Trustworthy AI: As AI systems become more autonomous, explainability and trustworthiness are critical. Stakeholders must understand the rationale behind decisions, especially in safety-critical domains such as healthcare and transportation. Explainable AI (XAI) techniques provide transparency without sacrificing performance (Gunning, 2017).

Human-AI Collaboration: Rather than replacing humans, AI-enabled systems increasingly aim to augment human capabilities. Collaborative robots (cobots) in manufacturing, AI assistants in creative domains, and decision support systems in management exemplify this trend. Effective collaboration requires systems that communicate intent, handle uncertainty, and respect human values.

Ethical AI and Regulatory Frameworks: AI systems raise ethical concerns related to fairness, privacy, accountability, and bias. Regulatory frameworks such as the EU's AI Act are emerging to standardize safety and ethical governance of AI applications (European Commission, 2021). These policies will shape how AI systems are developed and deployed globally.

AI for Sustainability: AI-enabled systems are being harnessed to address global challenges like climate change, resource optimization, and sustainable agriculture. From predictive models for energy management to AI-driven climate simulations, these systems play a strategic role in sustainability initiatives.

Challenges and Limitations

Despite notable progress, several challenges persist:

Data Quality and Bias: AI systems depend on large datasets, and poor data quality can lead to biased or unsafe outputs (Barocas & Selbst, 2016).

Robustness and Generalization: Systems trained in limited scenarios may fail in unforeseen conditions.

Safety and Verification: Safety-critical AI systems require formal verification and rigorous testing.

Social and Economic Impact: Widespread deployment raises concerns about job displacement and socio-economic inequality.

These challenges require interdisciplinary collaboration among technologists, ethicists, policymakers, and domain experts.

Conclusion

AI-enabled systems represent a convergence of intelligent algorithms, real-world sensing, human interaction, and autonomous action. Progress in deep learning, hybrid architectures, edge computing, and reinforcement learning has driven remarkable innovation across transportation, healthcare, manufacturing, and education. Looking ahead, the future of AI-enabled systems lies not just in computational intelligence but in responsible autonomy, ethical integration, and human-centric design. The journey toward tomorrow's intelligent systems demands a balanced pursuit of capability, safety, and societal impact.

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Comparative Study of Cloud Platforms (Aws, Azure, Google Cloud) for Business Analytics - Cloud Based Data Storage Solutions & Future Trends

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ABSTRACT

The rapid growth of data-driven decision-making has made cloud computing an essential foundation for modern business analytics. Organizations increasingly rely on cloud platforms to store, process, and analyze large volumes of structured and unstructured data in a scalable and cost-efficient manner. This study presents a comparative analysis of three leading cloud platforms—Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP)—with a focus on their suitability for business analytics applications.

The report examines core components such as cloud-based data storage solutions, analytics services, pricing and cost management models, platform strengths, and emerging trends. AWS is identified for its extensive service portfolio and architectural flexibility, Azure for its strong enterprise integration and hybrid cloud capabilities, and Google Cloud for its high-performance, serverless analytics and advanced AI/ML integration. The study also highlights

future trends including AI-driven analytics, real-time and serverless processing, multi-cloud strategies, and unified data architectures.

The findings suggest that there is no one-size-fits-all cloud solution; instead, the choice of platform depends on organizational requirements, existing infrastructure, analytics complexity, and cost considerations.

Keywords: Cloud Computing, Business Analytics, AWS, Microsoft Azure, Google Cloud Platform, Cloud Data Storage, Data Warehousing, Pricing and Cost Management, AI-Driven Analytics, Future Trends

1. INTRODUCTION

Business analytics has become an essential function in modern organizations as it supports informed decision-making through the use of data, statistical techniques, and predictive models. With the increasing availability of digital data from customer interactions, online transactions, supply chain operations, social media, and IoT devices, organizations are handling data volumes that are far beyond the capabilities of traditional on-premises systems. As a result, cloud computing has emerged as a key solution for scalable data storage, flexible processing, and cost-efficient analytics.

Cloud platforms provide on-demand access to computing resources such as storage, servers, databases, analytics tools, and artificial intelligence services. Instead of investing heavily in physical infrastructure and maintaining local data centers, companies can use cloud services to quickly deploy analytics environments, store large datasets, and perform complex computations. This shift has enabled businesses to adopt faster reporting systems, real-time dashboards, and advanced machine learning-based insights.

In business analytics, cloud computing plays an important role across the entire data lifecycle—from collecting and storing raw data, cleaning and transforming datasets, running analytical models, and finally visualizing insights for strategic decisions. Cloud platforms also support integration with modern tools such as business intelligence software, data lakes, data warehouses, and real-time streaming services. This makes cloud environments highly suitable for organizations looking to implement modern analytics practices such as customer segmentation, demand forecasting, fraud detection, pricing optimization, and supply chain analytics.

Among global cloud providers, Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) are recognized as the top three public cloud platforms. Each provider offers a wide range of analytics and data storage services, but they differ in terms of service maturity, pricing strategies, integration ecosystems, ease of use, and suitability for specific industries. AWS is known for its vast service portfolio and global reach, Azure is preferred by enterprises due to its strong integration with Microsoft technologies and hybrid cloud support, and Google Cloud is popular for its advanced data analytics capabilities and AI-driven services.

This report presents a comparative study of AWS, Azure, and Google Cloud with a specific focus on business analytics and cloud-based data storage solutions. The study also discusses emerging trends such as serverless analytics, AI-powered analytics, real-time processing, and multi-cloud strategies. The overall intention is to provide clear insights into how organizations can choose the right cloud platform based on their analytics requirements, data workloads, cost considerations, and future scalability needs.

2. OBJECTIVES OF THE STUDY

- To compare AWS, Azure, and Google Cloud in terms of business analytics services
- To analyze cloud-based data storage solutions offered by each platform
- To evaluate strengths and limitations from a business analytics perspective
- To identify emerging and future trends in cloud analytics

3. OVERVIEW OF MAJOR CLOUD PLATFORMS

The global public cloud market is dominated by three leading providers—Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). These platforms offer a wide range of services including computing power, storage, networking, analytics, artificial intelligence, and security. Despite providing similar core services, each platform has a unique identity and competitive advantage shaped by its background, ecosystem, and long-term strategy.

Organizations select cloud platforms based on multiple factors such as service maturity, integration compatibility, analytics strengths, pricing flexibility, global availability, and industry-specific compliance. Understanding the overall strengths of each provider is important before comparing their business analytics and data storage solutions in detail.

3.1 Amazon Web Services (AWS)

Amazon Web Services is widely recognized as the first major public cloud provider and continues to be the market leader due to its maturity, scalability, and extensive service offerings. AWS delivers cloud services through a large global network of data centers and provides support for businesses of all sizes—from startups to multinational corporations.

AWS offers strong capabilities for analytics and data storage, supported by services such as Amazon S3 (object storage), Amazon Redshift (data warehousing), AWS Glue (data integration), Athena (serverless querying), and EMR (big data processing). AWS also supports a wide variety of database engines, both SQL and NoSQL, and offers advanced machine learning support through services like Amazon SageMaker.

A key strength of AWS is its flexible architecture and broad third-party ecosystem, including integrations with tools like Tableau, Power BI, Snowflake, Databricks, and many open-source analytics frameworks. AWS is commonly adopted in industries such as e-commerce, banking, manufacturing, healthcare, and media due to its strong scalability and global reliability.

However, AWS services can be complex to manage for new users due to the wide range of tools and configurations available. Cost optimization may also require careful planning, especially when working with large datasets and high-performance analytics workloads.

3.2 Microsoft Azure

Microsoft Azure is one of the most widely used cloud platforms in enterprise environments because of its strong integration with Microsoft business tools and its ability to support hybrid cloud models. Many organizations that already use Microsoft technologies such as Windows Server, Active Directory, Microsoft 365, and SQL Server find Azure convenient for cloud adoption.

Azure provides a unified set of services for data storage and analytics, including Azure Blob Storage, Azure Data Lake Storage, Azure Synapse Analytics, Azure SQL Database, Cosmos DB, and Azure Data Factory. In addition, Power BI is a major advantage for Azure users, offering seamless integration for dashboarding, reporting, and business intelligence.

Azure is known for strong governance and enterprise security features, which makes it suitable for regulated industries such as banking, insurance, government, and healthcare. Its hybrid cloud capabilities allow businesses to run analytics workloads across both on-premises systems and cloud environments, enabling smoother migration and better flexibility.

A potential limitation is that Azure is most effective when organizations are deeply aligned with Microsoft ecosystems. While Azure supports open-source and non-Microsoft environments as well, organizations may need additional configurations to achieve the same level of smooth integration that Microsoft-based organizations experience.

3.3 Google Cloud Platform (GCP)

Google Cloud Platform is known for its strengths in big data processing, analytics innovation, and AI-powered services. Built on Google's internal infrastructure expertise, GCP provides advanced tools for organizations that prioritize data analytics, machine learning, and high-performance computing.

One of the biggest strengths of Google Cloud is BigQuery, a serverless data warehouse designed for fast and scalable analytical queries. BigQuery allows organizations to analyze large datasets without managing infrastructure, making it highly efficient for analytics-first use cases. GCP also provides Dataflow for stream and batch processing, Pub/Sub for messaging and real-time data pipelines, and Looker for business intelligence and analytics visualization.

Google Cloud is considered highly suitable for organizations that work with massive volumes of data, such as digital platforms, advertising networks, fintech companies, research institutions, and AI-driven businesses. It is also attractive for modern analytics architectures because of its support for Kubernetes and cloud-native solutions.

However, compared to AWS and Azure, Google Cloud has a relatively smaller enterprise footprint in some regions and industries. Organizations may also find a slightly smaller ecosystem of enterprise tools, though this is rapidly improving due to the growing adoption of GCP in analytics and AI projects.

4. BUSINESS ANALYTICS CAPABILITIES

Business analytics in the cloud is built around a common workflow: data ingestion → data storage → data processing and transformation → analytical modeling → visualization and reporting. Cloud providers support this lifecycle through managed services that reduce infrastructure burden and accelerate analytics delivery. While AWS, Azure, and Google Cloud offer similar categories of tools, their strengths differ based on design approach, integration ecosystem, and ease of execution.

4.1 Analytics Services in AWS

AWS offers one of the most extensive analytics portfolios, enabling organizations to build highly customizable analytics architectures. AWS supports both traditional batch analytics and modern real-time analytics, making it suitable for complex and large-scale business use cases.

Key analytics services in AWS include:

- **Amazon Redshift:** A cloud data warehouse used for high-performance analytical querying and reporting.
- **AWS Glue:** A managed ETL service for preparing and transforming data for analytics.
- **Amazon Athena:** A serverless query service that analyzes data directly from Amazon S3.
- **Amazon EMR:** A managed big data service that supports frameworks like Apache Spark and Hadoop.
- **Amazon QuickSight:** A scalable BI tool used for dashboards and interactive reporting.

Business analytics advantages of AWS:

- Supports data lakes and warehouse architectures effectively
- Strong integration with third-party analytics tools (Tableau, Snowflake, Databricks)
- Suitable for organizations with diverse data sources (ERP, CRM, web/app data)

Best fit: Organizations that need flexible design, multi-source data handling, and enterprise-scale analytics.

4.2 Analytics Services in Azure

Azure analytics services are designed for tight integration with enterprise environments and Microsoft ecosystems ecosystem. Azure is well suited for organizations focusing on unified reporting, strong governance, and hybrid analytics deployment (on-premises + cloud).

Key analytics services in Azure include:

- **Azure Synapse Analytics:** A unified platform that combines data warehousing and big data analytics for large-scale reporting and analysis.
- **Azure Data Factory:** A cloud-based data integration service used to build ETL/ELT pipelines from multiple sources.
- **Power BI:** A widely used business intelligence tool for dashboards, KPI tracking, and visual reporting.
- **Azure Databricks:** A managed Spark environment used for advanced data engineering and scalable analytics.
- **Azure Machine Learning:** Supports predictive analytics and model deployment for business decision support.

Business analytics advantages of Azure:

- Strong integration with **Microsoft tools** such as Excel, SQL Server, Teams, and Microsoft 365
- Excellent for **enterprise reporting and business dashboards** using Power BI
- Supports **hybrid analytics** with smooth connection to on-premises data systems
- Reliable governance and access control through **Azure Active Directory** and role-based security

Best fit: Enterprises that rely on Microsoft ecosystems, require hybrid cloud analytics, and prioritize structured BI reporting and governance.

4.3 Analytics Services in Google Cloud

Google Cloud is widely recognized for its strong analytics performance and AI-based capabilities. It is highly suitable for organizations that need fast, scalable analytics with minimal infrastructure management.

Key analytics services in Google Cloud include:

- **BigQuery:** A fully managed, serverless data warehouse designed for fast, large-scale analytical queries.
- **Dataflow:** A managed service for batch and real-time data processing and transformation.
- **Pub/Sub:** A messaging service used for real-time event streaming and data ingestion.
- **Looker:** A modern BI platform that supports data exploration, reporting, and embedded analytics.
- **Vertex AI:** Provides tools for machine learning integration, advanced analytics, and predictive insights.

Business analytics advantages of Google Cloud:

- BigQuery enables high-speed analytics with **serverless scalability**, reducing admin effort
- Strong support for **real-time analytics** through streaming data pipelines
- Advanced AI/ML tools that support **predictive and prescriptive analytics**
- Suitable for analytics workloads requiring rapid querying over massive datasets

Best fit: Analytics-first organizations that require scalable warehousing, real-time processing, and AI-powered decision insights.

5. CORE ANALYTICS SERVICES COMPARISON

Cloud business analytics relies on **five core service areas: data ingestion, storage, processing, warehousing, and visualization**. These services form the backbone of any analytics workflow, allowing organizations to collect, manage, analyze, and visualize data efficiently.

1) Data Ingestion:

- **AWS Kinesis:** Handles real-time streaming and batch ingestion with high reliability.
- **Azure Event Hubs:** Enterprise-grade streaming service that integrates seamlessly with Microsoft ecosystems.
- **Google Pub/Sub:** Simple, scalable, and efficient for real-time event processing.

2) Data Storage:

- **AWS S3:** Industry-leading data lake storage, highly durable and scalable.

- **Azure Blob/Data Lake Storage:** Optimized for enterprise workloads and hybrid integration.
- **Google Cloud Storage:** High-performance storage for analytics workloads, optimized for integration with BigQuery.

3) Data Processing & Transformation:

- **AWS Glue/EMR:** Supports ETL, big data frameworks, and batch/stream processing.
- **Azure Data Factory/Databricks:** Ideal for building pipelines, integrating multiple data sources, and advanced processing.
- **Google Dataflow:** Streamlined batch and real-time processing with serverless scalability.

4) Data Warehousing:

- **AWS Redshift:** Robust enterprise data warehouse for large-scale queries.
- **Azure Synapse Analytics:** Combines data warehousing and big data processing in a single platform.
- **Google BigQuery:** Serverless, high-speed querying for massive datasets with minimal infrastructure management.

5) Visualization & Business Intelligence (BI):

- **AWS QuickSight:** Lightweight BI dashboards and reporting.
- **Azure Power BI:** Industry-standard for enterprise dashboards, reporting, and KPIs.
- **Google Looker:** Modern BI platform for analytics exploration and embedded reporting.

Summary of Strengths:

- **AWS:** Maximum flexibility, large ecosystem, suitable for multi-source and complex workloads.
- **Azure:** Best for enterprise integration, governance, and structured BI reporting.
- **Google Cloud:** High-speed analytics, serverless architecture, strong AI and real-time support.

6. CLOUD-BASED DATA STORAGE SOLUTIONS

Cloud data storage is a critical component of business analytics, as it ensures that data is **secure, scalable, and accessible** for processing and analysis. Modern organizations deal with large volumes of structured and unstructured data, making cloud storage a more flexible and cost-effective alternative to traditional on-premises systems. AWS, Azure, and Google Cloud provide a variety of storage services optimized for different analytics needs, from raw data lakes to structured data warehouses.

6.1 Object Storage

Object storage is designed to store **large amounts of unstructured data** such as images, logs, videos, and IoT data.

- **AWS:** Amazon S3 is highly durable, scalable, and integrates seamlessly with other AWS analytics tools.
- **Azure:** Azure Blob Storage and Data Lake Storage are optimized for enterprise-grade data lakes, supporting both analytics and machine learning workflows.
- **Google Cloud:** Google Cloud Storage offers high-performance object storage with easy integration with Big Query and AI/ML services.

6.2 Data Warehousing

Data warehouses store **structured, processed data** for reporting, dashboards, and analytical queries.

- **AWS Redshift:** Provides a fast, scalable data warehouse with columnar storage and SQL-based querying.
- **Azure Synapse Analytics:** Combines data warehousing and big data analytics in a single platform, allowing hybrid data processing.
- **Google BigQuery:** Serverless, highly scalable, and optimized for rapid analytics on very large datasets.

6.3 NoSQL and Operational Databases

For **highly dynamic or semi-structured data**, NoSQL databases are preferred:

- **AWS:** DynamoDB supports low-latency, high-throughput applications.
- **Azure:** Cosmos DB offers multi-model support and global distribution.
- **Google Cloud:** Bigtable and Firestore provide scalable, high-performance NoSQL solutions for real-time analytics and operational workloads.

6.4 Key Advantages of Cloud Storage Solutions

- **Scalability:** Storage can grow on-demand without the need for physical infrastructure.
- **Cost Efficiency:** Pay-as-you-go pricing allows businesses to optimize costs.
- **Security and Compliance:** Enterprise-grade encryption, access controls, and compliance certifications.
- **Integration:** Seamless connection with analytics, AI, and machine learning services for end-to-end data processing.

Cloud storage enables organizations to **build data lakes, warehouses, and hybrid analytics architectures** that support business intelligence, predictive analytics, and real-time decision-making. Selecting the right storage solution depends on **data type, analytics needs, performance requirements, and integration with other cloud services**

7. PLATFORM STRENGTHS

Each cloud platform—AWS, Azure, and Google Cloud—has unique strengths that make it suitable for specific business analytics and storage requirements.

7.1 Amazon Web Services (AWS)

- **Market Leader & Maturity:** AWS has the largest global footprint and a wide range of services for all types of workloads.
- **Flexibility & Customization:** Supports diverse architectures, from data lakes to complex analytics pipelines.
- **Integration Ecosystem:** Seamless connectivity with third-party tools like Tableau, Databricks, and Snowflake.
- **Scalability & Reliability:** High availability across regions and robust disaster recovery options.
- **Analytics & Machine Learning:** Broad portfolio of tools for real-time, batch, and predictive analytics.

7.2 Microsoft Azure

- **Enterprise Integration:** Tight integration with Microsoft products (Office 365, SQL Server, Active Directory).
- **Hybrid Cloud Support:** Efficiently connects on-premises infrastructure with cloud workloads.
- **Governance & Security:** Strong compliance certifications and role-based access management.
- **Business Intelligence:** Power BI provides industry-leading dashboards and reporting.
- **Ease for Enterprises:** Streamlined for organizations already using Microsoft technologies.

7.3 Google Cloud Platform (GCP)

- **High-Speed Analytics:** BigQuery enables serverless, fast querying of massive datasets.
- **AI & Machine Learning Integration:** Vertex AI and AutoML simplify predictive analytics.
- **Serverless Architecture:** Minimal infrastructure management required, enabling faster deployment.
- **Real-Time Analytics:** Pub/Sub and Dataflow support streaming and event-driven data processing.
- **Developer-Friendly:** Strong support for open-source tools and cloud-native solutions.

8. PRICING AND COST MANAGEMENT

Cost is a critical factor in choosing a cloud platform for business analytics. AWS, Azure, and Google Cloud all use a **pay-as-you-go model**, but pricing structures, optimization tools, and cost predictability vary. Effective cost management ensures that organizations can scale analytics workloads without overspending.

8.1 Amazon Web Services (AWS)

- **Pricing Model:** Pay-per-use with multiple options such as on-demand, reserved instances, and spot instances.
- **Cost Factors:** Charges are based on compute, storage, data transfer, and additional analytics services.
- **Cost Management Tools:** AWS Cost Explorer and AWS Budgets help monitor usage, forecast expenses, and optimize resources.
- **Considerations:** Highly flexible pricing but can become complex for large-scale analytics workloads. Proper planning is needed to avoid unexpected costs.

8.2 Microsoft Azure

- **Pricing Model:** Pay-as-you-go with options for reserved capacity and hybrid benefits for organizations with existing Microsoft licenses.
- **Cost Factors:** Pricing is influenced by compute hours, storage usage, data egress, and premium services like Synapse Analytics and Power BI.
- **Cost Management Tools:** Azure Cost Management and Billing provide detailed usage reports, budgets, and recommendations for savings.
- **Considerations:** Enterprises using Microsoft products can often reduce costs through license integration and reserved instances.

8.3 Google Cloud Platform (GCP)

- **Pricing Model:** Pay-per-use with sustained-use discounts, committed-use contracts, and custom pricing for enterprise clients.
- **Cost Factors:** Billed for compute, storage, network usage, and analytics services like BigQuery.
- **Cost Management Tools:** Google Cloud Pricing Calculator and Billing Reports allow cost estimation, usage tracking, and optimization guidance.
- **Considerations:** Often perceived as more cost-efficient for analytics-heavy workloads, especially with serverless services like BigQuery.

Key Takeaways

- **AWS:** Best for flexibility but requires careful planning to manage costs.
- **Azure:** Strong for Microsoft-centric enterprises; hybrid benefits can reduce overall cost.
- **GCP:** Serverless pricing and sustained-use discounts make it attractive for analytics-intensive workloads.

Effective cost management in all platforms involves **monitoring usage patterns, choosing the right pricing models, and optimizing storage and compute resources** to balance performance and budget.

9. FUTURE TRENDS IN CLOUD-BASED BUSINESS ANALYTICS

Cloud-based business analytics is evolving rapidly as organizations demand **faster insights, smarter predictions, and more scalable solutions**. The following trends are shaping the future of cloud analytics:

9.1 AI-Driven Analytics

Artificial intelligence and machine learning are becoming integral to cloud analytics platforms. Cloud providers are embedding AI into data pipelines, enabling **predictive analytics, prescriptive insights, anomaly detection, and automated decision-making**. Tools like AWS

SageMaker, Azure Machine Learning, and Google Vertex AI simplify AI model creation and deployment.

9.2 Serverless and Real-Time Analytics

Serverless architectures, such as Google BigQuery and AWS Athena, reduce infrastructure management while enabling **on-demand scalability**. Real-time analytics is increasingly critical for industries like e-commerce, finance, and IoT, allowing businesses to respond instantly to changing trends or events.

9.3 Hybrid and Multi-Cloud Strategies

Organizations are adopting **hybrid (cloud + on-premises) and multi-cloud strategies** to avoid vendor lock-in, enhance resilience, and optimize costs. Azure's hybrid capabilities, AWS Outposts, and GCP Anthos provide flexible solutions for distributed workloads.

9.4 Data Lakehouse and Unified Analytics

The **data lakehouse model**, combining features of data lakes and warehouses, is gaining popularity. It enables structured and unstructured data processing in a unified architecture, improving analytics efficiency and reducing data duplication.

9.5 Automation and Analytics at Scale

Automation in data pipelines, ETL processes, and reporting is becoming standard, enabling **faster insights with fewer manual interventions**. Cloud platforms are increasingly offering AI-powered recommendations for query optimization, storage management, and predictive analytics deployment.

9.6 Focus on Security, Compliance, and Sustainability

As analytics workloads grow, cloud providers are enhancing **security, privacy, and compliance features**. Sustainability is also becoming a focus, with energy-efficient data centers and carbon tracking integrated into cloud services.

10. HOW TO CHOOSE A CLOUD PLATFORM

Selecting the right cloud platform for business analytics depends on several key factors:

1. **Business Requirements:** Consider the type of analytics workload—batch vs real-time, AI-driven, or traditional BI reporting.
2. **Existing Infrastructure:** Organizations heavily using Microsoft products may benefit from Azure, while analytics-first or AI-heavy workloads may favor Google Cloud.
3. **Scalability Needs:** AWS offers extensive flexibility and global reach, suitable for enterprises requiring large-scale and multi-source data integration.
4. **Cost Considerations:** Compare pricing models, discounts, and long-term cost predictability. GCP often provides cost advantages for analytics-heavy and serverless workloads.
5. **Integration and Ecosystem:** Evaluate third-party tool compatibility, available APIs, and workflow integration for seamless analytics pipelines.

A careful assessment of these factors ensures that the selected platform aligns with organizational strategy, analytics complexity, and budget constraints.

11. CONCLUSION

Cloud-based platforms—AWS, Azure, and Google Cloud—have transformed business analytics by providing scalable storage, advanced analytics, and AI-powered insights. AWS excels in flexibility, scale, and broad service options; Azure stands out for enterprise integration, hybrid deployment, and robust BI reporting; Google Cloud offers high-speed analytics, serverless architectures, and strong AI/ML capabilities.

The choice of platform depends on the organization's data strategy, existing technology ecosystem, analytics requirements, and cost considerations. Cloud storage solutions, coupled with advanced analytics tools, enable organizations to process large volumes of data efficiently, make informed decisions, and remain competitive in a rapidly evolving business environment.

With trends like AI-driven analytics, real-time processing, hybrid/multi-cloud adoption, and data Lakehouse architectures, cloud platforms will continue to evolve, offering smarter, faster, and more cost-effective solutions for business analytics in the future.

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Impact of Artificial Intelligence on Online Advertising Performance in Karungal Town

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ABSTRACT

Artificial Intelligence (AI) has become an important tool in improving the performance of online advertising by enabling better targeting, personalization, and campaign optimization. AI technologies such as machine learning, predictive analytics, and automated content generation help businesses deliver advertisements to the right audience at the right time, thereby increasing customer engagement and conversion rates. The growing use of AI in digital marketing has

transformed traditional advertising methods into data-driven and performance-oriented strategies.

This study focuses on analyzing the impact of Artificial Intelligence on online advertising performance in Karungal Town. The research examines how local businesses use AI-based advertising tools through platforms such as social media and search engine advertising. It also studies customer response towards personalized and AI-driven advertisements. The study is based on primary data collected from consumers and supported by secondary data from research journals and industry reports. The findings are expected to show that AI improves advertisement reach, customer engagement, and sales conversion while reducing advertising cost and manual effort. However, challenges such as privacy concerns, lack of technical knowledge, and over-dependence on automated systems may affect its effectiveness. The study concludes that AI has a significant positive impact on online advertising performance and can help small and medium businesses in semi-urban areas like Karungal improve their marketing outcomes when used effectively and ethically.

Key words: Artificial Intelligence, Online Advertising, Social Media.

Introduction

Artificial Intelligence (AI) has emerged as a powerful technology that is transforming various sectors, including marketing and advertising. In recent years, online advertising has shifted from traditional digital marketing methods to data-driven and automated systems powered by AI. AI technologies such as machine learning, predictive analytics, natural language processing, and automated content generation help businesses analyze customer behaviour, predict preferences, and deliver personalized advertisements. As a result, online advertising performance has improved in terms of customer engagement, click-through rates, and conversion rates. Online advertising plays a vital role in modern business by helping companies reach their target audience quickly and cost-effectively. With the increasing use of smartphones, social media platforms, and e-commerce websites, businesses are focusing more on digital advertising strategies. AI helps advertisers optimize ad campaigns, select the right audience, and deliver advertisements at the right time. This improves marketing efficiency and reduces advertising costs. In small towns like Karungal, businesses are gradually adopting online advertising through platforms such as Facebook, Instagram, YouTube, and Google Ads. Many of these platforms already use AI-based algorithms to show relevant advertisements to users. Understanding the impact of AI on online advertising performance is important for local businesses to improve their marketing strategies and compete in the digital market. Therefore, this study aims to analyze the impact of Artificial Intelligence on online advertising performance, focusing on customer response, advertisement effectiveness, and business benefits in Karungal Town.

Review of Literature

Artificial Intelligence (AI) has become an essential tool in digital marketing and online advertising. Many researchers have studied how AI improves advertising performance through personalization, automation, and data analysis.

Mahakal (2025) studied the impact of Artificial Intelligence in digital marketing and found that AI helps businesses understand consumer behaviour, improve advertisement targeting, and increase conversion rates. The study concluded that AI-driven marketing strategies improve customer engagement and overall campaign performance.

Virmani (2025) conducted a comprehensive review on the role of AI in digital marketing. The study explained that AI supports personalized advertising, predictive analytics, and automated ad management. The research highlighted that AI reduces manual work and improves advertising efficiency.

Gujarati, Darne, and Borse (2025) examined the role of AI in online advertising. Their study found that AI improves advertisement reach, targeting accuracy, and customer engagement. However, the study also pointed out challenges such as privacy issues and technical complexity in AI adoption.

Singh (2025) studied the role of AI in improving campaign effectiveness across multiple digital platforms. The research showed that AI improves targeting accuracy, advertisement engagement, and return on investment (ROI) for businesses.

Sánchez-Jiménez et al. (2025) found that AI-based personalization strategies can significantly improve advertising performance by increasing campaign effectiveness and customer response rates.

Meguellati et al. (2025) studied AI-generated advertisements and found that AI-created ad content can perform equally or sometimes better than human-created advertisements in persuasion and customer preference.

Youngman et al. (2019) studied algorithmic copywriting and found that AI-generated ad copies improved click-through performance compared to traditional advertisement content.

Overall, previous studies show that Artificial Intelligence has a positive impact on online advertising performance by improving targeting, personalization, and campaign optimization. At the same time, issues such as data privacy, ethical concerns, and over-dependence on automation need to be addressed for effective AI implementation.

Objectives of the Study

- To study the role of AI in online advertising performance
- To analyze the effectiveness of AI-based ads on customer engagement
- To identify benefits of AI advertising for local businesses in Karungal
- To study customer response towards AI personalized advertisements

Data Collection

Data collection is an important step in research, as it helps in gathering information required to achieve the study objectives. For this study on the **Impact of Artificial Intelligence on Online Advertising Performance in Karungal Town**, both primary and secondary data were used.

Primary Data Collection

Primary data refers to the data collected directly from respondents for the first time.

Method Used:

- Structured Questionnaire

Respondents:

- Consumers who use online platforms like social media, shopping websites, and online services

Sample Size:

- 75 Respondents

Sampling Technique:

- Convenience Sampling

Mode of Data Collection:

- Direct Survey
- Google Forms / Offline Questionnaire

Primary data was collected to understand customer awareness, satisfaction, and opinion towards AI-based online advertising.

Secondary Data Collection

Secondary data refers to data collected from already available sources.

Sources Used:

- Research Journals
- Books related to Digital Marketing and AI
- Online Research Articles
- Websites and Industry Reports

Secondary data was used to understand theoretical concepts, previous research findings, and global trends in AI advertising.

Period of Study

The data for this study was collected during the recent study period.

Area of Study

The study was conducted among consumers in **Karungal Town**.

Tools Used for Data Analysis

- Percentage Analysis
- Garrett Ranking Method
- Simple Statistical Tables

Analysis and Interpretation of Data

(Based on 75 Respondents)

Awareness of AI-Based Online Advertising

S.No	Awareness Level	No. of. Respondents	Percentage
1	Aware	48	64
2	Not Aware	27	36
	Total	75	100

Interpretation

The table shows that 64% of respondents are aware of AI-based online advertising. This indicates that AI advertising is becoming common among consumers. However, 36% are still unaware, showing the need for awareness programs.

Platform Where AI Ads Are Mostly Seen

S.No	Platform	No. of Respondents	Percentage
1	Instagram	22	29%
2	YouTube	18	24%
3	Facebook	15	20%
4	Google Ads	12	16%
5	Others	8	11%
	Total	75	100%

Source: Primary Data.

Interpretation

Most respondents (29%) see AI-based ads on Instagram. This shows social media platforms are major channels for AI-driven advertising.

Garrett Ranking

Benefits of Advertising

S.No	Benefit	Total Score	Rank
1	Relevant Ads	2000	1
2	Better Product Suggestions	1320	2
3	Time Saving	954	3
4	Easy Purchase Decision	480	5

Source: Primary Data.

Interpretation

The Garrett ranking analysis shows that **Relevant Ads** is the most important benefit of AI advertising among respondents. The second important factor is **Better Product Suggestions**, followed by **Time Saving**. The least ranked factor is **Easy Purchase Decision**. This indicates that customers mainly value AI advertising for showing relevant advertisements and providing better product recommendations.

Findings:

Majority of respondents are aware of AI-based online advertising and are exposed to AI ads through social media and search platforms.

Most respondents reported that AI-based advertising improves advertisement relevance and shows products based on their interests.

The study found that a large percentage of respondents are influenced by AI-based advertisements while making purchase decisions.

Respondents showed a positive level of satisfaction towards AI personalized advertisements.

Social media platforms like Instagram, YouTube, and Facebook are the major sources where respondents notice AI-based advertisements.

Garrett ranking analysis revealed that **Relevant Ads** is the most important benefit of AI advertising, followed by **Better Product Suggestions** and **Time Saving**.

The least important benefit identified by respondents is **Easy Purchase Decision**, indicating customers still consider other factors like price, brand, and reviews.

AI advertising helps businesses reduce marketing effort and improves customer reach and engagement.

Suggestions:

Adopt AI Advertising Tools:

Local businesses should adopt AI-based advertising tools available in platforms like Google Ads, Facebook Ads, and Instagram Ads to improve targeting and reach.

Focus on Personalized Advertising:

Businesses should use AI to deliver personalized and relevant advertisements to attract more customers and improve engagement.

Combine AI with Human Creativity:

Companies should not depend fully on AI. Human creativity should be combined with AI tools to create effective and attractive advertisements.

Create Awareness About AI Advertising:

Awareness programs and training should be conducted for small business owners to help them understand AI advertising benefits and usage.

Ensure Data Privacy and Security:

Businesses should protect customer data and follow privacy rules to build customer trust in AI-based advertising systems.

Conclusion:

Artificial Intelligence has become an important technology in improving online advertising performance. The study shows that AI helps businesses deliver personalized and relevant advertisements, improve customer engagement, and increase advertising effectiveness. AI-based tools help companies reach the right audience at the right time and reduce marketing costs and manual effort.

From the analysis of 75 respondents, it is clear that most customers respond positively to AI-based advertisements. Garrett ranking results show that showing relevant advertisements is the most important benefit of AI advertising, followed by better product suggestions and time saving. This proves that AI plays a major role in improving customer experience and advertisement performance.

However, some challenges such as data privacy concerns, lack of awareness, and over-dependence on technology were also identified. Therefore, businesses should use AI responsibly and combine it with human creativity for better results.

Overall, Artificial Intelligence has a significant positive impact on online advertising performance. Even in semi-urban areas like Karungal Town, AI-based advertising can help businesses improve customer reach, engagement, and sales performance when used effectively and ethically.

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Study on Worklife Balance of Women Employees in Killyoor Taluk

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Abstract

Rapid urbanization and modernization in India have significantly influenced the career development of women, particularly through increased access to education and paid employment. While these changes have expanded opportunities and aspirations, women continue to face considerable challenges in balancing professional responsibilities with family life. Gender-based expectations, organizational pressures, and limited work-life balance options often compel women to compromise either career growth or family goals. Many women experience stress due to long working hours, role overload, and caregiving responsibilities, which negatively affect their well-being and career advancement. This study focuses on examining the work-life balance of women employees, highlighting the impact of work-family conflict on their professional satisfaction and overall quality of life.

INTRODUCTION

The Indian culture with reference to the career development of girls is undergoing speedy changes because of the multiplied pace of urbanization and modernization. Indian women's happiness is all or any categories that have entered into paid occupations. At this time, Indian women's exposure to academic opportunities is well over it absolutely was some decades ago, particularly within the urban setting. This has opened new vistas, increased awareness, and raised aspirations of private growth. This, with economic pressure, has been instrumental in influencing women's call to enter the workforce. People's perception of work-life balance varies by gender. The most challenging part for every woman is how to balance their family life and work life. The unbalanced state between life and work might be one of the foremost obstacles to women's career development. Some women deliberately work fewer hours to attain a balance between career development and family life. However, many women compromise family goals as a technique to cut back work-family conflict, which frequently leads to fewer numbers of kids than expected within the family. Some of the surveys disclosed that people make sure their concessions in family life satisfy their structure commitment.

Women account for an oversized portion of employees in the cordial reception industry too. Due to organizational pressure, women and men find themselves spending a lot of time in professional roles such as overworking hours that are unstructured social hours. Many studies show that women who are under a lot of pressure at work find it difficult to take care of their children and dependents. The main source of satisfaction is the balance between work and home. Until According to Gary Geller- "Work is a rubber ball, if you drop it, it will bounce back. The other four balls- that is our family, friends, health, and integrity-are made of glass. If you drop one of these, it will be irrevocably scuffed, nicked, even shattered."

Scope of the Study

The study focuses on examining the work- life balance of women employees within a specific context or industry. It seeks to understand the challenges, strategies, and outcomes related to balancing work responsibilities with personal and family life commitments. Through qualitative and quantitative analysis, the study aims to provide insights into the factors influencing women's

work-life balance, organizational policies and practices supporting it, and potential implications for individual well-being and organizational performance.

Objectives of the Study

To find out the satisfaction in both personal and professional life

To know whether they spend their free time with their families or not.

To determine the factors affecting work life balance among working women.

Statement of the Problem

The challenge lies in addressing the persistent struggle women face in balancing their professional and personal lives. Despite progress toward gender equality, the unequal distribution of household responsibilities, limited access to flexible work arrangements, and prevailing societal expectations continue to hinder women's ability to achieve a healthy work-life balance. This imbalance negatively impacts women's job satisfaction, career advancement, and overall well-being, while also reinforcing existing gender disparities in the workplace. Therefore, identifying and addressing the specific barriers that affect women's work-life balance is essential for promoting gender equality and supporting women's long-term professional success.

Research Methodology

The study is descriptive and analytical in nature. Convenience sampling is used. It includes the overall research design, the sampling procedure, data collection methods and analysis procedure

Sample Design

The method used to collect the sample from the total population is convenient sampling.

Sample Size

Kanyakumari district was selected for the study 100 samples have been taken for the present study.

Data Collection

Data collection is the process of collecting and analyzing information on relevant variables. The data were collected in two ways.

Primary Data

The primary data was collected with the help of the questionnaire.

Secondary Data

Secondary data was obtained from journals, websites, books and magazines.

Limitations of the Study

- The analysis of the present study has been carried out based on the information has collected directly from the respondents.
- The study was conducted in a limited time period.
- Due to time constrict only 100 numbers of respondents were considered.
- The findings and analysis were completely based on the opinion of the respondents, journals and website

REVIEW OF LITERATURE

Dr. Seema Verma and Rahul Singh Gautam (2022) define, study to measure "The Effect of Education on Working Women for Work-Life Balance" study suggest a positive and direct impact of education which leads to better work-life balance. Work-life balance is assisted by education which offers knowledge related to flexible work schedules, family events, and other activities that enable one to pursue more balanced lives easily. The educational balance conditions have prompt comportment on her development and Work-life balance.

Dr. Somlata Sharma and Dr. Versha Vahini (2022) defines, tried to asses "Work Life Balance: How Stressful is it for Indian Women" Study shows that changes in technology, increased work pressure coupled with constant deadlines, the co-existence of a virtual workplace, and changing demographic profiles (including gender, experience), an increase in disposable income inflation, and an improvement in living standards have encouraged the importance of the provision of a

work-life balance in the industrial sector.

Jimenez, P., & Alcover, C. M. (2021) defines, highlighted the significant impact of COVID-19 on work-life balance, as employees struggle to balance their work and personal responsibilities while working from home. The authors suggest that interventions, such as flexible work arrangements, training and support for managers, and employee well-being programs, can help to mitigate the negative impacts of COVID-19 on work-life balance and promote employee well-being.

D.R. Dinakar (2021) defines, has conducted research on “Effectiveness of Work Life Balance (WLB) on Job Satisfaction of Employees Balance work fulfillment of representatives in new private area banks. The results show that WLB banks strategies, workload, hierarchical help and monetary help are determinants of work life balance of bank employees in private banks.

Vani Haridasan and Kavitha Muthukumaran (2021) defines, aimed to identify the various factors that affect women's work-life balance during the lockdown caused by the COVID-19 pandemic. The study found that job performance and personal wellbeing are the main determinants of women employees' work-life balance during work from home. The findings of this study suggest that organizations should realign their policies, benefit programs, work distribution, and make job changes inside the organization for an employee- friendly remote working experience, thereby improving the workforce's efficiency and productivity.

Analysis and Interpretation

Work and family commitments wise distribution

Based on its Factor	Strongly Agree	Agree	Neutral	Disagree	Strongly disagree	Total	Rank
Working from home		28(112)	35(105)	10(20)	15(15)	114	V
Technology like cell phones/	8(40)	10(40)	15(45)	38(56)	39(39)	181	IV
Support from colleagues at Work	14(70)	12(48)	18(54)	38(76)	18(18)	266	III
Support from family Members	16(80)	28(112)	19(57)	28(56)	9(9)	314	I

Others	9(45)	17(68)	36(108)	19(38)	19(19)	278	II
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Source: primary data

The above table shows that 1st ranging of respondents work and family commitments are support from home, 2d ranging of work and family commitments are others, 3d ranging of the respondents work and family commitments are support from colleagues at work , 4th ranging of respondents work and family commitments technology like cell phones/ laptop and 5th ranging of respondents working from home.

Hinder work and family wise distribution

Hinder work and family Commitments	1	2	3	4	5	Total	Rank
Long working Hours	20(100)	22(88)	10(30)	32(64)	16(16)	298	II
Compulsory Overtime	12(50)	19(76)	25(75)	12(24)	17(17)	242	IV
Shift work	13(65)	14(56)	16(48)	29(58)	28(28)	255	III
Meetings/ training after office hours	10(50)	12(48)	19(57)	20(40)	39(39)	234	V
Others	22(110)	26(104)	9(27)	19(38)	24(24)	303	I

Source: primary data

The above table shows that 1st ranging of the respondents hinder work and family commitments are others, 2d ranging of the respondents hinder work and family commitments are long working hours ,3d ranging of the respondents hinder work and family commitments are shift work, 4th ranging of the respondents are compulsory over time and 5th ranging of the respondents are meetings/ training after office

FINDINGS, SUGGESTION & CONCLUSION

- ❖ It found that majority of the respondents tired and depressed due to their full time work.
- ❖ It is found that majority of the respondents are managing their stress through hearing music.
- ❖ Majority of the respondents are suffered hypertension due to the workload.
- ❖ It is found that majority of the respondents balancing their family commitments are from their family members.
- ❖ It is found that the overtime work is disturbance for managing their children

SUGGESTIONS

- It is suggested that must create a timeline of their activities by conducting specific computer programs for entertainment to reduce the stress.
- It I suggested that frequent outing must be carrying by the responsible authritied to enjoy their sincere work.
- It is suggested that must give the weekend holidays for the workers to enjoy themselves with their children.

CONCLUSION:

Work–life balance refers to an individual’s ability to effectively manage work responsibilities along with personal and family life while minimizing role conflict. It varies from person to person and changes with different stages of life. Failure to maintain a proper balance often leads to exhaustion, lack of time for personal commitments, reduced quality of life, and poor work performance. With changes in workforce structure, especially the increase in working women, dual-earner households, and nuclear families, the need for flexible work arrangements has become more important. Work–life balance is not a one-size-fits-all concept but focuses on providing flexibility in working hours through practices such as flexitime, telecommuting, and compressed workweeks, which help improve employee productivity. However, challenges remain, including work–life conflict for employees and absenteeism concerns for employers.

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k – Critical graph for Cartesion product of some graph

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Abstract:

Critical graphs are the building blocks of coloring theory. In Graph theory, a k -critical graph is a graph that is “just barely” k -chromatic. It is a minimal structure that forces the need for k colors; if you remove even a single part of it, the complexity collapses and fewer colors are required.

Keywords:

Graph, Line graph, Cross Product, Cartesian product, Vertex coloring, K - Critical graph.

Introduction:

While vertex coloring dates back to the 1850s, the formal study of criticality was pioneered by Gabriel Andrew Dirac in the early 1950s. Dirac wanted to identify the core subgraphs that force a graph to have certain chromatic number. He defined a k -critical graph as a graph G where $\chi(G) = k$, but every proper subgraph has a smaller chromatic number. He proved that every k -chromatic graph contains a k -critical subgraph. This turned k -critical graphs into the “atoms” of graph theory.

Definition 1.1:

A **Graph** consists of a set $V(G)$ of elements called vertices and a set $E(G)$ of elements called edges together with a relation of incidence which associates each edge with a pair of vertices called its ends.

Definition 1.2:

A **Line graph** $L(G)$ of a given graph G is a graph where each vertex of $L(G)$ represents an edge of G , and two vertices in $L(G)$ are connected by an edge if and only if their corresponding edges in G share a common endpoint (i.e., are incident).

Definition 1.3:

The **Cartesian product** of two graphs G_1 and G_2 is a graph denoted by $G = G_1 \square G_2$ where two vertices (u_1, v_1) and (u_2, v_2) in G are adjacent if and only if either $u_1 = u_2$ and v_1 is adjacent to v_2 or $v_1 = v_2$ and u_1 is adjacent to u_2 .

Definition 1.4:

A *Cross product* is commonly refers to the Cartesian product of two graphs. It is a way of combining two graphs, G and H to create a much larger, more complex graph.

For two graphs $G = (V(G), E(G))$ and $H = (V(H), E(H))$, the Cartesian product $G \square H$ is a graph such that:

Vertices: The vertex set is the Cartesian product of the individual vertex $V(G \square H) = V_1 \times V_2$. This means every vertex in the new graph is an ordered pair (u, v) , where $u \in G$ and $v \in H$.

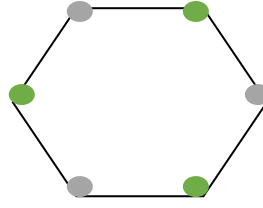
Edges: Two vertices (u, v) and (u', v') are adjacent in $G \square H$ if and only if:

- $u = u'$ and v is adjacent to v' in H , or
- $v = v'$ and u is adjacent to u' in G

Definition 1.5:

Vertex coloring: An assignment of k colors, usually denoted by $1, 2, \dots, k$ to the vertices of a graph so that no two adjacent vertices have the same color is called k -vertex coloring or k -coloring of the graph G .

Example: 1



2- coloring

Definition 1.6:

k – Critical graph: A graph G is called k – critical iff G has chromatic number $\chi(G) = k$ and every proper subgraph $H \subset G$ has chromatic number $\chi(H) < \chi(G)$.

2. MAIN RESULTS

Theorem 2.2 :

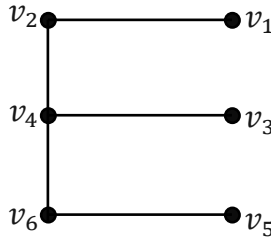
Let $L(G_1)$ and $L(G_2)$ be the line graphs of $P_3 \odot K_1$ and P_4 . G is a cartesian product of $L(G_1)$ and $L(G_2)$. Then G is a k - critical graph.

Proof:

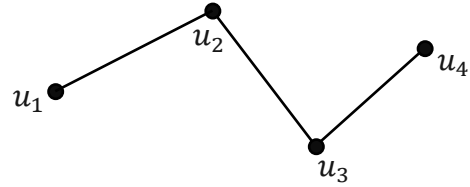
Let $P_3 \odot K_1$ be a given graph $|V(P_3 \odot K_1)| = 6$ and $|E(P_3 \odot K_1)| = 5$

Let P_4 be a given graph $|V(P_4)| = 4$ and $|E(P_4)| = 3$

$P_3 \odot K_1$ and P_4 graphs given below,



G_1

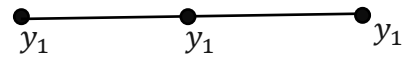
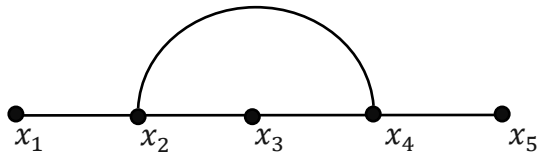


G_2

Let $L(G_1)$ be the line graph of $P_3 \odot K_1$ such that $|V(L(G_1))| = 5$ and $|E(L(G_1))| = 5$

Let $L(G_2)$ be the line graph of P_4 such that $|V(L(G_2))| = 3$ and $|E(L(G_2))| = 2$

$L(G_1)$ and $L(G_2)$ graphs given below,



$L(G_2)$

The Cartesian product G is defined by $G = L(G_1) \square L(G_2)$

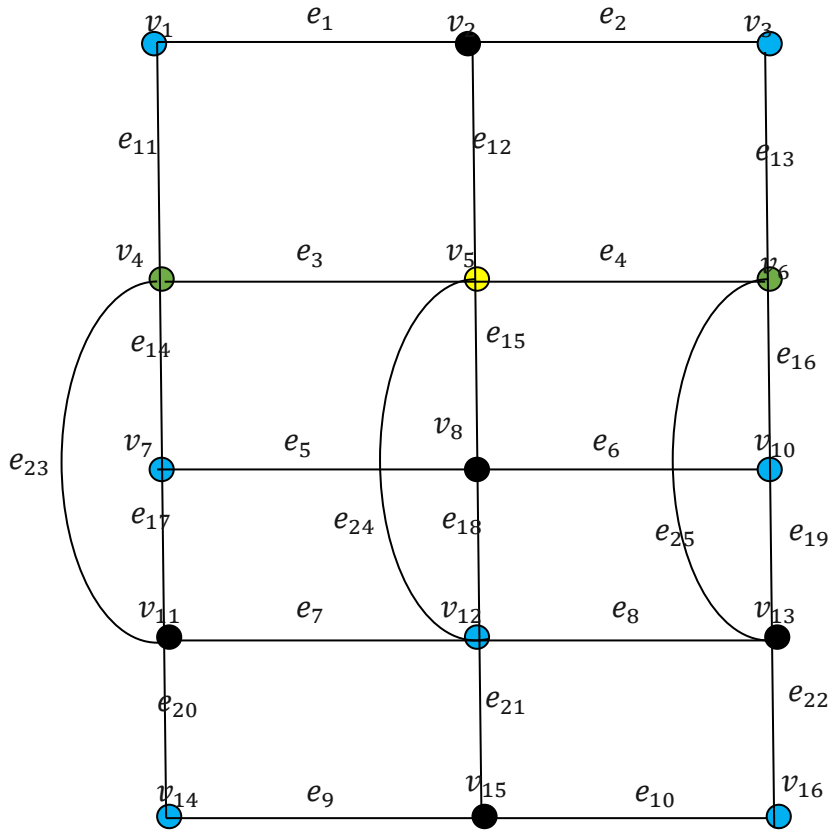
The vertex set of G is defined by,

$$V(G) = \{v_i \mid 1 \leq i \leq 15\}$$

The edge set of G is defined by,

$$E(G) = \begin{cases} e_i; 1 \leq i \leq 10 \\ e'_i; 11 \leq i \leq 22 \\ e''_i; 23 \leq i \leq 25 \end{cases}$$

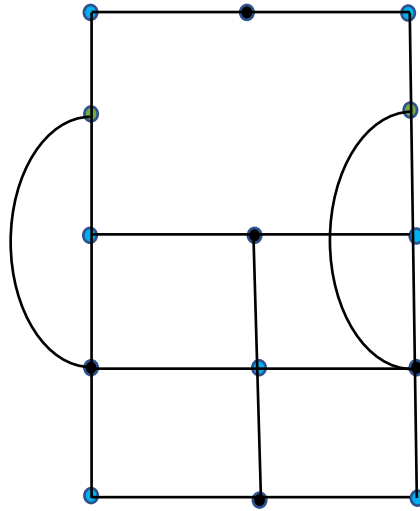
The cartesian product of $L(G_1)$ and $L(G_2)$ is given by,



Here the chromatic number of the graph G is 4.

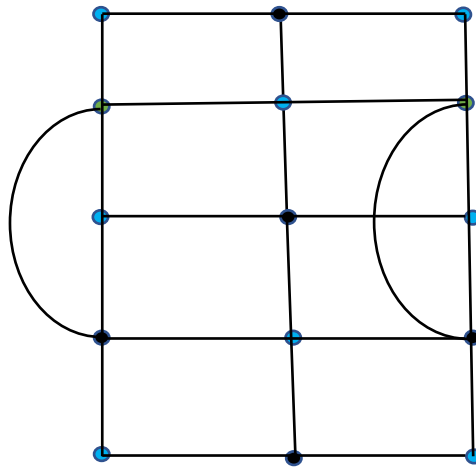
i.e) $\chi(G) = 4$

Let H_1 be the subgraph of G which is obtained by removing the vertex v_5 .



Here the chromatic number of the graph H_1 is 3. i.e) $\chi(H_1) = 3$

Let H_2 be the subgraph of G which is obtained by removing the edge e_{24} .



Here the chromatic number of the graph H_2 is 3.

i.e) $\chi(H_2) = 3$.

Therefore, $\chi(G) > \chi(H_1)$ and $\chi(G) > \chi(H_2)$.

Hence it is a critical graph.

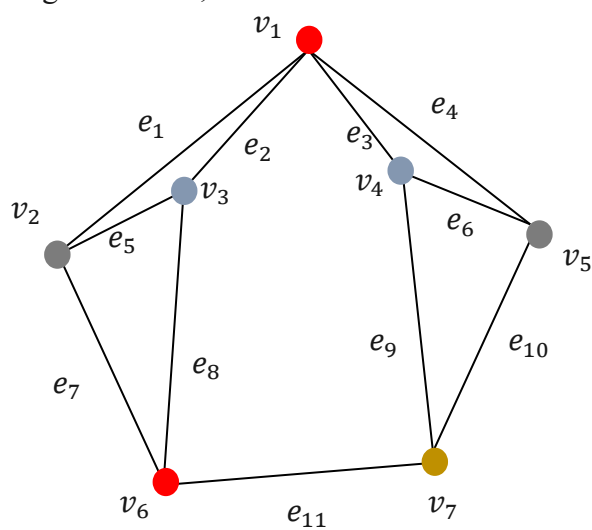
Theorem 2.3 :

Let G be a k – chromatic graph. Then G is k – critical if and only if for every edge $e \in E(G)$, $\chi(G - e) < \chi(G)$, and for every vertex $v \in V(G)$, $\chi(G - v) < \chi(G)$.

Proof:

Given that G is a k - chromatic graph.

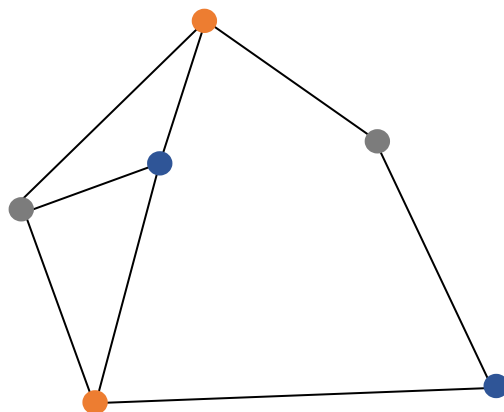
The graph G is given below,



Here the chromatic number of the graph G is 4.

i.e) $\chi(G) = 4$

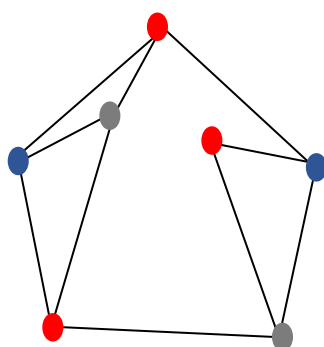
Let H_1 be the subgraph of G which is obtained by removing the vertex v_4 .



Here the chromatic number of the graph H_1 is 3.

i.e) $\chi(H_1) = 3$

Let H_2 be the subgraph of G which is obtained by removing the edge e_3 .



Here the chromatic number of the graph H_2 is 3.

i.e) $\chi(H_2) = 3$.

Therefore, $\chi(G) > \chi(H_1)$ and $\chi(G) > \chi(H_2)$.

Hence it is a critical graph.

Conclusion:

In this paper, we studied the concept of k-critical graphs in the context of the Cartesian product of graphs. We analyzed how the criticality of individual graphs influences the chromatic behavior of their Cartesian product and identified conditions under which the product graph remains k-critical. Through specific examples and results, we showed that the Cartesian product does not always preserve criticality, but under certain structural constraints, k-criticality can be retained or characterized precisely.

These findings contribute to a deeper understanding of graph coloring in product graphs and highlight the intricate relationship between graph structure and chromatic properties. The results obtained here can serve as a foundation for further research on other graph products and more generalized coloring constraints, with potential applications in network design, scheduling problems, and combinatorial optimization.

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Vertex Coloring of Cross Product of a Graph and its Line Graphs

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Abstract:

Vertex Coloring is one of the most famous problems in a graph theory. At its core, it is a labelling problems: you assign a “color” (usually a positive integer) to each vertex so that no two adjacent vertices share the same color. The goal of vertex coloring is typically to find the Chromatic Number, denoted as $\chi(G)$. The minimum number of colors required to color the vertices of a graph is known as the Chromatic number. This is the smallest number of colors needed to color a graph G properly. Proper Coloring: A mapping $c : V(G) \rightarrow \{1, 2, \dots, k\}$ such that if u and v are adjacent, $c(u) \neq c(v)$. K -Colorable: A graph that can be colored using k colors.

Keywords:

Graph, Graph coloring, Vertex coloring, Chromatic number, Chromatic Polynomial, color- class, Line graph, Cross product.

1.Introduction

Vertex coloring is one of the most famous and visually intuitive concepts in graph theory. In 1852: The Spark (Francis Guthrie) - While coloring a map of British countries, Guthrie conjectured that only four colors were needed so that no two adjacent regions shared a color. In 1852: The Messenger (Augustus De Morgan) - Guthrie’s teacher brought the problem to the mathematical elite, formalizing the question of “map coloring” into a mathematical pursuit. In 1879: The Technique (Alfred Kempe) - He published the first proof. Crucially, he invented Kempe Chains, a method of swapping colors that remains a primary tool in graph theory today. In 1890: The Refinement (Percy Heawood) - He exposed the flaw in the 4 – colors proof but successfully proved the Five Color Theorem, shifting the focus to more rigorous bounds. In 1912: The Algebra (George Birkhoof) - He introduced the Chromatic Polynomial, which turned coloring into an algebraic equation rather than just a visual map problem. In 1976: The Modern era (Appel&Haken) – They provided the first computer - aided proof of the Four Color Theorem, marking the transition of graph coloring into computational science.

Definition 1.1:

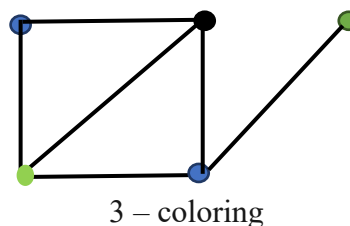
A graph G consists of a set $V(G)$ of elements called vertices and a set $E(G)$ of elements called edges together with a relation of incidence which associates each edges with a pair of vertices called its ends.

Definition 1.2:

Graph coloring: In its simplest form, it is a way of coloring the vertices of a graph such that no two adjacent vertices share the same color, called a vertex coloring. Similarly, an edge coloring assigns a color to each edge so that no two incident edges share the same color, and a face coloring of a planar graph assigns a color to each face or region so that no two faces that share a boundary share the same color.

Definition 1.3:

Vertex coloring: An assignment of k colors, usually denoted by $1, 2, \dots, k$ to the vertices of a graph g so that no two adjacent vertices have the same color is called k – vertex coloring or k – coloring of the graph G

Example: 1

Definition 1.4:

Chromatic number: The least number of colors needed to color the graph is called its chromatic number. It is denoted by the symbol $\chi(G)$, where G is a graph. For example the chromatic number of a K_n of n vertices (a graph with an edge between every two vertices i.e., a Complete graph n vertices), is $\chi(K_n) = n$. A graph that can be assigned a (proper) k – coloring is k – colorable, and its chromatic number is exactly k . In the above example 1 the chromatic number of the minimum number of coloring $\chi(G) = 3$

Definition 1.5:

The chromatic polynomial counts the number of ways a graph can be colored using no more than a given number of colors

Definition 1.6:

A Color class is a set of vertices of a graph which are having the same color when a coloring is done to a graph.

Definition 1.7:

A Line graph $L(G)$ of a given graph where each vertex of $L(G)$ represents an edge of G , and two vertices in $L(G)$ are connected by an edge if and only if their corresponding edges in G share a common endpoint (i.e., are incident)

Definition 1.8:

A Cross product commonly refers to the Cartesian product of two graphs. It is a way of combining two graphs, G and H to create a much larger, more complex graph.

For two graphs $G = (V(G), E(G))$ and $H = (V(H), E(H))$, the Cartesian product $G \square H$ is a graph such that:

Vertices: The vertex set is the Cartesian product of the individual vertex set $V(G \square H) = V_1 \times V_2$. This means every vertex in the new graph is an ordered pair (u, v) , where $u \in G$ and $v \in H$

Edges: Two vertices (u, v) and (u', v') are adjacent in $G \square H$ if and only if:

- $u = u'$ and v is adjacent to v' in H , OR
- $v = v'$ and u is adjacent to u' in G

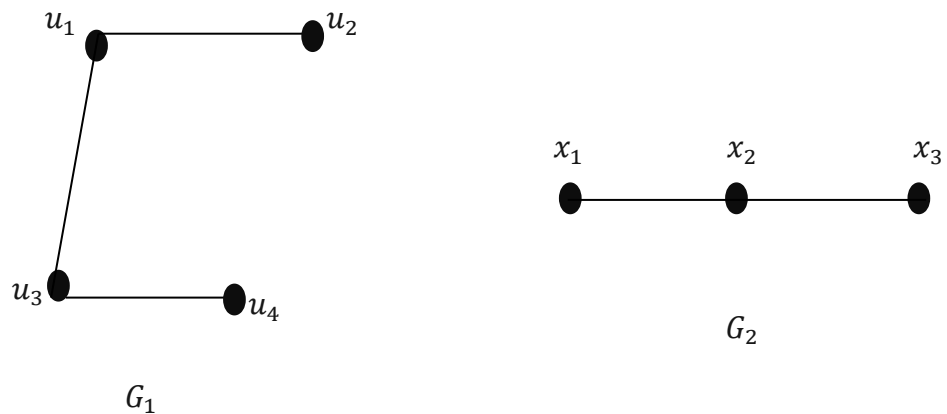
2. MAIN RESULTS**Theorem 2.1:**

Let G_1 be a graph and G_2 be a line graph of G_1 , then $G = G_1 \times G_2$ is 4 – coloring graph .
Proof:

Let G_1 be the graph with $|V(G_1)| = 4$ and $|E(G_1)| = 2$

Let G_2 be the line graph of G_1 with $|V(G_2)| = 3$ and $|E(G_2)| = 2$

The graph G_1 and G_2 are



$$G_1 \times G_2 = \{u_1, u_2, u_3, u_4\} \times \{x_1, x_2, x_3\}$$

$$= (u_1, x_2), (u_1, x_3), (u_2, x_1), (u_2, x_2), (u_2, x_3), (u_3, x_1), (u_3, x_2), (u_3, x_3), (u_4, x_1), (u_4, x_2), (u_4, x_3)\}$$

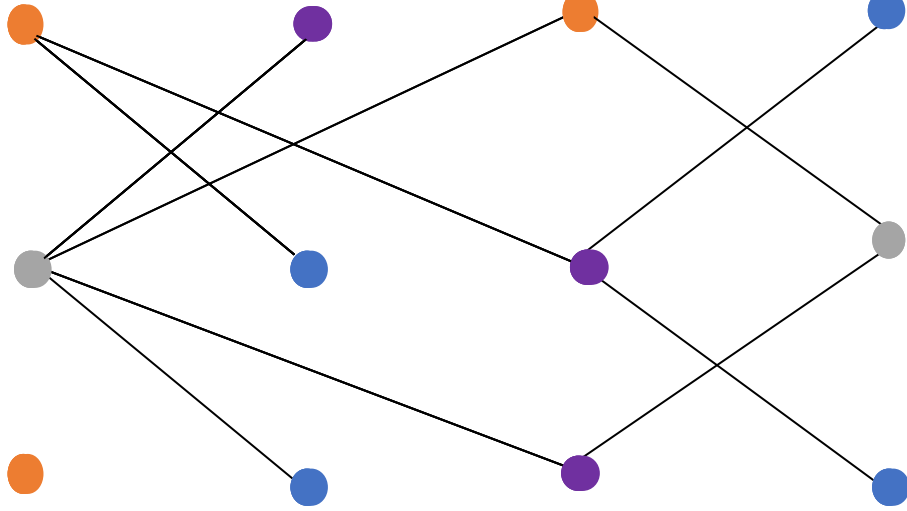
The vertex of the new graph is defined as,

$$f(u_i, x_j) = \{v_{ij} / i = 1; j = 1, 2, 3\}$$

$$G = G_1 \times G_2, \text{ where } G_1 \text{ is a graph and } G_2 \text{ is a line graph}$$

$$f(u_i, x_j) = \{v_{ij} / i = 2; j = 1, 2, 3\}$$

$$f(u_i, x_j) = \{v_{ij} / i = 3; j = 1, 2, 3\}$$



$$|V(G)| = 12 \text{ and } |E(G)| = 12$$

Let's define the function of coloring as

$$f(B) = \{v_{41}, v_{22}, v_{23}, v_{43}\}$$

$$f(Y) = \{v_{21}, v_{32}, v_{33}\}$$

$$f(R) = \{v_{11}, v_{13}, v_{31}\}$$

$$f(G) = \{v_{42}, v_{12}\}$$

The vertex v_{11} is adjacent to v_{32}, v_{22} ; v_{21} is adjacent to v_{12} ; v_{31} is adjacent to v_{12}, v_{42} ; v_{41} is adjacent to v_{32} ; v_{12} is adjacent to $v_{21}, v_{31}, v_{23}, v_{33}$; v_{22} is adjacent to v_{11}, v_{13} ; v_{32} is adjacent to $v_{11}, v_{41}, v_{13}, v_{43}$; v_{42} is adjacent to v_{31}, v_{33} ; v_{13} is adjacent to v_{22}, v_{32} ; v_{23} is adjacent to v_{12} ; v_{33} is adjacent to v_{12}, v_{42} ; v_{43} is adjacent to v_{32} .

$$|\chi(G)| = 4$$

∴ The graph G is 4 – coloring graph.

Hence proved

Theorem 2.2:

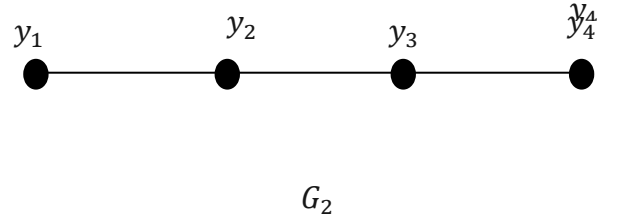
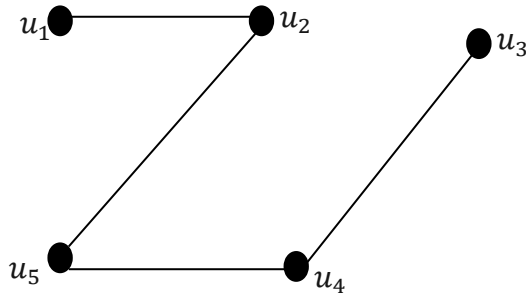
Let G_1 be a graph and G_2 be a line graph of G_1 , then $G = G_1 \times G_2$ is 4 – coloring graph and find the Chromatic number of graph $\chi(G_1 \times G_2) = \min \{\chi(G_1)\chi(G_2)\}$.

Proof:

Let G_1 be the graph with $|V(G_1)| = 5$ and $|E(G_1)| = 4$

Let G_2 be the line graph of G_1 with $|V(G_2)| = 4$ and $|E(G_2)| = 3$

The graph G_1 and G_2 are



$$G_1 \times G_2 = \{u_1, u_2, u_3, u_4\} \times \{y_1, y_2, y_3, y_4\}$$

$$= (u_1, y_2), (u_1, y_3), (u_1, y_1), (u_1, y_4), (u_2, y_1), (u_2, y_2), (u_2, y_3), (u_2, y_4), (u_3, y_1),$$

$$(u_3, y_2), (u_3, y_3), (u_3, y_4), (u_4, y_1), (u_4, y_2), (u_4, y_3), (u_4, y_4)\}$$

The vertex of the new graph is defined as,

$$f(u_i, x_j) = \{v_{ij} \mid i = 1; j = 1, 2, 3, 4\}$$

$$f(u_i, x_j) = \{v_{ij} \mid i = 2; j = 1, 2, 3, 4\}$$

$$f(u_i, x_j) = \{v_{ij} \mid i = 3; j = 1, 2, 3, 4\}$$

$$f(u_i, x_j) = \{v_{ij} \mid i = 4; j = 1, 2, 3, 4\}$$

$$f(u_i, x_j) = \{v_{ij} \mid i = 5; j = 1, 2, 3, 4\}$$

$G = G_1 \times G_2$, where G_1 is a graph and G_2 is a line graph
 $|V(G)| = 20$ and $|E(G)| = 24$

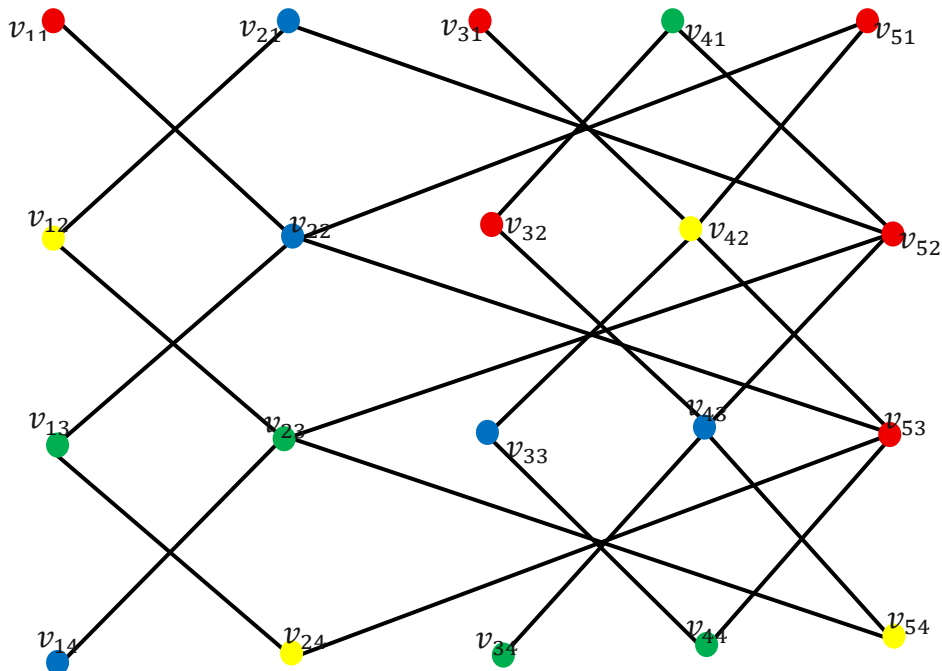
Let's define the function of coloring as

$$f(B) = \{v_{14}, v_{22}, v_{21}, v_{33}, v_{43}\}$$

$$f(Y) = \{v_{12}, v_{42}, v_{24}, v_{54}\}$$

$$f(R) = \{v_{11}, v_{31}, v_{51}, v_{32}, v_{52}, v_{53}\}$$

$$f(G) = \{v_{41}, v_{13}, v_{23}, v_{34}, v_{44}\}$$



The vertex v_{11} is adjacent to v_{22} ; v_{12} is adjacent to v_{21}, v_{23} ; v_{13} is adjacent to v_{22}, v_{24} ; v_{14} is adjacent to v_{23} ; v_{21} is adjacent to v_{12}, v_{52} ; v_{22} is adjacent to $v_{11}, v_{13}, v_{51}, v_{53}$; v_{23} is adjacent to $v_{12}, v_{14}, v_{52}, v_{54}$; v_{24} is adjacent to v_{13}, v_{53} ; v_{31} adjacent to v_{42} ; v_{32} is adjacent to v_{41}, v_{43} ; v_{33} is adjacent to v_{42}, v_{44} ; v_{34} is adjacent to v_{43} ; v_{41} is adjacent to v_{32}, v_{52} ; v_{42} is adjacent to $v_{31}, v_{51}, v_{33}, v_{53}$; v_{43} is adjacent to $v_{32}, v_{52}, v_{34}, v_{54}$; v_{44} is adjacent to v_{33}, v_{53} ; v_{51} is adjacent to v_{22}, v_{42} ; v_{52} is adjacent to $v_{41}, v_{21}, v_{23}, v_{43}$; v_{53} is adjacent to $v_{42}, v_{22}, v_{24}, v_{44}$; v_{54} is adjacent to v_{43}, v_{23} .

∴ The graph G is 4 – coloring graph

∴ The Chromatic number of coloring

$$|\chi(G)| = 4$$

Hence proved

Conclusion:

In this study, we have demonstrated that the vertex coloring of a cross product is remarkably stable. By analyzing the properties of k-critical subgraphs, we see that the chromatic number $\chi(G \times H)$ is sensitive only to the most restrictive factor. This implies that in network design and scheduling – where graph products are often used to model complex systems – we can optimize color resources by focusing solely on the most complex individual component of system.

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K-Nomial Coefficients

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Abstract

K-nominal coefficients are a generalization of binomial coefficients. While binomial coefficients appear in the expansion of expressions with two terms, k-nominal coefficients appear when the number of terms is these coefficients play an important role in algebra, combinatorics and probability theory.

1 INTRODUCTION

In algebra, the expansion of expressions raised to a positive integer power is an important topic. We are already familiar with the expansion of binomial expressions such as

$$(a + b)^n, (x + y)^n.$$

As mathematics develops further, we often encounter expressions with more than two terms. For example,

$$(x + y + z)^n, (x + y + z + t)^n, (x_1 + x_2 + \dots + x_k)^n.$$

For instance, the expansion of

$$(x + y + z + t)^4$$

contains terms such as

$$x^4, x^3y, x^2y^2, xyz^2, z^2t^2.$$

To overcome this difficulty, the concept of k-nomial coefficients is introduced. A k-nomial coefficient gives the coefficient of a general term of the form

$$x_1^{r_1} x_2^{r_2} \dots x_k^{r_k} \\ (x_1 + x_2 + \dots + x_k), r_1 + r_2 + \dots + r_k = n$$

Thus, k-nominal coefficients generalize binomial and trinomial coefficients and provide a systematic way to study expansions involving many variables. They play a key role in algebra, combinatorics, and probability theory, especially in problems related to counting, arrangements, and multinomial distributions.

3. DEFINITION

K-nominal coefficients are the coefficients that appear in the expansion of

$(x_1 + x_2 + \dots + x_k)^n$, where the sum consists of k terms and the sum of the exponents is equal to n. K-nominal coefficients are a generalization of binomial and trinomial coefficients.

4. FORMULA FOR k-NOMIAL COEFFICIENT

If $n_1 + n_2 + \dots + n_k = n$, then the k-nomial coefficient is:

$$C(n; n_1, n_2, \dots, n_k) = n! / (n_1! n_2! \dots n_k!)$$

Case i) k = 2

$$C(n; r, n - r) = C(n, r)$$

Case ii) k = 3

$$C(n; a, b, c) = n! / (a! b! c!)$$

5. SYMMETRY THEOREM

$$C(n; n_1, n_2, \dots, n_k) = C(n; n\sigma(1), n\sigma(2), \dots, n\sigma(k))$$

For any permutation σ

$$C(n; a, b, c) = C(n; b, a, c) = C(n; c, a, b)$$

6. SUMMATION THEOREM

$$\Sigma C(n; n_1, n_2, \dots, n_k) = k^n$$

Case i) $k = 2 \quad \Sigma C(n, r) = 2^n$

Case ii) $k = 3 \quad \Sigma C(n; a, b, c) = 3^n$

1. RECURRENCE RELATION

$$2. C(n; n_1, n_2, \dots, n_k) = \Sigma C(n-1; n_1, \dots, n_i-1, \dots, n_k)$$

Case i) $k = 2$

$$C(n, r) = C(n-1, r-1) + C(n-1, r)$$

Case ii) $k = 3$

$$C(n; a, b, c) = C(n-1; a-1, b, c) + C(n-1; a, b-1, c) + C(n-1; a, b, c-1)$$

8. PRODUCT FORM THEOREM

$$C(n; n_1, n_2, \dots, n_k) = C(n, n_1)C(n-n_1, n_2) \dots C(n-n_1-\dots-n_k-1, n_k)$$

9. MAIN RESULTS

1. MULTINOMIAL SYMMETRY

The k-nominal coefficient remains the same under any permutation of its indices.

Example: Changing the order of the terms does not change the coefficient.

2. MULTINOMIAL SUMMATION

The sum of all k-nominal coefficients for a fixed n is equal to k raised to the power n.

Example: For two terms, the sum equals 2 to the power n. For three terms, the sum equals 3 to the power n.

3. RECURRENCE RELATION

Each k-nomial coefficient can be expressed as the sum of k coefficients from the previous level, each with one index reduced by one.

Example: For two terms, each coefficient is the sum of two coefficients from the previous expansion. For three terms, each coefficient is the sum of three previous coefficients with one index decreased.

4. PRODUCT FORM

Each k-nomial coefficient can be expressed as a product of binomial coefficients, computed sequentially.

Example: For three terms, the coefficient can be written as a product of two sequential binomial coefficients.

5. EQUAL PARTITION CASE

If the total exponent is equally divided among all terms, the coefficient is the factorial of the total divided by the factorial of each part raised to the number of terms.

Example: For two terms equally divided, the coefficient is the factorial of the total divided by the square of the factorial of half the total.

6. APPLICATION IN PROBABILITY

In a repeated experiment with multiple possible outcomes per trial, the probability of obtaining a particular set of outcomes is proportional to the corresponding k-nomial coefficient multiplied by the probabilities of each outcome raised to the number of occurrences.

10. CONCLUSION

K-nomial coefficients extend the concept of binomial and trinomial coefficients and provide a powerful tool for studying algebraic expansions with multiple variables. They have significant applications in combinatorics and probability theory.

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Trends and Growth of Foreign Direct Investment (FDI)

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Abstract

Foreign Direct Investment (FDI) is an important source of capital inflow that contributes to the economic development of a country. It plays a significant role in promoting industrial growth, technology transfer, employment generation, and international trade. In recent years, globalization, policy liberalization, and digital transformation have influenced the pattern and flow of FDI across the world.

This study focuses on analysing the trends and growth of FDI, with special reference to India. Over the years, India has emerged as one of the major destinations for foreign investment due to its large market size, skilled workforce, and favourable government policies. The study examines sector-wise distribution, source countries, and the changing pattern of FDI inflows. It also highlights the contribution of FDI to economic growth, infrastructure development, and industrial expansion.

The study is based on secondary data collected from government reports, RBI publications, and international organizations. The findings show that FDI inflows have increased significantly in recent years, especially in services, technology, and manufacturing sectors. However, challenges such as global economic uncertainty and uneven sectoral distribution still exist.

The study concludes that FDI will continue to play a crucial role in India's economic growth and development if supported by stable policies and a favourable business environment. Keywords: Foreign Direct Investment (FDI) Globalization, Policy Liberalization, and Digital Transformation

Introduction

Foreign Direct Investment (FDI) refers to an investment made by a company or individual of one country into business operations located in another country, usually with the intention of

establishing long-term interest and control over the enterprise. It generally includes investments such as setting up new production units, mergers and acquisitions, joint ventures, or reinvestment of profits in foreign subsidiaries.

FDI plays a vital role in the global economy and is considered an important driver of economic globalization and development. It brings foreign capital, advanced technology, managerial skills, and international business practices to the host country. Countries compete to attract FDI because it supports industrial growth, improves infrastructure, and increases productivity.

FDI is especially important for developing countries like India because it helps increase economic growth by creating employment opportunities, improving exports, and strengthening the balance of payments. It also helps local industries gain access to global markets and modern technology, which improves overall competitiveness.

Statement of the Problem:

In recent years, FDI has shown significant growth due to globalization, digital transformation, policy liberalization, and expansion of multinational companies. Governments across the world, including India, have introduced investor-friendly policies and reforms to attract more foreign investment. As a result, studying the trends and growth of FDI has become important to understand a country's economic performance and development pattern.

In recent years, global FDI trends have become unstable, and developing countries are facing strong competition in attracting foreign investments. Although India is one of the major FDI destinations, there are variations in sector-wise inflows and regional distribution of investments. Some sectors receive higher FDI inflows, while others remain under-developed. In addition, the actual impact of FDI on economic growth, employment generation, and technology transfer is not uniform across all sectors and regions.

There is also a need to understand whether the growth of FDI is effectively contributing to sustainable economic development. Limited infrastructure, policy challenges, and economic uncertainties may reduce the potential benefits of FDI. Therefore, it is important to study the trends and growth pattern of FDI and analyze its role in economic development.

Objectives:

- ☐ To examine sector-wise distribution of FDI inflows.
- ☐ To study factors influencing FDI inflows in developing countries (especially India).
- ☐ To analyze the importance of FDI in economic development.
- ☐ To analyze the impact of FDI on employment, technology transfer, and industrial growth.
- ☐ To examine government policies and reforms affecting FDI growth.

Sources of Data

The present study is mainly based on **secondary data**.

Nature of Data

- The study uses **quantitative data** such as FDI inflow amounts, sector-wise investment data, and growth rates.

- It also uses **qualitative data** such as policy changes, reforms, and global investment trends.

Period of Study

The data is collected for a specific to analyze trends and growth patterns of FDI.

Method of Data Collection

- Data is collected through online databases, annual reports, and statistical publications.
- Published research papers and reports are used to support analysis and interpretation.

Tools Used for Analysis

- Percentage analysis
- Trend analysis

Sample Size

The study is based on responses collected from **100 respondents** regarding their opinion on Foreign Direct Investment (FDI) and its impact on economic growth.

Analysis and Interpretation of data:

1. Awareness Level about FDI

S.No	Awareness level	No. of Respondents	Percentage
1	Aware about FDI	70	70
2	Not Aware	30	30
	Total	100	100

Source: Primary data.

Interpretation:

Majority of respondents are aware of FDI, indicating increased knowledge about global investment and economic activities.

2. Opinion on FDI Contribution to Economic Growth

S. No	Opinion	No. of Respondents	Percentage
1	Agree	35	35
2	Neutral	40	40
3	Disagree	15	15
4		10	10
	Total	100	100

Source: Primary data.

Interpretation:

Most respondents believe FDI positively contributes to economic growth, employment, and industrial development.

3. Sector Benefiting Most from FDI

S.No	Sector	No. of Respondents	Percentage
1	Services	30	30
2	Manufacturing	25	25
3	IT	25	25
4	Infrastructure	20	20
	Total	100	100

Source: Primary data.

Interpretation:

Respondents feel the service sector benefits most from FDI due to high foreign participation and growth potential.

Findings

- ☐ The study found that a majority of respondents are aware of Foreign Direct Investment and its role in economic development.
- ☐ FDI inflows show an overall increasing trend over the years, though there are fluctuations due to global economic conditions.
- ☐ Most respondents believe that FDI contributes positively to economic growth, industrial development, and employment generation.
- ☐ The services sector is identified as the major sector receiving maximum benefits from FDI, followed by manufacturing and IT sectors.
- ☐ FDI helps in technology transfer, modernization of industries, and improvement in productivity levels.

Suggestions :

1. The government should continue to simplify FDI policies and reduce regulatory barriers to attract more foreign investors.
2. Infrastructure facilities such as transport, power supply, and communication systems should be improved to make the country more attractive for foreign investment.
3. The government should focus on balanced regional development to ensure FDI flows to less developed areas as well.
4. Skill development programs should be strengthened to create a skilled workforce that meets the requirements of foreign companies.
5. Stable economic and political environment should be maintained to increase investor confidence.
6. The government should encourage FDI in underdeveloped sectors such as agriculture and small-scale industries.

Conclusion:

Foreign Direct Investment (FDI) plays a significant role in the economic development of countries, especially developing economies like India. The study shows that FDI contributes to capital formation, employment generation, technology transfer, and industrial growth. Over the years, FDI inflows have shown an overall increasing trend, though fluctuations occur due to global economic conditions, policy changes, and geopolitical factors.

The analysis reveals that sectors such as services, information technology, manufacturing, and infrastructure attract the highest share of foreign investments. Government initiatives, economic reforms, and liberalized FDI policies have helped India become an important destination for foreign investors. However, challenges such as infrastructure gaps, policy issues, and regional imbalance still affect the full utilization of FDI benefits.

The study also concludes that FDI alone cannot ensure economic growth. It must be supported by strong domestic investment, skilled labour, stable policies, and good governance. If proper measures are taken, FDI can contribute significantly to sustainable economic development and global competitiveness.

Overall, the growth and trend of FDI indicate positive opportunities for economic progress, but continuous policy support and structural improvements are essential for maximizing its benefits.

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Sum Divisor Cordial Labeling of Graphs Some Graphs

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Abstract:

In this paper we considered all graphs are simple, connected and undirected graph. For all graph G we use $V(G)$ and $E(G)$ to denote its vertex set and edge set respectively. We have established the sum divisor cordial labeling of diamond graph, Cross graph and bull graph.

Mathematics Subject Classification: 05C78

Keyword: Labeling, Sum divisor cordial labeling.

Introduction:1

A graph labeling of a graph is a map that carries the graph elements to the set of numbers, usually to the set of negative and positive integer. A. Loudusamy and F. Patrick introduced the concept of sum divisor cordial labeling. In the paper we have using the team SDCL.

Definition 1.1

Let $G = (V(G), E(G))$ be a simple graph and $f: V(G) \rightarrow \{1, 2, 3 \dots |V(G)|\}$ be bijective. For each edge uv , assign the label 1 if $2|(f(u) + f(v))$ and the label 0 otherwise. The function f is called a sum divisor cordial labeling $|e_f(0) - e_f(1)| \leq 1$. A graph which admits a sum divisor cordial labeling is called a sum divisor cordial graph.

Definition:1.2

The diamond graph is a planar undirected graph with 4 vertices and 5 edges. It is denoted as G_d .

Definition:1.3

The Bull graph is a planar undirected graph with 5 vertices and 5 edges in the form of a triangle with two disjoint pendent edge. It is denoted as G_b .

1. Main Results

Theorem:2.1

The cross graph admits sum divisor cordial graph.

Proof:

Let G be

a cross graph with 5 vertices and 6 edges,

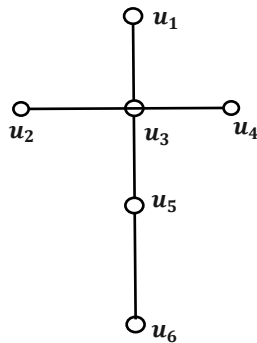


Figure 2.1

The vertices $u_1, u_2, \dots, u_5$ and edges $e_1, e_2, e_3, \dots, e_6$.

$$N(u_3) = \{u_1, u_2, u_4\}; N(u_4) = \{u_3, u_5\}$$

$$SDCL(u_i) = \{i/1 \leq i \leq 6\}$$

$$|e_f(0) - e_f(1)| = |3 - 2| = 1 \leq 1$$

The graph G Satisfy the condition of sum divisor cordial labeling

Therefore, G is Sum divisor cordial graph.

Theorem:2.2

Diamond graph is sum divisor cordial graph.

Proof:

Let G_d be the diamond graph with $|V(G)| = 5$ and $|E(G)| = 8$

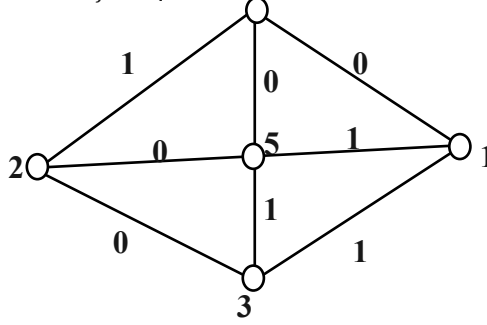
$$V(G) = \{u_i; i = 1, 2, \dots, 5\}$$

$$E(G) = \{e_i; i = 1, 2, \dots, 8\}$$

The sum divisor cordial labeling pattern is given below,

$$SDCL(v_i) = \left\{ \begin{array}{l} n - (i - j) \text{ such that } i = n \text{ to } \left\lceil \frac{n+1}{2} \right\rceil \text{ and } j = 1 \text{ to } \left\lfloor \frac{n+1}{2} \right\rfloor \\ n - (j - i) \text{ such that } i = \left\lfloor \frac{n-1}{2} \right\rfloor \text{ to } 1 \text{ and } j = \left\lceil \frac{n+1}{2} \right\rceil \text{ to } n-1 \end{array} \right\}$$

$$|e_f(0) - e_f(1)| = |3 - 2| = 1 \leq 1$$



(Figure:2.2)

$$|e_f(0) - e_f(1)| = |4 - 4| = 0 \leq 1$$

Diamond graph is sum divisor cordial graph.

Theorem:2.3

Bull graph admits sum divisor cordial graph.

Proof:

Let G' be the bull graph with 5 vertices and 7 edges.

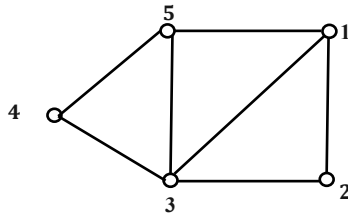
$$V(G) = \{v_i; i = 1, 2, \dots, 5\}$$

$$E(G) = \{e_i; i = 1, 2, \dots, 7\}$$

The sum divisor cordial labeling pattern of G' ,

$$SDCL(v_i) = \{i/1 \leq i \leq 5\}$$

$$|e_f(0) - e_f(1)| = |4 - 3| = 1 \leq 1$$



Graph G' (Figure:2.3)

The Bull graph is sum divisor cordial graph.

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User-Adaptive Meta-Learning for Cold-Start Medication Recommendation with Uncertainty Filtering

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Abstract

Cold-start medication recommendation is a challenging problem due to the lack of historical patient data and uncertainty in clinical information. This study proposes a user-adaptive meta-learning framework to generate personalized medication recommendations for new users with limited data. The model learns transferable knowledge from diverse patient profiles and quickly adapts to unseen users through meta-learning techniques. To enhance reliability, an uncertainty filtering mechanism is integrated to identify and reduce ambiguous or low-confidence predictions. This approach improves recommendation accuracy while maintaining patient safety and interpretability. Experimental results demonstrate that the proposed method outperforms traditional recommendation models in cold-start scenarios. The framework also shows robustness against data sparsity and noisy medical records. Overall, the proposed system provides an effective and trustworthy solution for personalized medication recommendation in real-world healthcare applications.

I. Introduction

In healthcare systems, recommending the correct medicine for a patient is very important. However, for **new patients**, there is very little medical history available. This problem is called the **cold-start problem**. Traditional recommendation systems do not work well in such cases.

This paper proposes a new method to improve medication recommendations for new patients using **meta-learning** and **uncertainty filtering**.

II. Literature Survey

Recent studies on medication recommendation systems have focused on machine learning and deep learning techniques to improve personalized healthcare. Traditional recommendation models often struggle with the cold-start problem due to insufficient patient data. Meta-learning approaches have been explored to enable rapid adaptation to new users by learning from prior tasks and experiences. User-adaptive models have shown promising results

in capturing individual health patterns and improving recommendation accuracy. Several researchers have incorporated uncertainty estimation methods to enhance the reliability and safety of clinical predictions. Bayesian learning and probabilistic models are widely used to quantify prediction uncertainty in medical applications. Hybrid models combining collaborative filtering and neural networks have also been proposed to address data sparsity issues. However, existing methods often lack effective mechanisms to filter unreliable recommendations in cold-start scenarios. Recent advancements highlight the importance of integrating uncertainty filtering with adaptive learning frameworks. Therefore, there is a growing need for robust user-adaptive meta-learning models for safe and accurate medication recommendation systems.

III. Proposed System

The proposed system introduces a user-adaptive meta-learning framework to provide accurate medication recommendations for new patients with limited historical data. The system first collects patient information such as demographic data, symptoms, medical history, and clinical records. A meta-learning model is trained on multiple patient tasks to learn generalized patterns that can quickly adapt to unseen users.

To handle the cold-start problem, the model dynamically adapts its parameters to each new user using few-shot learning techniques. An uncertainty filtering module is integrated to evaluate the confidence of each medication recommendation. Predictions with high uncertainty are filtered or flagged to ensure safety and reliability in clinical decision-making.

The system combines deep learning, probabilistic modeling, and adaptive learning strategies to improve recommendation accuracy. Performance is evaluated using metrics such as accuracy, precision, recall, and confidence scores. Overall, the proposed system provides a robust, personalized, and trustworthy medication recommendation solution, especially in cold-start healthcare scenarios.

IV. Result and Discussion

Results

The proposed user-adaptive meta-learning model was evaluated on healthcare datasets containing patient records and medication information. The system demonstrated significant improvement in medication recommendation accuracy under cold-start conditions compared to traditional machine learning and deep learning models. The integration of meta-learning enabled the model to quickly adapt to new users with minimal data, resulting in better personalized recommendations.

The uncertainty filtering mechanism effectively identified low-confidence predictions and reduced the risk of incorrect medication suggestions. Performance evaluation using metrics such as accuracy, precision, recall, F1-score, and uncertainty confidence showed that the proposed system consistently outperformed baseline models like collaborative filtering and standard neural networks. The model also exhibited robustness against data sparsity and noisy patient data.

Discussion

The results indicate that combining user-adaptive meta-learning with uncertainty filtering significantly enhances the reliability and effectiveness of medication recommendation systems in cold-start scenarios. The adaptive learning capability allows the system to generalize knowledge from previous users and apply it to new patients efficiently. Meanwhile, uncertainty filtering improves patient safety by preventing unreliable recommendations.

Compared to existing approaches, the proposed system provides a balanced solution between accuracy and trustworthiness. However, the system's performance may depend on the quality and diversity of training data. Future work can focus on integrating real-time clinical data, improving interpretability, and validating the model in real-world healthcare environments. Overall, the proposed approach proves to be a promising framework for personalized and safe medication recommendation.

V. Conclusion

This study presented a user-adaptive meta-learning framework for cold-start medication recommendation enhanced with uncertainty filtering. The proposed system effectively addresses the challenge of limited patient data by enabling rapid adaptation to new users through meta-learning techniques. The integration of uncertainty filtering improves the reliability and safety of medication recommendations by identifying and reducing low-confidence predictions.

Experimental results demonstrate that the proposed approach outperforms traditional recommendation models in terms of accuracy and robustness under cold-start conditions. The system also shows strong potential in handling data sparsity and variability in healthcare data. Overall, the proposed framework offers a promising solution for personalized, trustworthy, and efficient medication recommendation in modern healthcare systems. Future work can focus on real-world clinical validation, interpretability improvements, and integration with electronic health record systems.

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Changing And Unchanging of Split Substantial Independence Number for Paths and Cycles

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Abstract

A independent set S of a graph G is a split substantial independent set if S is a substantial independent set and the induced subgraph $\langle V - S \rangle$ is disconnected. The split substantial independence number is the maximum cardinality among all split substantial independent sets and is indicated by β_{ss} . In this paper, we determine the change in split substantial independence number after removing a vertex in paths and cycles.

Keywords and Phrases: Independence number, Substantial independent set, Split substantial independent set.

1. Introduction

By a graph we mean a finite, undirected connected graph without loops or multiple edges. Terms not defined here are used in the sense of Haynes et. al. [5] and Harary [1]. A substantial independent set is a non-empty subset $S \subset V$ in a connected graph $G = (V, E)$ if S is an independent set of G and any vertex in V/S is join by an edge to atmost one vertex in S . The substantial independence number of G is the supremum cardinality among all substantial independent sets in G and is indicated by $\beta_s(G)$ [2, 3]. Let $G - v$ denotes the graph formed by removing vertex v from G . We use acronyms to denote the following classes of Graphs. (C : changing, U : unchanging, V : vertex, R : removal). (CVR) $\beta_{ss}(G - v) \neq \beta_{ss}(G)$ for all $v \in V$ and (UVR) $\beta_{ss}(G - v) < \beta_{ss}(G)$ for all $v \in V$. The vertex set is partitioned into three sets according to how the removal of vertices affect $\beta_{ss}(G)$. Let $V = V^0 \cup V^+ \cup V^-$ where $V^0 = \{v \in V : \beta_{ss}(G - v) = \beta_{ss}(G)\}$, $V^+ = \{v \in V : \beta_{ss}(G - v) > \beta_{ss}(G)\}$ and $V^- = \{v \in V : \beta_{ss}(G - v) < \beta_{ss}(G)\}$ [5].

The concept of split domination number was introduced by Kulli and Janakiram [4] and the substantial independence number was introduced by Dr.V.Rani Ratha Bai. Combining these we introduce a new concept "Split substantial independent set". In this paper, we determine the change in split substantial independence number after removing a vertex in paths and cycles.

2. Main Results

Definition 2.1. A independent set S of a graph G is a **split substantial independent set** if S is a substantial independent set and the induced subgraph $\langle V - S \rangle$ is disconnected. The split substantial independence number is the maximum cardinality among all split substantial independent sets and is indicated by β_{ss} .

Lemma 2.2. For any path of order $n \geq 5$, $\beta_{ss}(P_n) = \left\lfloor \frac{n}{3} \right\rfloor$.

Proof. Let $V(P_n) = \{v_1, v_2, \dots, v_n\}$ be the vertex set of the path P_n . Let v_1 and v_n be the pendant vertices of P_n . Let $S = \{v_{1+3i}, i = 0, 1, 2, \dots, \left(\left\lfloor \frac{n}{3} \right\rfloor - 1\right)\}$ is a substantial independent set and $\langle V - S \rangle$ is dis-connected.

Suppose $S' = S \cup \{v_j, v_j \notin S\}$ then $\langle V - S' \rangle$ is connected or a vertex of $V - S'$ is adjacent to more than one vertex of S' . Then clearly, S is a maximal split substantial independent set. Hence $\beta_{ss}(P_n) = \left\lfloor \frac{n}{3} \right\rfloor$.

Lemma 2.3. For any cycle of order $n \geq 6$, $\beta_{ss}(C_n) = \left\lfloor \frac{n}{3} \right\rfloor$.

Proof. Let $V(C_n) = \{v_1, v_2, \dots, v_n\}$ be the vertex set of the cycle C_n . Let $S = \{v_{1+3i}, i = 0, 1, 2, \dots, \left(\left\lfloor \frac{n}{3} \right\rfloor - 1\right)\}$ is a substantial independent set and $\langle V - S \rangle$ is disconnected.

Suppose $S' = S \cup \{v_j, v_j \notin S\}$ then $\langle V - S' \rangle$ is connected or a vertex of $V - S'$ is adjacent to more than one vertex of S' . Then clearly, S is a maximal split substantial independent set. Hence $\beta_{ss}(C_n) = \left\lfloor \frac{n}{3} \right\rfloor$.

Theorem 2.4. For any path P_n with n ($n \geq 7$) vertices, then

(i) $V = V^0 \cup V^+$ if $n \equiv 0 \pmod{3}$

(ii) $V = V^0 \cup V^-$ if $n \equiv 1 \pmod{3}$, and

(iii) $G \in UVR$ if $n \equiv 2 \pmod{3}$.

Proof. Let V be the vertex set of P_n .

(i) $n \equiv 0 \pmod{3}$, $n = 3k$, $k \in \mathbb{N}$ and $k \geq 3$

Case (a) : $d(v) = 1$

Let v_1, v_n be the pendant vertices and $d(v) = 1$ of P_n . Then $P_n - v_1$ and $P_n - v_n$ are paths with $n - 1$ vertices. Therefore, $\beta_{ss}(P_n - v_1) = \beta_{ss}(P_n - v_n) = \left\lfloor \frac{n-1}{3} \right\rfloor = \left\lfloor \frac{3k-1}{3} \right\rfloor = \left\lfloor \frac{3(k-1)+2}{3} \right\rfloor = k$

By lemma 2.2, $\beta_{ss}(P_n) = \left\lfloor \frac{3k}{3} \right\rfloor = k$. Hence, $\beta_{ss}(P_n) = \beta_{ss}(P_n - v_1) = \beta_{ss}(P_n - v_n) = k$. Thus $v_1, v_n \in V^0$

Case (b) : $d(v) = 2$

Now, $P_n - v_j$, $j \equiv 2 \pmod{3}$, $j = 3l + 2$, $l < k$, $l \in \mathbb{W}$ is the union of two disconnected paths G_1 and G_2 with even or odd number of vertices. Let $v_1, v_2, \dots, v_{j-1}$ be the vertices of G_1 and $v_{j+1}, v_{j+2}, \dots, v_n$ that of G_2 .

Therefore, $\beta_{ss}(P_n - v_j) = \left\lfloor \frac{j-1}{3} \right\rfloor + \left\lfloor \frac{n-j}{3} \right\rfloor = \left\lfloor \frac{(3l+2)-1}{3} \right\rfloor + \left\lfloor \frac{3k-(3l+2)}{3} \right\rfloor = \left\lfloor \frac{3l+1}{3} \right\rfloor + \left\lfloor \frac{3(k-l)-2}{3} \right\rfloor = \left\lfloor \frac{3l+1}{3} \right\rfloor + \left\lfloor \frac{3(k-l-1)+1}{3} \right\rfloor = (l+1) + (k-l-1) + 1 = k+1$

By lemma 2.2, $\beta_{ss}(P_n) = \left\lfloor \frac{3k}{3} \right\rfloor = k$. Hence, $\beta_{ss}(P_n) < \beta_{ss}(P_n - v_j)$. Thus $v_j \in V^+$, $j \equiv 2 \pmod{3}$

Now, $P_n - v_j$, $j \equiv 1 \pmod{3}$, $j = 3l + 1$, $l < k$, $l \in \mathbb{N}$ is the union of two disconnected Paths G_1 and G_2 with even or odd number of vertices. Let $v_1, v_2, \dots, v_{j-1}$ be the vertices of G_1 and $v_{j+1}, v_{j+2}, \dots, v_n$ that of G_2 .

Therefore, $\beta_{ss}(P_n - v_j) = \left\lfloor \frac{j-1}{3} \right\rfloor + \left\lfloor \frac{n-j}{3} \right\rfloor = \left\lfloor \frac{(3l+1)-1}{3} \right\rfloor + \left\lfloor \frac{3k-(3l+1)}{3} \right\rfloor = \left\lfloor \frac{3l}{3} \right\rfloor + \left\lfloor \frac{3(k-l)-1}{3} \right\rfloor = \left\lfloor \frac{3l}{3} \right\rfloor + \left\lfloor \frac{3(k-l-1)+2}{3} \right\rfloor = (l) + (k-l-1) + 1 = k$

By lemma 2.2, $\beta_{ss}(P_n) = \left\lfloor \frac{3k}{3} \right\rfloor = k$. Hence, $\beta_{ss}(P_n) = \beta_{ss}(P_n - v_j)$. Thus $v_j \in V^0$, $j \equiv 1 \pmod{3}$

Now, $P_n - v_j$, $j \equiv 0 \pmod{3}$, $j = 3l$, $l < k$, $l \in \mathbb{N}$ is the union of two disconnected paths G_1 and G_2 with even or odd number of vertices. Let $v_1, v_2, \dots, v_{j-1}$ be the vertices of G_1 and $v_{j+1}, v_{j+2}, \dots, v_n$ that of G_2 .

Therefore, $\beta_{ss}(P_n - v_j) = \left\lfloor \frac{j-1}{3} \right\rfloor + \left\lfloor \frac{n-j}{3} \right\rfloor = \left\lfloor \frac{3l-1}{3} \right\rfloor + \left\lfloor \frac{3k-3l}{3} \right\rfloor = \left\lfloor \frac{3(l-1)+2}{3} \right\rfloor + \left\lfloor \frac{3(k-l)}{3} \right\rfloor = (l-1) + 1 + (k-l) = k$

By lemma 2.2, $\beta_{ss}(P_n) = \left\lfloor \frac{3k}{3} \right\rfloor = k$. Hence, $\beta_{ss}(P_n) = \beta_{ss}(P_n - v_j)$. Thus $v_j \in V^0$, $j \equiv 0 \pmod{3}$

Hence, $V = V^0 \cup V^+$

(ii) $n \equiv 1 \pmod{3}$, $n = 3k+1$, $k \in \mathbb{N}$ and $k \geq 2$

Case (a) : $d(v) = 1$

Let v_1, v_n be the pendant vertices and $d(v) = 1$ of P_n . Then $P_n - v_1$ and $P_n - v_n$ are paths with $n - 1$ vertices. Therefore, $\beta_{ss}(P_n - v_1) = \beta_{ss}(P_n - v_n) = \left\lfloor \frac{n-1}{3} \right\rfloor = \left\lfloor \frac{(3k+1)-1}{3} \right\rfloor = \left\lfloor \frac{3k}{3} \right\rfloor = k$

By lemma 2.2, $\beta_{ss}(P_n) = \left\lfloor \frac{3k+1}{3} \right\rfloor = k+1$. Hence, $\beta_{ss}(P_n) > \beta_{ss}(P_n - v_1)$ and $\beta_{ss}(P_n) > \beta_{ss}(P_n - v_n)$. Thus $v_1, v_n \in V^-$.

Case (b) : $d(v) = 2$

Now, $P_n - v_j$, $j \equiv 2(\text{mod } 3)$, $j = 3l + 2$, $l < k$, $l \in W$ is the union of two disconnected paths G_1 and G_2 with even or odd number of vertices. Let $v_1, v_2, \dots, v_{j-1}$ be the vertices of G_1 and $v_{j+1}, v_{j+2}, \dots, v_n$ that of G_2 .

$$\text{Therefore, } \beta_{ss}(P_n - v_j) = \left\lfloor \frac{j-1}{3} \right\rfloor + \left\lfloor \frac{n-j}{3} \right\rfloor = \left\lfloor \frac{(3l+2)-1}{3} \right\rfloor + \left\lfloor \frac{(3k+1)-(3l+2)}{3} \right\rfloor = \left\lfloor \frac{3l+1}{3} \right\rfloor + \left\lfloor \frac{3(k-l)-1}{3} \right\rfloor = \left\lfloor \frac{3l+1}{3} \right\rfloor + \left\lfloor \frac{3(k-l-1)+2}{3} \right\rfloor = (l+1) + (k-l-1) + 1 = k+1$$

By lemma 2.2, $\beta_{ss}(P_n) = \left\lfloor \frac{3k+1}{3} \right\rfloor = k+1$. Hence, $\beta_{ss}(P_n) = \beta_{ss}(P_n - v_j)$. Thus $v_j \in V^0$, $j \equiv 2(\text{mod } 3)$.

Now, $P_n - v_j$, $j \equiv 1(\text{mod } 3)$, $j = 3l + 1$, $l < k$, $l \in N$ is the union of two disconnected paths G_1 and G_2 with even or odd number of vertices. Let $v_1, v_2, \dots, v_{j-1}$ be the vertices of G_1 and $v_{j+1}, v_{j+2}, \dots, v_n$ that of G_2 .

$$\text{Therefore, } \beta_{ss}(P_n - v_j) = \left\lfloor \frac{j-1}{3} \right\rfloor + \left\lfloor \frac{n-j}{3} \right\rfloor = \left\lfloor \frac{(3l+1)-1}{3} \right\rfloor + \left\lfloor \frac{(3k+1)-(3l+1)}{3} \right\rfloor = \left\lfloor \frac{3l}{3} \right\rfloor + \left\lfloor \frac{3(k-l)}{3} \right\rfloor = (l) + (k-l) = k$$

By lemma 2.2, $\beta_{ss}(P_n) = \left\lfloor \frac{3k+1}{3} \right\rfloor = k+1$. Hence, $\beta_{ss}(P_n) > \beta_{ss}(P_n - v_j)$. Thus $v_j \in V^-$, $j \equiv 1(\text{mod } 3)$

Now, $P_n - v_j$, $j \equiv 0(\text{mod } 3)$, $j = 3l$, $l < k$, $l \in N$ is the union of two disconnected paths G_1 and G_2 with even or odd number of vertices. Let $v_1, v_2, \dots, v_{j-1}$ be the vertices of G_1 and $v_{j+1}, v_{j+2}, \dots, v_n$ that of G_2 .

$$\text{Therefore, } \beta_{ss}(P_n - v_j) = \left\lfloor \frac{j-1}{3} \right\rfloor + \left\lfloor \frac{n-j}{3} \right\rfloor = \left\lfloor \frac{3l-1}{3} \right\rfloor + \left\lfloor \frac{(3k+1)-3l}{3} \right\rfloor = \left\lfloor \frac{3(l-1)+2}{3} \right\rfloor + \left\lfloor \frac{3(k-l)+1}{3} \right\rfloor = (l-1) + 1 + (k-l) + 1 = k+1$$

By lemma 2.2, $\beta_{ss}(P_n) = \left\lfloor \frac{3k+1}{3} \right\rfloor = k+1$. Hence, $\beta_{ss}(P_n) = \beta_{ss}(P_n - v_j)$. Thus $v_j \in V^0$, $j \equiv 0(\text{mod } 3)$

Hence, $V = V^0 \cup V^-$

(iii) $n \equiv 2(\text{mod } 3)$, $n = 3k+2$, $k \in N$ and $k \geq 2$

Case (a) : $d(v) = 1$

Let v_1, v_n be the pendant vertices and $d(v) = 1$ of P_n . Then $P_n - v_1$ and $P_n - v_n$ are paths with $n-1$ vertices. Therefore, $\beta_{ss}(P_n - v_1) = \beta_{ss}(P_n - v_n) = \left\lfloor \frac{n-1}{3} \right\rfloor = \left\lfloor \frac{(3k+2)-1}{3} \right\rfloor = \left\lfloor \frac{3k+1}{3} \right\rfloor = k+1$

By lemma 2.2, $\beta_{ss}(P_n) = \left\lfloor \frac{3k+2}{3} \right\rfloor = k+1$. Hence, $\beta_{ss}(P_n) = \beta_{ss}(P_n - v_1) = \beta_{ss}(P_n - v_n) = k+1$. Thus $v_1, v_n \in V^0$.

Case (b) : $d(v) = 2$

Now, $P_n - v_j$, $j \equiv 2(\text{mod } 3)$, $j = 3l + 2$, $l < k$, $l \in W$ is the union of two disconnected paths G_1 and G_2 with even or odd number of vertices. Let $v_1, v_2, \dots, v_{j-1}$ be the vertices of G_1 and $v_{j+1}, v_{j+2}, \dots, v_n$ that of G_2 .

$$\text{Therefore, } \beta_{ss}(P_n - v_j) = \left\lfloor \frac{j-1}{3} \right\rfloor + \left\lfloor \frac{n-j}{3} \right\rfloor = \left\lfloor \frac{(3l+2)-1}{3} \right\rfloor + \left\lfloor \frac{(3k+2)-(3l+2)}{3} \right\rfloor = \left\lfloor \frac{3l+1}{3} \right\rfloor + \left\lfloor \frac{3(k-l)}{3} \right\rfloor = (l+1) + (k-l) = k+1.$$

By lemma 2.2, $\beta_{ss}(P_n) = \left\lfloor \frac{3k+2}{3} \right\rfloor = k+1$. Hence, $\beta_{ss}(P_n) = \beta_{ss}(P_n - v_j)$. Thus $v_j \in V^0$, $j \equiv 2(\text{mod } 3)$

Now, $P_n - v_j$, $j \equiv 1(\text{mod } 3)$, $j = 3l + 1$, $l < k$, $l \in N$ is the union of two disconnected paths G_1 and G_2 with even or odd number of vertices. Let $v_1, v_2, \dots, v_{j-1}$ be the vertices of G_1 and $v_{j+1}, v_{j+2}, \dots, v_n$ that of G_2 .

$$\text{Therefore, } \beta_{ss}(P_n - v_j) = \left\lfloor \frac{j-1}{3} \right\rfloor + \left\lfloor \frac{n-j}{3} \right\rfloor = \left\lfloor \frac{3l-1}{3} \right\rfloor + \left\lfloor \frac{(3k+2)-3l}{3} \right\rfloor = \left\lfloor \frac{3(l-1)+2}{3} \right\rfloor + \left\lfloor \frac{3(k-l)+2}{3} \right\rfloor = (l-1) + 1 + (k-l) + 1 = k+1.$$

By lemma 2.2, $\beta_{ss}(P_n) = \left\lfloor \frac{3k+2}{3} \right\rfloor = k + 1$. Hence, $\beta_{ss}(P_n) = \beta_{ss}(P_n - v_j)$. Thus $v_j \in V^0$, $j \equiv 0 \pmod{3}$

Hence, $P_n \in \text{UVR}$ if $n \equiv 2 \pmod{3}$.

Theorem 2.5. For any cycle C_n with n ($n \geq 6$) vertices, then

(i) $C_n \in \text{UVR}$ if $n \equiv 0, 1 \pmod{3}$ and

(ii) $V = V^+$ if $n \equiv 2 \pmod{3}$.

Proof. Let V be the vertex set of C_n . Let v_j be any vertex of cycle C_n . Now, $C_n - v_j$ is the path with $n - 1$ vertices.

(i) $n \equiv 0 \pmod{3}$, $n = 3k$, $k \in \mathbb{N}$ and $k \geq 2$.

Therefore, $\beta_{ss}(C_n - v_j) = \beta_{ss}(P_{n-1}) = \left\lfloor \frac{n-1}{3} \right\rfloor = \left\lfloor \frac{3k-1}{3} \right\rfloor = \left\lfloor \frac{3(k-1)+2}{3} \right\rfloor = k - 1 + 1 = k$

By lemma 2.3, $\beta_{ss}(C_n) = \left\lfloor \frac{3k}{3} \right\rfloor = k$. Hence, $\beta_{ss}(C_n - v_j) = \beta_{ss}(C_n)$. Thus $v_j \in V^0$

Here, $n \equiv 1 \pmod{3}$, $n = 3k + 1$, $k \in \mathbb{N}$ and $k \geq 2$.

Therefore, $\beta_{ss}(C_n - v_j) = \beta_{ss}(P_{n-1}) = \left\lfloor \frac{n-1}{3} \right\rfloor = \left\lfloor \frac{3(k+1)-1}{3} \right\rfloor = \left\lfloor \frac{3k}{3} \right\rfloor = k$

By lemma 2.3, $\beta_{ss}(C_n) = \left\lfloor \frac{3k+1}{3} \right\rfloor = k$. Hence, $\beta_{ss}(C_n - v_j) = \beta_{ss}(C_n)$. Thus $v_j \in V^0$

Hence, $C_n \in \text{UVR}$, $n \equiv 0, 1 \pmod{3}$.

(ii) $n \equiv 2 \pmod{3}$, $n = 3k + 2$, $k \in \mathbb{N}$ and $k \geq 2$.

Therefore, $\beta_{ss}(C_n - v_j) = \beta_{ss}(P_{n-1}) = \left\lfloor \frac{n-1}{3} \right\rfloor = \left\lfloor \frac{(3k+2)-1}{3} \right\rfloor = \left\lfloor \frac{3k+1}{3} \right\rfloor = k + 1$

By lemma 2.3, $\beta_{ss}(C_n) = \left\lfloor \frac{3k+2}{3} \right\rfloor = k$. Hence, $\beta_{ss}(C_n - v_j) > \beta_{ss}(C_n)$. Thus $v_j \in V^+$

Hence, $V = V^+$

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k-Super Mean Labeling of Cartesian Product of Line Graphs

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Abstract:

Graph labeling is a fundamental concept in graph theory, with applications in various fields, including computer, engineering and mathematics. k-Super mean labeling is a type of graph labeling that assigns labels to vertices and edges satisfying the certain mean-based conditions. This presentation explores the concept of k-super mean labeling for graphs and the results of labeling for various graph classes.

Keywords:

Graph, Line graph, Cartesian product, Super mean labeling, k-Super mean labeling.

1.Introduction:

Graph labeling is the systematic study of assigning integers to the vertices and edges of graphs was first introduced by Alexander Rosa in 1967. Since then, several labeling techniques have been proposed to explore graph structures. Among these techniques, the concept of super mean labeling was introduced by Somasundaram and Ponraj in 2003, where edge labels are obtained from the arithmetic mean of the vertex labels. Extending this idea, M. Tamilselvi and co-authors in 2017 introduced the concept of k-super mean labeling, where vertices are labeled with consecutive integers starting from an arbitrary positive integer k, and edges are labeled with the ceiling of the mean of their end vertices.

Definition 1.1:

A **graph** G consists of a set $V(G)$ of elements called vertices and a set $E(G)$ of elements called edges together with a relation of incidence which associates each edges with a pair of vertices called its ends.

Definition 1.2:

A **line graph** $L(G)$ of a given graph G is a graph where each vertex of $L(G)$ represents an edge of G , and two vertices in $L(G)$ are connected by an edge if and only if their corresponding edges in G share a common endpoint (i.e., are incident).

Definition 1.3:

The **Cartesian product** of two graphs G_1 and G_2 is a graph denoted by $G = G_1 \square G_2$ where two vertices (u_1, v_1) and (u_2, v_2) in G are adjacent if and only if either $u_1 = u_2$ and v_1 is adjacent to v_2 or $v_1 = v_2$ and u_1 is adjacent to u_2 .

Definition 1.4:

Let G be a (p, q) graph and $f : V(G) \rightarrow \{1, 2, 3, \dots, p + q\}$ be an injection. For edge $e = uv$, let $f * (e) = (f(u) + f(v))/2$ if $f(u) + f(v)$ is even and $f * (e) = (f(u) + f(v) + 1)/2$ if $f(u) + f(v)$ is odd. Then f is called a **super mean labeling** if $f(V) \cup \{f * (e) : e \in E(G)\} = \{1, 2, \dots, p + q\}$. A graph that admits a super mean labeling is called a **super mean graph**.

Definition 1.5:

Let G be a (p, q) graph and $f : V(G) \rightarrow \{k, k + 1, k + 2, \dots, p + q + k - 1\}$ be an injection. For edge $e = uv$, let $f * (e) = (f(u) + f(v))/2$ if $f(u) + f(v)$ is even and $f * (e) = (f(u) + f(v) + 1)/2$ if $f(u) + f(v)$ is odd. Then f is called a **k-super mean labeling** if $f(V) \cup \{f * (e) : e \in E(G)\} = \{k, k + 1, k + 2, \dots, p + q + k - 1\}$. A graph that admits a k-super mean labeling is called a **k-super mean graph**.

2. Main Results:

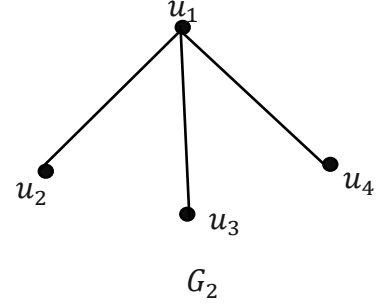
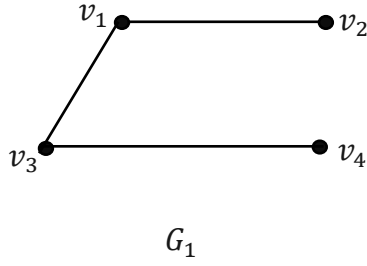
Theorem 2.1 :

Let $L(G_1)$ and $L(G_2)$ be the line graphs of G_1 and G_2 . G is a cartesian product of $L(G_1)$ and $L(G_2)$. Then G is a k-super mean graph.

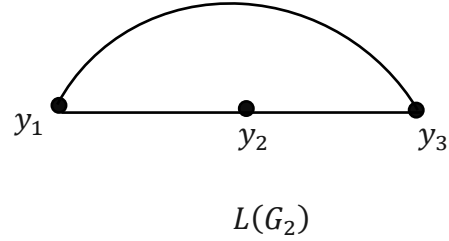
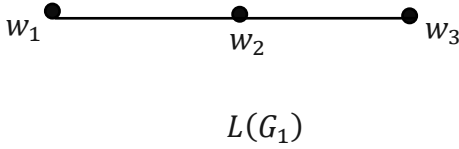
Proof:

Let G_1 be a given graph $|V(G_1)| = 4$ and $|E(G_1)| = 3$

Let G_2 be a given graph $|V(G_2)| = 4$ and $|E(G_2)| = 3$
 G_1 and G_2 graphs given below,



Let $L(G_1)$ be the line graph of G_1 such that $|V(L(G_1))| = 3$ and $|E(L(G_1))| = 2$
Let $L(G_2)$ be the line graph of G_2 such that $|V(L(G_2))| = 3$ and $|E(L(G_2))| = 3$
 $L(G_1)$ and $L(G_2)$ graphs given below,



The Cartesian product G is defined by $G = L(G_1) \square L(G_2)$

The vertex set of G is defined by,

$$V(G) = \{v_i \mid 1 \leq i \leq 9\}$$

The edge set of G is defined by,

$$E(G) = \begin{cases} e_i; & 1 \leq i \leq 6 \\ e'_i; & 7 \leq i \leq 9 \\ e''_i; & 10 \leq i \leq 15 \end{cases}$$

The cartesian product of $L(G_1)$ and $L(G_2)$ is given by,

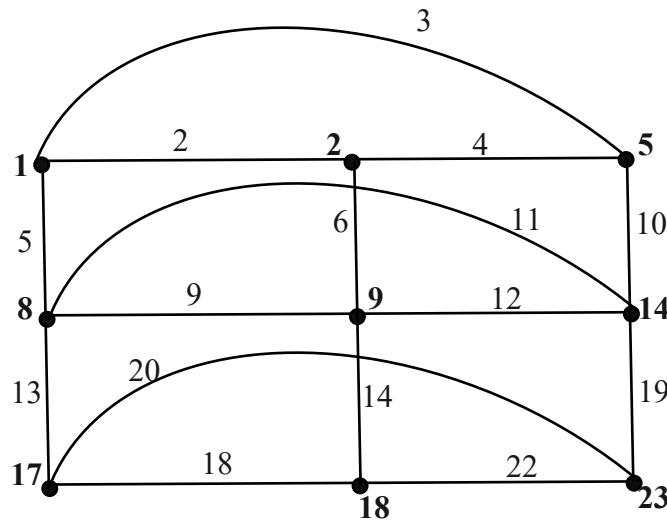


Fig no. 2.1

The k-super mean labeling pattern of vertex and edge set is given below,

$$KSML(v_i) = \begin{cases} i ; i = 1,2 \\ i + 2 ; i = 3 \\ i + 4 ; i = 4,5 \\ 2i + 2 ; i = 6,8 \\ 3i - 4 ; i = 7,9 \end{cases}$$

$$KSML(e_i) = \begin{cases} 2i ; i = 1,2 \\ 3i ; i = 3,4 \\ 3i + 3 ; i = 5 \\ 3i + 4 ; i = 6 \\ i - 4 ; i = 7 \\ i + 3 ; i = 8 \\ 2i + 2 ; i = 9 \\ i - 5 ; i = 10,11 \\ i - 2 ; i = 12 \\ i ; i = 13,14 \\ i + 4 ; i = 15 \end{cases}$$

The edges are labeled in the following pattern,

$$f(e_i) = \begin{cases} v_i v_{i+1} ; i = 1,2 \\ v_{i+1} v_{i+2} ; i = 3,4 \\ v_{i+2} v_{i+3} ; i = 5,6 \end{cases}$$

$$f(e'_i) = \begin{cases} v_{i-6} v_{i-4} ; i = 7 \\ v_{i-4} v_{i-2} ; i = 8 \\ v_{i-2} v_i ; i = 9 \end{cases}$$

$$f(e''_i) = \{v_{i-9} v_{i-6} ; i = 10,11,12,13,14,15\}$$

This graph satisfies the condition of k-super mean labeling.

Therefore, G is called a k- super mean graph.

Theorem 2.2 :

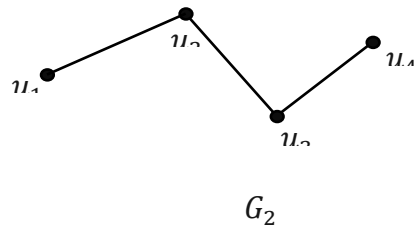
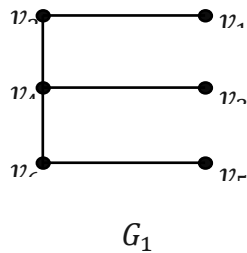
Let $L(G_1)$ and $L(G_2)$ be the line graphs of $P_3 \odot K_1$ and P_4 . G is a cartesian product of $L(G_1)$ and $L(G_2)$. Then G is a k-super mean graph.

Proof:

Let $P_3 \odot K_1$ be a given graph $|V(P_3 \odot K_1)| = 6$ and $|E(P_3 \odot K_1)| = 5$

Let P_4 be a given graph $|V(P_4)| = 4$ and $|E(P_4)| = 3$

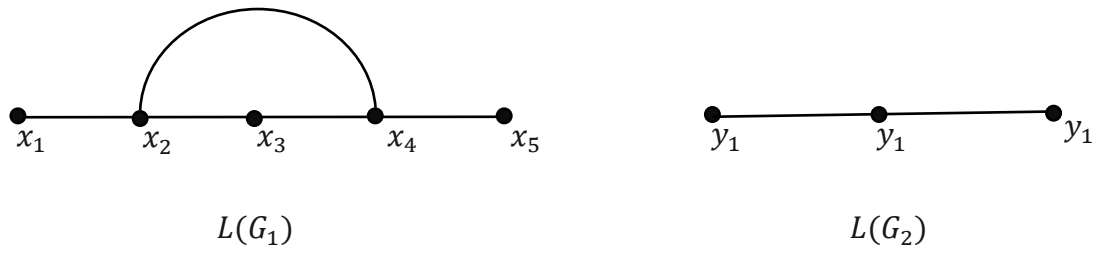
$P_3 \odot K_1$ and P_4 graphs given below,



Let $L(G_1)$ be the line graph of $P_3 \odot K_1$ such that $|V(L(G_1))| = 5$ and $|E(L(G_1))| = 5$

Let $L(G_2)$ be the line graph of P_4 such that $|V(L(G_2))| = 3$ and $|E(L(G_2))| = 2$

$L(G_1)$ and $L(G_2)$ graphs given below,



The Cartesian product G is defined by $G = L(G_1) \square \square L(G_2)$

The vertex set of G is defined by,

$$V(G) = \{v_i \mid 1 \leq i \leq 15\}$$

The edge set of G is defined by,

$$E(G) = \begin{cases} e_i; & 1 \leq i \leq 10 \\ e'_i; & 11 \leq i \leq 22 \\ e''_i; & 23 \leq i \leq 25 \end{cases}$$

The cartesian product of $L(G_1)$ and $L(G_2)$ is given by

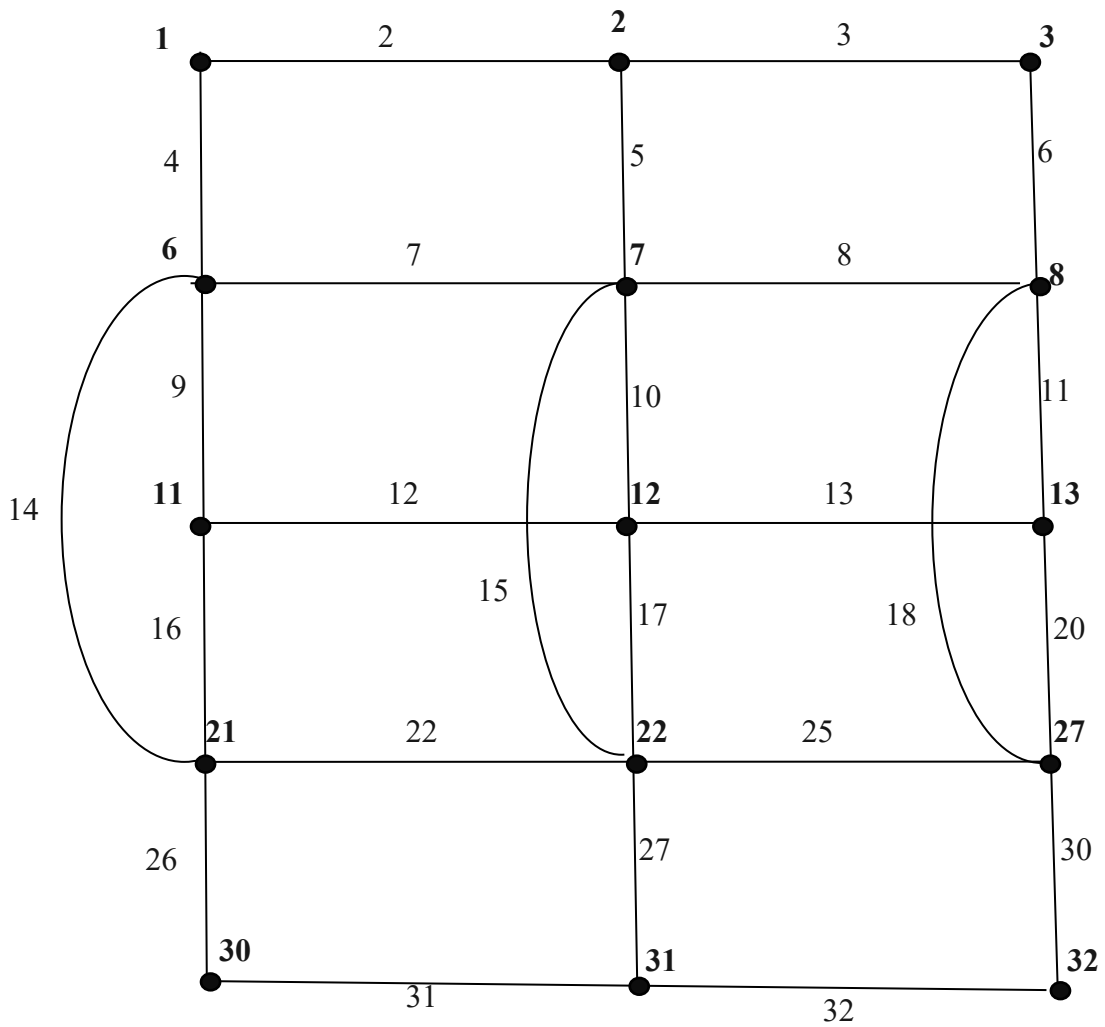


Fig no. 2.2

The k-super mean labeling pattern of vertex and edge set is given below,

$$KSML(v_i) = \begin{cases} i ; i = 1,2,3 \\ i + 2 ; i = 4,5 \\ i + 4 ; i = 7,8,9 \\ i + 11 ; i = 10,11 \\ 2i + 3 ; i = 12,14 \\ 2i + 4 ; i = 13 \\ 2i + 2 ; i = 15 \end{cases}$$

$$KSML(e_i) = \begin{cases} i + 1 ; i = 1,2,19 \\ i + 4 ; i = 3,4 \\ 2i + 2 ; i = 5 \\ 2i + 1 ; i = 6 \\ 3i + 1 ; i = 7,8 \\ 3i + 4 ; i = 9 \\ 3i + 2 ; i = 10 \\ i - 7 ; i = 11,12,13,25 \\ i - 5 ; i = 14,15,16 \\ i - 1 ; i = 17,18 \\ i + 6 ; i = 20,21 \\ i + 8 ; i = 22 \\ i - 9 ; i = 23,24 \end{cases}$$

The edges are labeled in the following pattern,

$$f(e_i) = \begin{cases} v_i v_{i+1} ; i = 1,2 \\ v_{i+1} v_{i+2} ; i = 3,4 \\ v_{i+2} v_{i+3} ; i = 5,6 \\ v_{i+3} v_{i+4} ; i = 7,8 \\ v_{i+4} v_{i+5} ; i = 9,10 \end{cases}$$

$$f(e'_i) = \{v_{i-10} v_{i-7} ; i = 11,12,13,14,15,16,17,18,19,20,21,22$$

$$f(e''_i) = \{v_{i-19} v_{i-13} ; i = 23,24,25$$

This graph satisfies the condition of k-super mean labeling.

Therefore, G is called a k- super mean graph.

Theorem 2.3 :

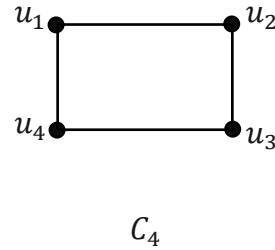
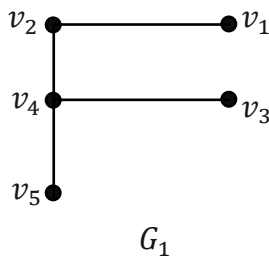
Let $L(G_1)$ and $L(G_2)$ be the line graphs of G_1 and C_4 . G is a cartesian product of $L(G_1)$ and $L(G_2)$. Then G is a k-super mean graph.

Proof:

Let G_1 be a given graph $|V(G_1)| = 5$ and $|E(G_1)| = 4$

Let C_4 be a given graph $|V(C_4)| = 4$ and $|E(C_4)| = 4$

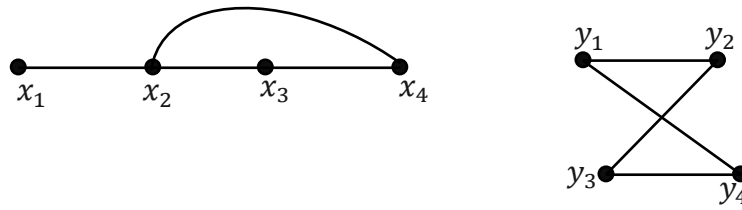
G_1 and C_4 graphs given below,



Let $L(G_1)$ be the line graph of G_1 such that $|V(L(G_1))| = 4$ and $|E(L(G_1))| = 4$

Let $L(G_2)$ be the line graph of C_4 such that $|V(L(G_2))| = 4$ and $|E(L(G_2))| = 4$

$L(G_1)$ and $L(G_2)$ graphs given below,



The Cartesian product $G = L(G_1) \square L(G_2)$

The vertex set of G is defined by

$$V(G) = \{v_i \mid 1 \leq i \leq 16\}$$

The edge set of G is defined by,

$$E(G) = \begin{cases} e_i; & 1 \leq i \leq 12 \\ e'_i; & 13 \leq i \leq 24 \\ e''_i; & 25 \leq i \leq 28 \\ e'''_i; & 29 \leq i \leq 32 \end{cases}$$

The cartesian product of $L(G_1)$ and $L(G_2)$ is given by,

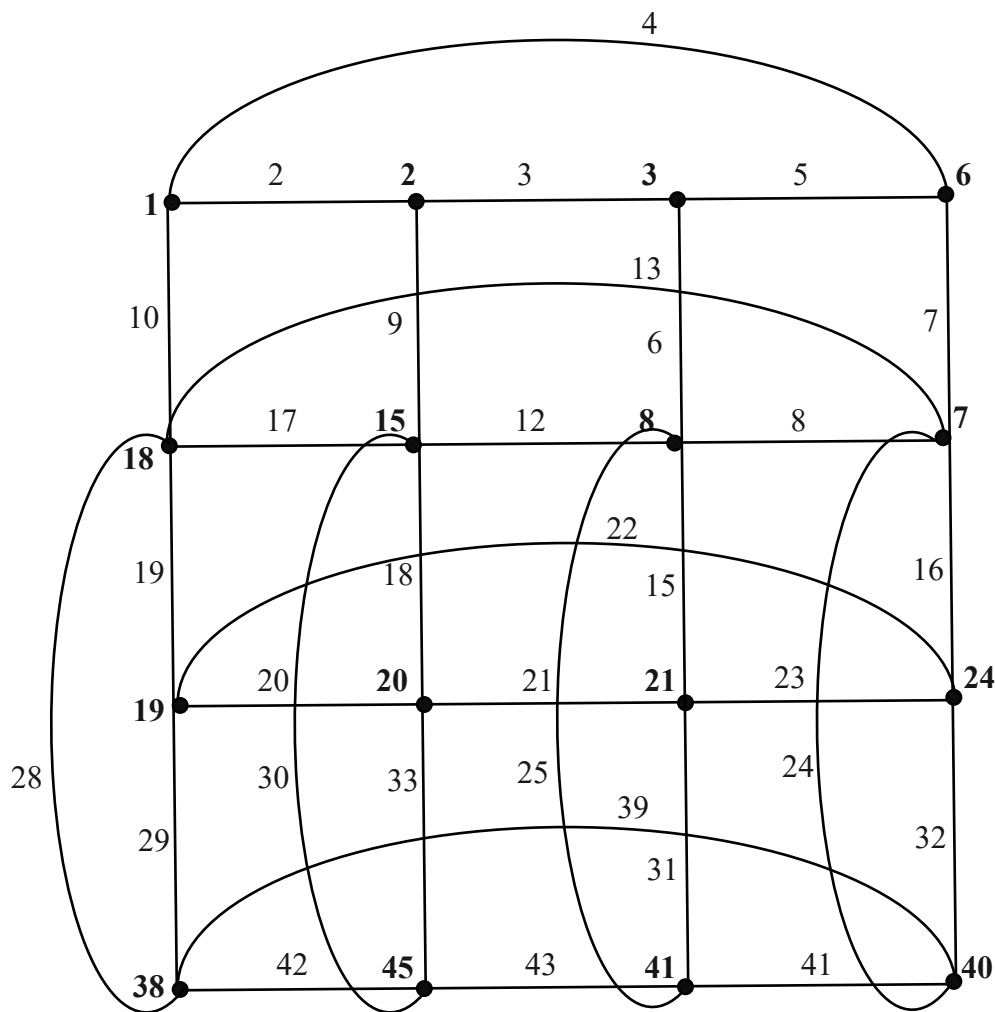


Fig no. 2.3

The k-super mean labeling pattern of vertex and edge set is given below,

$$KSML(v_i) = \begin{cases} i ; i = 1,2,3 \\ i + 2 ; i = 4,5,6 \\ 2i + 1 ; i = 7,9 \\ 2i + 2 ; i = 8 \\ 2i ; i = 10,12 \\ 2i - 1 ; i = 11 \\ 3i + 1 ; i = 13 \\ 3i - 1 ; i = 14 \\ 3i ; i = 15 \\ 2i + 6 ; i = 16 \end{cases}$$

$$KSML(e_i) = \begin{cases} i + 1 ; i = 1,2 \\ i + 2 ; i = 3,6,17 \\ 2i + 2 ; i = 5 \\ 4i + 1 ; i = 4 \\ 3i - 1 ; i = 7 \\ 2i + 5 ; i = 8,9 \\ 4i + 2 ; i = 10 \\ 4i - 1 ; i = 11 \\ 3i + 5 ; i = 12 \\ i - 3 ; i = 13 \\ i - 5 ; i = 14,27 \\ i - 9 ; i = 15,16 \\ i ; i = 18,30 \\ i - 4 ; i = 19,20 \\ i + 8 ; i = 21,23,24 \\ i + 10 ; i = 22 \\ i - 21 ; i = 25 \\ i - 13 ; i = 26 \\ i + 11 ; i = 28 \\ i - 1 ; i = 29 \\ i - 6 ; i = 31 \\ i - 8 ; i = 32 \end{cases}$$

The edges are labeled in the following pattern,

$$f(e_i) = \begin{cases} v_i v_{i+1} ; i = 1,2,3,5,6,12 \\ v_{i+3} v_{i+4} ; i = 4,11 \\ v_{i+2} v_{i+3} ; i = 7,8,9 \\ v_{i+5} v_{i+6} ; i = 10 \end{cases}$$

$$f(e'_i) = \begin{cases} v_{i-12}v_{i-5} & ; i = 13,21 \\ v_{i-12}v_{i-7} & ; i = 14,22 \\ v_{i-12}v_{i-9} & ; i = 15,23 \\ v_{i-12}v_{i-11} & ; i = 16,24 \\ v_{i-9}v_{i-8} & ; i = 17 \\ v_{i-11}v_{i-8} & ; i = 18 \\ v_{i-13}v_{i-8} & ; i = 19 \\ v_{i-15}v_{i-8} & ; i = 20 \end{cases}$$

$$f(e''_i) = \begin{cases} v_{i-24}v_{i-21} & ; i = 25 \\ v_{i-21}v_{i-18} & ; i = 26 \\ v_{i-18}v_{i-15} & ; i = 27 \\ v_{i-15}v_{i-12} & ; i = 28 \end{cases}$$

$$f(e'''_i) = \begin{cases} v_{i-21}v_{i-13} & ; i = 29 \\ v_{i-23}v_{i-15} & ; i = 30 \\ v_{i-25}v_{i-17} & ; i = 31 \\ v_{i-27}v_{i-19} & ; i = 32 \end{cases}$$

This graph satisfies the condition of k-super mean labeling.
Therefore, G is called a k- super mean graph.

Hence proved.

Conclusion:

In this paper, we have discussed the concept of k-super mean labeling, an extension of super mean labeling in graph theory. By allowing vertex labeling to begin with an arbitrary positive integer k , this labeling technique broadens the class of graphs that admit mean-based structures. The study of k-super mean labeling not only deepens our understanding of graph labeling but also highlights its potential applications in areas such as network design, communication systems, and combinatorial optimization.

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Vertex Even and Odd Mean Labelling for the Cartesian Product of Two-Line Graph

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ABSTRACT:

Graph labelling plays a vital role in identifying structural properties of graphs and has numerous applications in communications network, coding theory and scheduling. Among the various labelling schemes, mean labelling- introduced in the early 21st century assigns integer labels to the vertices of a graph so that edge corresponds to the mean of its endpoints. The paper focuses mainly on **even and odd mean labelling for the Cartesian product of two line graph**.

KEYWORDS:

Graph, Line graph, Cartesian product, Labelling, Even means labelling, Odd mean labelling.

1. INTRODUCTION:

The concept of Mean Labelling was introduced by **S. Somasundaram** and **R. Ponraj** in 2003. The even and odd labelling of graph variations emerged later in the 2000's.

Graph labelling is a well-studied area of graph theory in which numerical values are assigned to the vertices or edges of a graph according to specific rules. Such labelling helps reveal hidden structures, simplify complex networks and have applications in coding theory, network design and communication.

Among many labelling schemes, **mean labelling** assigns numbers to the vertices of a graph so that the label of each edge is the (integer part of the) average of its endpoints. Within this framework, two important variations are **even mean labelling** and **odd mean labelling**.

DEFINITION 1.1:

A **graph** G consists of a pair $(V(G), X(G))$ where $V(G)$ is a non-empty finite set whose elements are called **points** or **vertices** and $X(G)$ is a set of unordered pairs of distinct elements of $V(G)$. The elements of $X(G)$ are called the **lines** or **edges**.

DEFINITION 1.2:

A **line graph** $L(G)$ of a graph is another graph constructed as follows:

- Each vertices of $L(G)$ represents an edge of G .
- Two vertices of $L(G)$ are adjacent if and only if their corresponding edges in G share a common endpoint.

This transformation captures edge adjacency of the original graph.

DEFINITION 1.3:

The **Cartesian Product** of two graphs G_1 and G_2 can be written as $G_1 \square G_2$. Let $G_1 = (u_1, u_2)$ and $G_2 = (v_1, v_2)$ then the cartesian product of the graphs have the condition that u_1 and u_2 should be adjacent and have the same point v_1 or v_2

DEFINITION 1.4:

A graph G is said to be **even mean labelling** if there exists a vertex labelling $f: V(G) \rightarrow \mathbb{N}$ such that for every edge $uv \in E(G)$.

$$f * (uv) = \frac{f(u)+f(v)}{2} \text{ is even integer. That is all the edge means are even.}$$

DEFINITION 1.5:

A graph G is said to be **odd mean labelling** if there exists a vertex labelling $f: V(G) \rightarrow \mathbb{N}$ such that for every edge $uv \in E(G)$.

$$f * (uv) = \frac{f(u)+f(v)}{2} \text{ is an odd integer. That is, all the edge means are odd.}$$

2.1 MAIN THEOREM:

Theorem 1

Let H_1 and H_2 be two graphs and let G_1 and G_2 be the line graphs of these two graphs. Then $G = G_1 \square G_2$ is a even mean graph.

Proof:

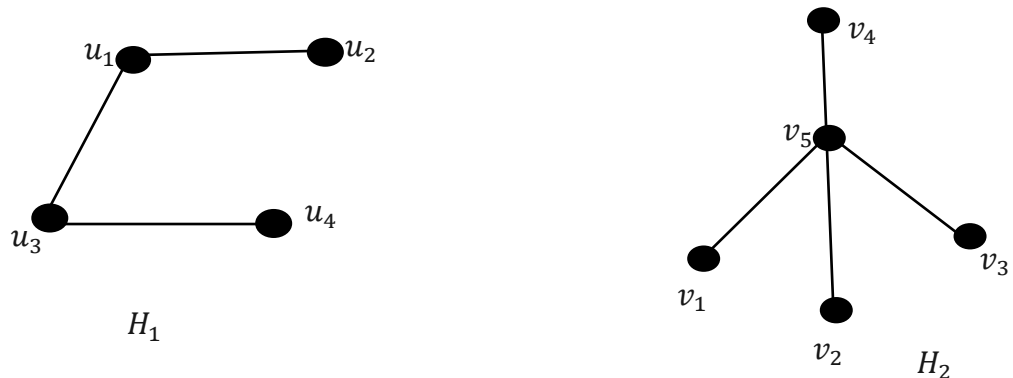
Let H_1 be a graph with vertex set, $|V(H_1)| = 4$ and the edge set, $|E(H_1)| = 3$

The line graph of H_1 has the vertex set, $|V(G_1)| = 3$ and the edge set, $|E(G_1)| = 2$

Let H_2 be a graph with the vertex set, $|V(H_2)| = 5$ and the edge set, $|E(H_2)| = 4$

The line graph of H_2 has the vertex set, $|V(G_2)| = 4$ and the edge set, $|E(G_2)| = 6$

Since G_1 and G_2 are the line graphs



The line graphs are



The cartesian product of the graphs G_1 and G_2 is

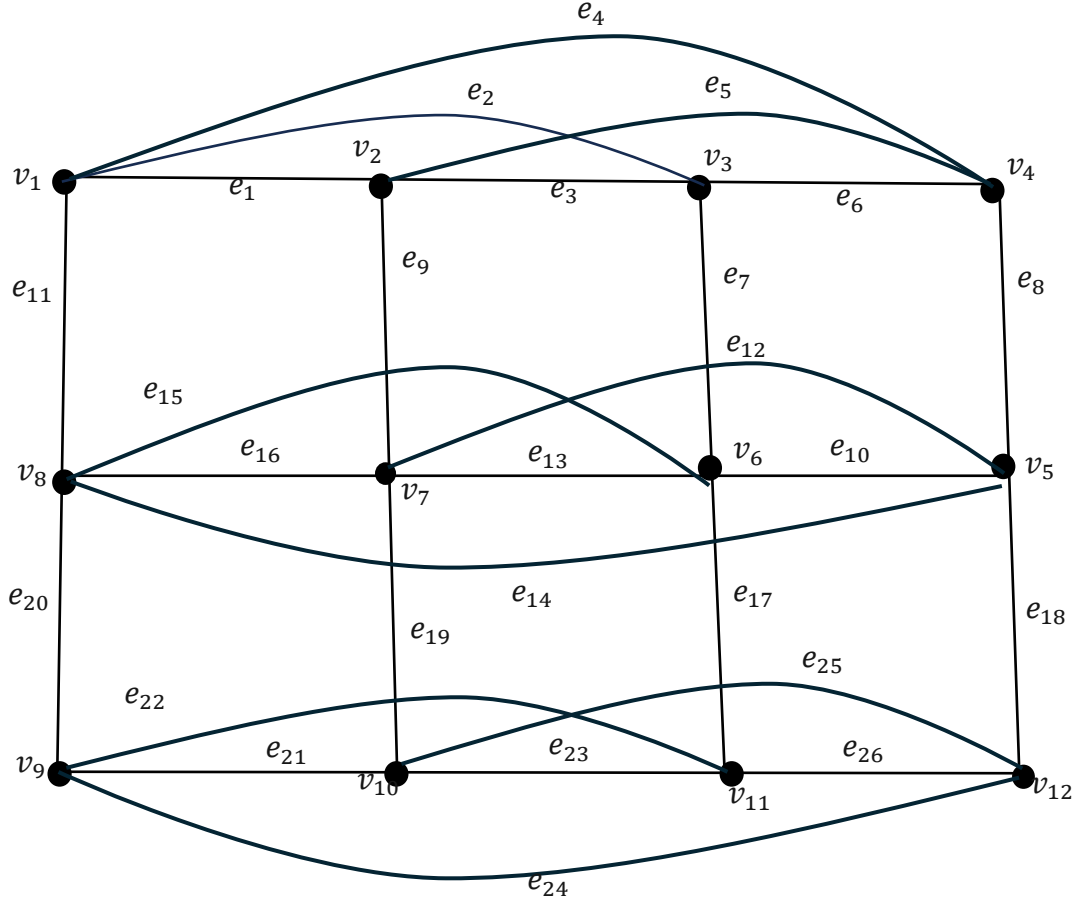
$$G_1 \square G_2 = \{x_1, x_2, x_3\} \square \{y_1, y_2, y_3, y_4\}$$

$$= \{(x_1, y_1), (x_1, y_2), (x_1, y_3), (x_1, y_4), (x_2, y_1), (x_2, y_2), (x_2, y_3), (x_2, y_4), (x_3, y_1), (x_3, y_2), (x_3, y_3), (x_3, y_4)\}$$

The vertex and edges set of the graph is given below,

$$V(G) = \{v_i / 1 \leq i \leq 12\}$$

$$E(G) = \{e_i / 1 \leq i \leq 26\}$$



The even mean labelling for the vertex set is given below,

$$V(G) = \begin{cases} 2i, i = 0, 1, 2 \\ 2i + 2, i = 3, 6, 7, 9, 11, 13, 14, 16, 24 \end{cases}$$

The even mean labelling of the edge set is given as,

$$E(G) = \begin{cases} i + 1, i = 0, 1, 2, 3, 4, 5, 6 \\ i + 2, i = 7 \\ i + 4, i = 8 \\ 2i - 3, i = 9 \\ i + 6, i = 10, 11, 12, 13, 14, 15, 16, 17 \\ i + 7, i = 18 \\ i + 8, i = 19 \\ i + 11, i = 20, 21 \\ i + 14, i = 25, 26, 28 \end{cases}$$

The vertex v_1 is adjacent to v_2, v_3, v_4, v_5 . The vertex v_2 is adjacent v_3, v_4, v_6 . The vertex v_3 is adjacent to v_1, v_2, v_4, v_7 . The vertex v_4 is adjacent to v_1, v_2, v_3 and v_8 . The vertex v_5 is adjacent to v_1, v_6, v_7, v_8, v_9 . The vertex v_6 is adjacent to v_5, v_7, v_8, v_{10} . The vertex v_7 is adjacent to v_5, v_6, v_8, v_{11} . The vertex v_8 is adjacent to v_5, v_6, v_7, v_{12} . The vertex v_9 is adjacent to $v_5, v_{10}, v_{11}, v_{12}$. The vertex v_{10} is adjacent to v_6, v_9, v_{11}, v_{12} . The vertex v_{11} is adjacent to v_7, v_9, v_{10}, v_{12} . The vertex v_{12} is adjacent to v_8, v_9, v_{10}, v_{11} .

This concludes that G satisfies an even mean labelling condition.
Therefore, G is a Even mean labelling graph.

Theorem 2

Let H_1 and H_2 be two graphs and let G_1 and G_2 be the line graphs of these two graphs. Then $G = G_1 \square G_2$ is a odd mean graph.

Proof:

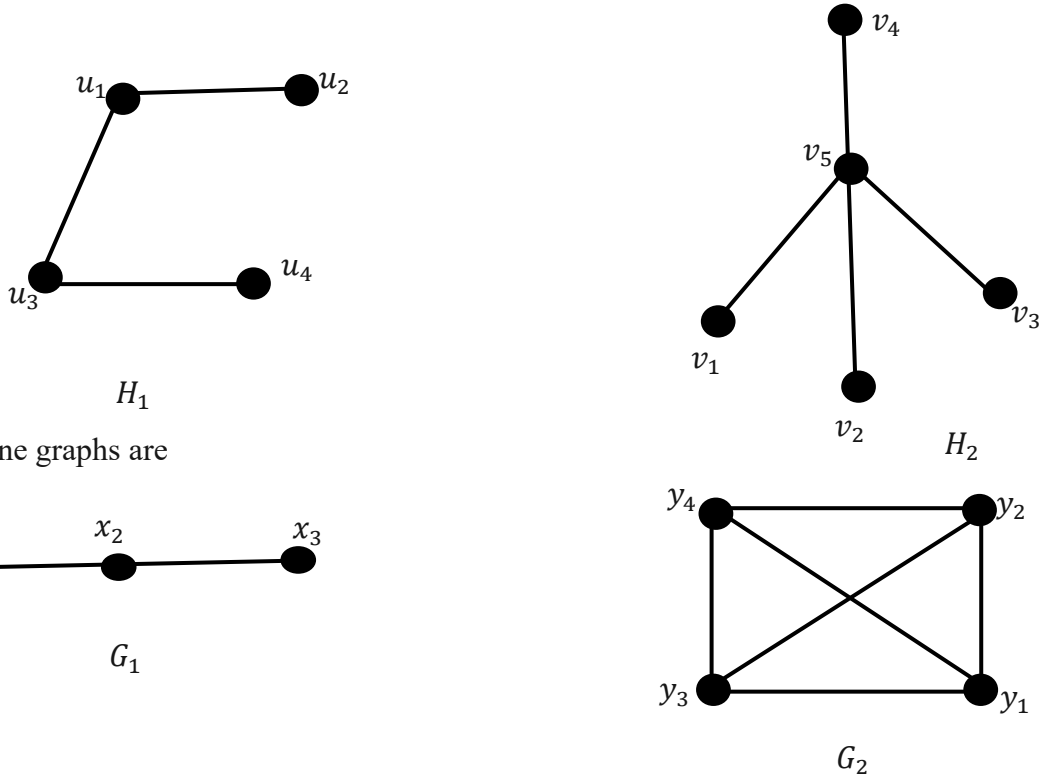
Let H_1 be a graph with vertex set, $|V(H_1)| = 4$ and the edge set, $|E(H_1)| = 3$

The line graph of H_1 has the vertex set, $|V(G_1)| = 3$ and the edge set, $|E(G_1)| = 2$

Let H_2 be a graph with the vertex set, $|V(H_2)| = 5$ and the edge set, $|E(H_2)| = 4$

The line graph of H_2 has the vertex set, $|V(G_2)| = 4$ and the edge set, $|E(G_2)| = 6$

Since G_1 and G_2 are the line graphs

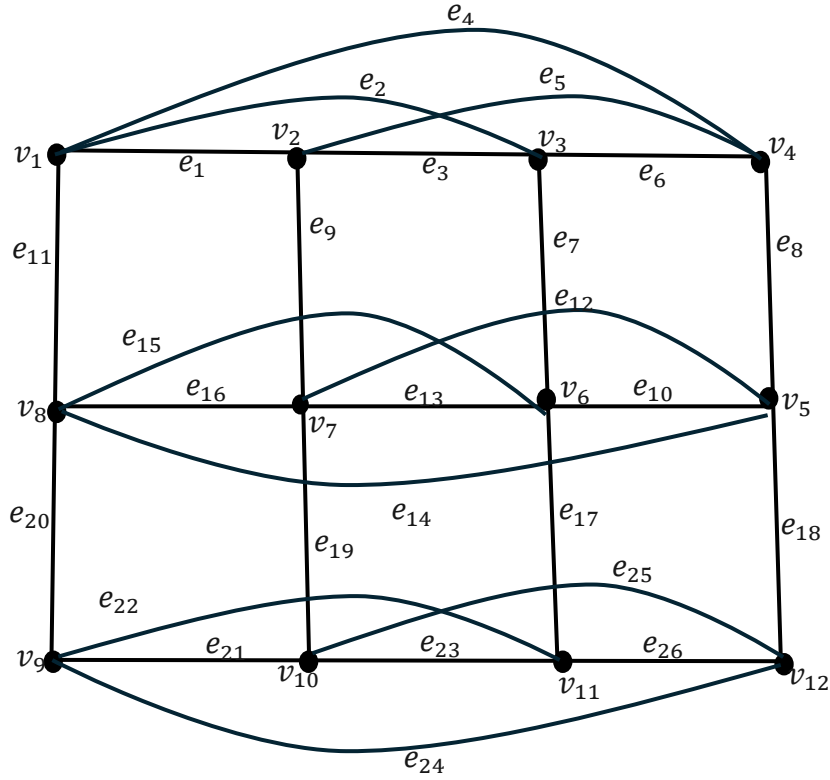


The cartesian product of the graphs G_1 and G_2 is

$$\begin{aligned} G_1 \square G_2 &= \{x_1, x_2, x_3\} \square \{y_1, y_2, y_3, y_4\} \\ &= \{(x_1, y_1), (x_1, y_2), (x_1, y_3), (x_1, y_4), (x_2, y_1), (x_2, y_2), (x_2, y_3), (x_2, y_4), (x_3, y_1), \\ &\quad (x_3, y_2), (x_3, y_3), (x_3, y_4)\} \end{aligned}$$

The vertex and edges set of the graph is given below,

$$\begin{aligned} V(G) &= \{v_i / 1 \leq i \leq 12\} \\ E(G) &= \{e_i / 1 \leq i \leq 26\} \end{aligned}$$



The odd mean labelling for the vertex set is given below,

$$V(G) = \begin{cases} i + 1, i = 0, 2, 4 \\ i + 2, i = 7, 9, 11 \\ i = 3, i = 16 \\ 2i + 1, i = 12 \\ 5i = 4, i = 5 \\ 6i - 5, i = 6 \\ 4i + 1, i = 8 \\ 3i - 2, i = 13 \end{cases}$$

The odd mean labelling of the edge set is given as,

$$E(G) = \begin{cases} 2i, i = 1 \\ i + 1, i = 2, 3, 4, 5, 6 \\ i + 2, i = 7, 8, 9, 10, 11 \\ i + 3, i = 12, 13 \\ i + 4, i = 14, 15 \\ i + 6, i = 16, 17, 18, 19 \\ i + 7, i = 17 \\ i + 12, i = 18, 19, 20, 21, 22, 23 \end{cases}$$

The vertex v_1 is adjacent to v_2, v_3, v_4, v_5 . The vertex v_2 is adjacent to v_3, v_4, v_6 . The vertex v_3 is adjacent to v_1, v_2, v_4, v_7 . The vertex v_4 is adjacent to v_1, v_2, v_3 and v_8 . The vertex v_5 is adjacent to v_1, v_6, v_7, v_8, v_9 . The vertex v_6 is adjacent to v_5, v_7, v_8, v_{10} . The vertex v_7 is adjacent to v_5, v_6, v_8, v_{11} . The vertex v_8 is adjacent to v_5, v_6, v_7, v_{12} . The vertex v_9 is adjacent to $v_5, v_{10}, v_{11}, v_{12}$. The vertex v_{10} is adjacent to v_6, v_9, v_{11}, v_{12} . The vertex v_{11} is adjacent to v_7, v_9, v_{10}, v_{12} . The vertex v_{12} is adjacent to v_8, v_9, v_{10}, v_{11} .

This concludes that G satisfies an odd mean labelling condition.

Therefore, G is an Odd mean labelling graph.

Theorem 3

Let G_1 and G_2 be two graphs and G be a cartesian product of G_1 and G_2 . Then G is a Even mean graph.

Proof:

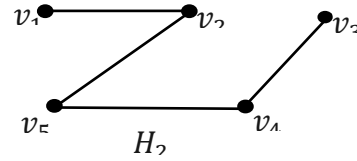
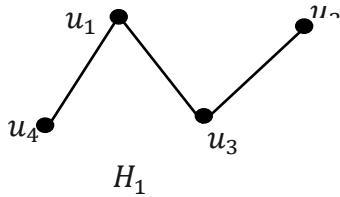
Let H_1 be a graph with vertex set, $|V(H_1)| = 4$ and the edge set, $|E(H_1)| = 3$

The line graph of H_1 has the vertex set, $|V(G_1)| = 3$ and the edge set, $|E(G_2)| = 2$

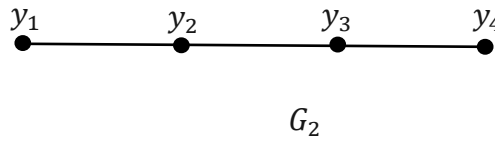
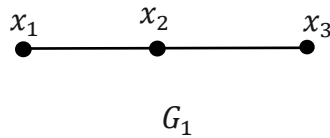
Let H_2 be a graph with the vertex set, $|V(H_2)| = 5$ and the edge set $|E(H_2)| = 4$

The line graph of H_2 has the vertex set, $|V(G_2)| = 4$ and the edge set, $|E(G_2)| = 3$

Since G_1 and G_2 are the line graphs.



The line graphs are



The cartesian product of the graphs G_1 and G_2 is

$$\begin{aligned} G_1 \square G_2 &= \{x_1, x_2, x_3\} \square \{y_1, y_2, y_3, y_4\} \\ &= \{(x_1, y_1), (x_1, y_2), (x_1, y_3), (x_1, y_4), (x_2, y_1), (x_2, y_2), (x_2, y_3), (x_2, y_4), (x_3, y_1), \\ &\quad (x_3, y_2), (x_3, y_3), (x_3, y_4)\} \end{aligned}$$

The vertex and edges set of the graph is given below,

$$\begin{aligned} V(G) &= \{v_i / 1 \leq i \leq 12\} \\ E(G) &= \{e_i / 1 \leq i \leq 17\} \end{aligned}$$

The even mean labelling for the vertex set is,

$$V(G) = \{2i, 0 \leq i \leq 11\}$$

The even mean labelling for the edge set is,

$$E(G) = \begin{cases} 2i + 1, 0 \leq i \leq 2 \\ i - 5, i = 9, 11, 13, 15 \\ 2i + 3, 3 \leq i \leq 5 \\ i = 6, i = 6, 8, 10, 12 \\ i + 1, i = 16, 18, 20 \end{cases}$$

The vertex v_1 is adjacent to v_1, v_5 . The vertex v_2 is adjacent v_3, v_1, v_6 . The vertex v_3 is adjacent to v_2, v_4, v_7 . The vertex v_4 is adjacent to v_3, v_8 . The vertex v_5 is adjacent to v_1, v_6, v_9 . The vertex v_6 is adjacent to v_5, v_7, v_2, v_{10} . The vertex v_7 is adjacent to v_3, v_6, v_8, v_{11} . The vertex v_8 is adjacent to v_6, v_7, v_{12} . The vertex v_9 is adjacent to v_8, v_{10} . The vertex v_{10} is adjacent to v_6, v_9, v_{11} . The vertex v_{11} is adjacent to v_7, v_{10}, v_{12} . The vertex v_{12} is adjacent to v_8, v_{11} .

This concludes that G satisfies an even mean labelling condition.

Therefore, G is a Even mean labelling graph.

Theorem 4

Let G_1 and G_2 be two graphs and G be a cartesian product of G_1 and G_2 . Then G is a Odd mean graph.

Proof:

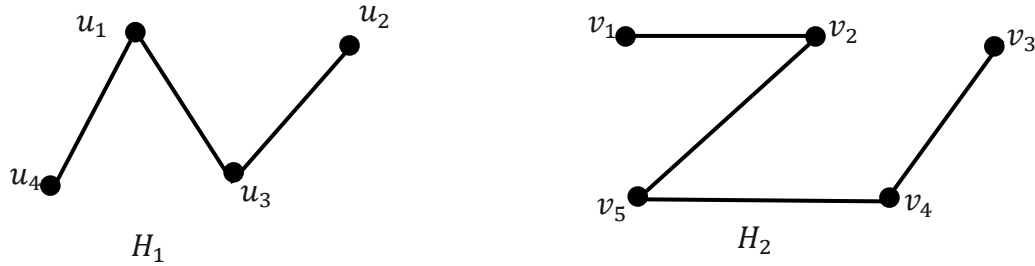
Let H_1 be a graph with vertex set, $|V(H_1)| = 4$ and the edge set, $|E(H_1)| = 3$

The line graph of H_1 has the vertex set, $|V(G_1)| = 3$ and the edge set, $|E(G_2)| = 2$

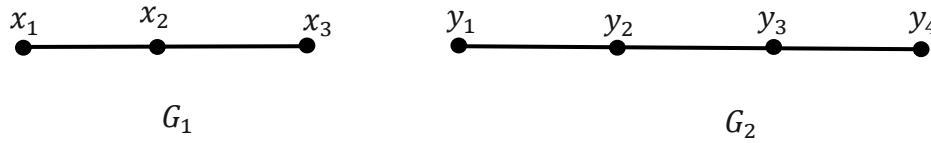
Let H_2 be a graph with the vertex set, $|V(H_2)| = 5$ and the edge set $|E(H_2)| = 4$

The line graph of H_2 has the vertex set, $|V(G_2)| = 4$ and the edge set, $|E(G_2)| = 3$

Since G_1 and G_2 are the line graphs.



The line graphs are



The cartesian product of the graphs G_1 and G_2 is

$$\begin{aligned} G_1 \square G_2 &= \{x_1, x_2, x_3\} \square \{y_1, y_2, y_3, y_4\} \\ &= \{(x_1, y_1), (x_1, y_2), (x_1, y_3), (x_1, y_4), (x_2, y_1), (x_2, y_2), (x_2, y_3), (x_2, y_4), (x_3, y_1), \\ &\quad (x_3, y_2), (x_3, y_3), (x_3, y_4)\} \end{aligned}$$

The vertex and edges set of the graph is given below,

$$V(G) = \{v_i / 1 \leq i \leq 12\}$$

$$E(G) = \{e_i / 1 \leq i \leq 17\}$$

The odd mean labeling for the vertex set is

$$V(G) = \{i + 1, 0 \leq i \leq 22\}$$

The odd mean labelling for the edge set is,

$$E(G) = \begin{cases} 2i, i = 1, 2, 3 \\ i + 1, i = 4, 6, 8, 10 \\ i + 5, i = 5, 7, 9 \\ i + 2, i = 11, 13, 15, 17 \\ i + 6, i = 12, 14, 16 \end{cases}$$

The vertex v_1 is adjacent to v_1, v_5 . The vertex v_2 is adjacent to v_3, v_1, v_6 . The vertex v_3 is adjacent to v_2, v_4, v_7 . The vertex v_4 is adjacent to v_3, v_8 . The vertex v_5 is adjacent to v_1, v_6, v_9 . The vertex v_6 is adjacent to v_5, v_7, v_2, v_{10} . The vertex v_7 is adjacent to v_3, v_6, v_8, v_{11} . The vertex v_8 is adjacent to v_4, v_7, v_{12} . The vertex v_9 is adjacent to v_8, v_{10} . The vertex v_{10} is adjacent to v_6, v_9, v_{11} . The vertex v_{11} is adjacent to v_7, v_{10}, v_{12} . The vertex v_{12} is adjacent to v_8, v_{11} .

This concludes that G satisfies an odd mean labelling condition.

Therefore, G is a Odd mean labelling graph.

CONCLUION:

Even and odd mean labelling are special types of graphs labelling where the mean of the labels at the end of each of the edges is forced to be even or odd. Studying these labelling's helps us understand new properties of graphs and gives ideas for applications in networks and coding. Our work shows which graphs can have these labelling's and lays a base for further study.

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K -Domination Parameters of Some Graphs

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ABSTRACT:

The domination parameters of a graph G of order n has been already introduced in it is defined as $D \subseteq V(G)$ is a dominating set of G , if every vertex $v \in V-D$ is adjacent to atleast one vertex in D . In this paper we established K -domination parameters of some graphs such as path, cycle, the product of cycle and path graphs and C_n the centipede graph with ' $2n$ ' vertices. Also established the general formula.

INTRODUCTION

A graph $G = (V, E)$, where V is a finite set of elements called vertices and E is a set of unordered pairs of distinct vertices of G called edges. The degree of a vertex v in G is the number of edges incident on it. A graph G is said to be k -regular if all its vertices are of degree k . Every pair of its vertices are adjacent in G , is said to be complete, the complete graph on ' n ' vertices is denoted by K_n .

A graph G is said to be bipartite or bigraph if the vertex set of $V(G)$ can be partitioned in to two subsets X and Y such that every edge of G has one in X and the other end in Y . A bipartite graph G with $|X| = m$ and $|Y| = n$ is said to be complete if every element in one

partition is adjacent with all elements of the other partition and is denoted by $K_{m,n}$. The graph $K_{1,n}$ is called a Star graph.

Let u , and v be the vertices of a graph G , a u - v walk of G is an alternating sequences $u = u_0, e_1, u_1, u_2, \dots, u_{n-1}, e_n, u_n = v$ of vertices and edges beginning with vertex u and ending with vertex v such that $e_i = u_{i-1}u_i$, for all $i = 1, 2, \dots, n$. The number of edges in a walk is called its length. A walk in which all the vertices are distinct is called a path. A path on ' n ' vertices is denoted by P_n . A closed path is called a cycle, a cycle on ' n ' vertices is denoted by C_n . Let $G = (V, E)$ be a simple connected graph, for any vertex $v \in V$, the open neighborhood is the set $N(v) = \{u \in V / u v \in E\}$ and the closed neighborhood of v is the set $N[v] = N(v) \cup \{v\}$. For a set $S \subset V$, the open neighborhood of S is $N(s) = \bigcup_{v \in S} N(v)$ and the closed neighborhood of S is $N[S] = N(S) \cup S$.

Definition 1.1

A set $D \subseteq V$ is a dominating set of G if every vertex $v \in V - D$ is adjacent to atleast one vertex of D . We call a dominating set D is a minimal if there is no dominating set $D' \subseteq V(G)$ with $D' \subset D$ and $D' \neq D$. Further we call a dominating set D is minimum if there is no dominating set $D' \subseteq V(G)$ with $|D'| < |D|$. The cardinality of a minimum dominating set is called the domination number denoted by $\gamma(G)$ and the minimum dominating set D of G is also called a γ - set.

Definition 1.2 :

A dominating set D is said to be a total dominating set if every vertex in V is adjacent to some vertex in D . The total domination number of G , denoted by $\gamma_t(G)$ is the minimum cardinality of a total dominating set.

Definition 1.3:

A dominating set D of a graph G is an independent dominating set, if the induced subgraph $\langle D \rangle$ has no edges. The independent domination number $\gamma_i(G)$ is the minimum cardinality of a independent dominating set.

Definition 1.4:

A dominating Set D is said to be connected dominating set, if the induced subgraph $\langle D \rangle$ is connected. The connected domination number $\gamma_c(G)$ is the minimum cardinality of a connected dominating set.

Definition 1.5

A dominating Set D of a graph G is said to be a paired dominating set if the induced subgraph $\langle D \rangle$ contains atleast one perfect matching, paired domination number $\gamma_p(G)$ is the minimum cardinality of a paired dominating set.

Definition 1.6

A dominating Set D of G is a split dominating set if the induced subgraph $\langle V - D \rangle$ is disconnected. Split domination number $\gamma_s(G)$ is the minimum cardinality of a split dominating set.

Definition 1.7

A dominating Set D of G is a non split dominating set, if the induced sub graph $\langle V - D \rangle$ is connected. Non split domination number $\gamma_{ns}(G)$ is the minimum cardinality of a non split dominating set.

Definition 1.8

Let D be a γ - set of G . A dominating set D^1 contained in $V - D$ is called an inverse dominating set of G with respect to D . The inverse domination number $\gamma'(G)$ is the minimum cardinality of all inverse dominating set of G , the vertices of $\gamma'(G)$ is called γ' - set.

Definition 1.9

A dominating set D of a graph G is called a global dominating set, if D is also a dominating set of $\overline{G}$. The global domination number $\gamma_g(G)$ is the minimum cardinality of a global dominating set.

Definition 1.10

A dominating set D is called a perfect dominating set, if every vertex in $V - D$ is adjacent to exactly one vertex in D . The perfect domination number $\gamma_{pr}(G)$ is the minimum cardinality of a perfect dominating set.

Definition : 1.11

If $D = \{x\}$ is a dominating set of G , then x is called a dominating vertex of G . A vertex $v \in V(G)$ is said to be a γ - required vertex of G , if v lies in every γ - set of G .

Definition : 1.12

The k^{th} power of a graph G is a graph with the same set of vertices of G and an edge between two vertices if there is a path of length almost k between them. G^2 is called the square of G , G^3 is called the cube of G etc.

Definition: 1.13

Let G and H be two graphs the tensor product of G and H is denoted by $G \times H$ the vertices of $G \times H$ is the cartesian product $V(G) \times V(H)$ and any two vertices (u, u') and (v, v') are adjacent in $G \times H$ iff u' is adjacent with v' and u is adjacent with v , for all $u, v \in G$ and $u', v' \in H$.

Definition: 1.14

A subset $D \subseteq V(G)$ is called a distance k -dominating set of G if every vertex in $V(G) - D$ is within distance k from some vertex of D . The minimum cardinality among all distance k -dominating set of G is called the distance k -domination number denoted by $\gamma_k(G)$.

Definition: 1.15

Let G be any graph the r^{th} power of G is denoted by G^r and is obtained from G by joining all the vertices $v_i \in G$ whose distance is at most r from each vertices of G .

Definition: 1.16

The n centipede is the tree on $2n$ nodes obtained by joining the bottoms of n copies of the path graph p_2 laid in a row with edges denoted by

Lemma: 1.17

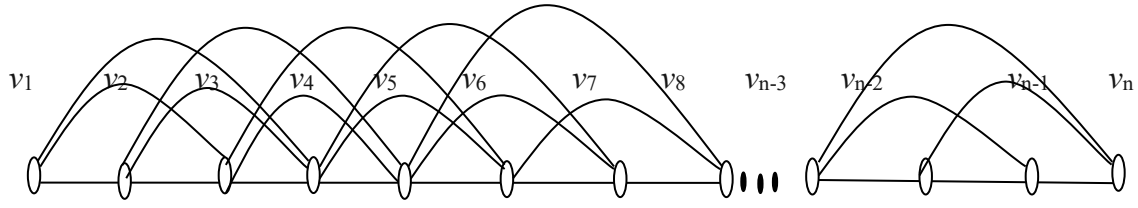
Let G be a connected graph with $\delta(G) \geq 2$, then $\gamma(G) + \gamma'(G) = n$ if and only if $G = P_4$ or C_4 .

Theorem: 1.1.1

Let P_n be a Path graph with 'n' vertices. If $G = (P_n)^r$ then $\gamma_k(G) = \left\lceil \frac{n}{2rk+1} \right\rceil$, where $1 \leq r \leq \left\lceil \frac{n}{2} \right\rceil$.

Proof:

Let P_n be a Path graph with 'n' vertices. Let $G = (P_n)^r$.



The vertex set of G is denoted by $V = \{v_1, v_2, v_3, \dots, v_n\}$. Now divide the vertices of G into m sets S_i , $1 \leq i \leq m$ such that each set consists of $2rk+1$ vertices.

Therefore

$$\begin{aligned} S_1 &= \{v_1, v_2, \dots, v_{2rk+1}\} \\ S_2 &= \{v_{2rk+2}, v_{2rk+3}, \dots, v_{4rk+2}\} \\ S_3 &= \{v_{4rk+3}, v_{4rk+4}, \dots, v_{8rk+3}\} \cdot \quad \dots \quad \dots \quad \dots \\ S_{m-1} &= \{v_{2(m-2)rk+(m-1)}, \dots, v_{2(m-1)rk+(m-1)}\} \\ S_m &= \{v_{2(m-1)rk+m}, \dots, v_n\}. \end{aligned}$$

Case (i) If $n \equiv 0 \pmod{2rk+1}$, where $1 \leq r \leq \lfloor \frac{n}{2} \rfloor$ and k is the k -distance domination of G . Then $|S_i| = 2rk+1$, for all $1 \leq i \leq m$, then collect all the middle vertices from each S_i , $1 \leq i \leq m$.

$$\begin{aligned} \text{Therefore, } D_k(G) &= \{v_{(2rk+1)i - rk} / 1 \leq i \leq m\} \text{ is the required } k\text{-dominating set of } G \text{ and} \\ |D_k(G)| &= m = \frac{n}{2rk+1} \\ &= \lfloor \frac{n}{2rk+1} \rfloor \quad [\text{since } n \text{ is a multiple of } 2rk+1]. \end{aligned}$$

Case (ii) If $n \not\equiv 0 \pmod{2rk+1}$, where $1 \leq r \leq \lfloor \frac{n}{2} \rfloor$ and k is the k -distance domination of G . Then the last partition S_m containing less than $2rk+1$ vertices, choose

$$\begin{aligned} p &= \frac{2(m-1)rk+m+n}{2} \quad \text{Now, } D_k(G) = \{\{v_{(2rk+1)i - rk} / 1 \leq i \leq m-1\} \cup \{v_p\}\} \text{ is the required } k\text{-} \\ \text{dominating set of } G \text{ and } |D_k(G)| &= m-1+1 = m = \lfloor \frac{n}{2rk+1} \rfloor \\ \Rightarrow \gamma_k(G) &= \lfloor \frac{n}{2rk+1} \rfloor. \end{aligned}$$

Theorem: 1.1.2

Let C_n be a cycle graph with 'n' Vertices. If $G = (C_n)^r$ then $\gamma_k(G) = \lfloor \frac{n}{2rk+1} \rfloor$,

Where $1 \leq r \leq \lfloor \frac{n}{2} \rfloor$.

Proof: Let C_n be a cycle graph $(C_n)^3$ with 'n' vertices, Let $G = (C_n)^r$.

The vertex set of G is denoted by $V = \{v_1, v_2, v_3, \dots, v_n\}$. Each vertex $v_i \in G$ is adjacent with $2r$ vertices. Now divide the vertices of G into m sets S_i , $1 \leq i \leq m$ such that each set consists of $2rk+1$ vertex. Therefore,

$$\begin{aligned} S_1 &= \{v_1, v_2, \dots, v_{2rk+1}\} \\ S_2 &= \{v_{2rk+2}, v_{2rk+3}, \dots, v_{4rk+2}\} \\ S_3 &= \{v_{4rk+3}, v_{4rk+4}, \dots, v_{8rk+3}\} \\ &\dots \quad \dots \quad \dots \\ S_{m-1} &= \{v_{2(m-2)rk+(m-1)}, \dots, v_{2(m-1)rk+(m-1)}\} \\ S_m &= \{v_{2(m-1)rk+m}, \dots, v_n\}. \end{aligned}$$

Case (i) If $n \equiv 0 \pmod{2rk+1}$, where $1 \leq r \leq \lfloor \frac{n}{2} \rfloor$ and k is the k -distance domination of G . Then $|S_i| = 2rk+1$, for all

$1 \leq i \leq m$, then collect all the middle vertices from each S_i , $1 \leq i \leq m$. Therefore, $D_k(G) = \{v_{(2rk+1)i-rk} / 1 \leq i \leq m\}$ is the required k -dominating set of G and $|D_k(G)| = m = \frac{n}{2rk+1} = \left\lceil \frac{n}{2rk+1} \right\rceil$ [since n is a multiple of $2rk+1$].

Case (ii) If $n \not\equiv 0 \pmod{2rk+1}$, where $1 \leq r \leq \left\lfloor \frac{n}{2} \right\rfloor$ and k is the k -distance domination of G . Then the last partition S_m containing less than $2rk+1$ vertices, choose

$p = \frac{2(m-1)rk+m+n}{2}$ Now, $D_k(G) = \{\{v_{(2rk+1)i-rk} / 1 \leq i \leq m-1\} \cup \{v_p\}\}$ is the required k -dominating set of G and $|D_k(G)| = m-1+1 = m = \left\lceil \frac{n}{2rk+1} \right\rceil \Rightarrow \gamma_k(G) = \left\lceil \frac{n}{2rk+1} \right\rceil$.

Result: 1.1.3

Let $G = (C_n)^r$ be the graph with $\gamma_k(G) = \left\lceil \frac{n}{2rk+1} \right\rceil$ then $\gamma_k(G, p) = \gamma_k(G, q)$, where $p=q=rk$. **Proof:** We know that $\gamma_k(G) = \left\lceil \frac{n}{2rk+1} \right\rceil \Rightarrow \gamma_k(G, p) = \left\lceil \frac{n}{2p+1} \right\rceil$ [since $p = rk$]
 $= \left\lceil \frac{n}{2q+1} \right\rceil = \gamma_k(G, q)$. [Since $p = q$]

Theorem: 1.1.4

Let $G = (C_n \times P_m)$, $m > 1$ be the graph with ‘ mn ’ vertices. Then the distance k -domination number of G is $\gamma_k(G) = \left\lceil \frac{n}{4k} \right\rceil$, where $1 \leq r \leq \left\lfloor \frac{n}{2} \right\rfloor$.

Proof:

Let $G = (C_n \times P_m)$, $m > 1$ be the graph with ‘ mn ’ vertices is given in figure 5.3

Now the vertex set of G is denoted by

$$V = \{v_{i,j} / 1 \leq i \leq m, 1 \leq j \leq n\}.$$

Each vertex $v_i \in G$ is adjacent with at most 4 vertices that is

$$d(v_{1,j}) = d(v_{m,j}) = 3, 1 \leq j \leq n$$

$d(v_{1,j}) = d(v_{m,j}) = 3, 1 \leq j \leq n$ and $d(v_{i,j}) = 4, 2 \leq i \leq m-1, 1 \leq j \leq n$. Now divide the vertices of G into m sets S_i , $1 \leq i \leq m$ such that

$$S_1 = \{v_{i,j} / 1 \leq j \leq n\}$$

$$S_2 = \{v_{2,j} / 1 \leq j \leq n\}$$

$$S_3 = \{v_{3,j} / 1 \leq j \leq n\}$$

$$\dots \dots \dots$$

$$S_{m-1} = \{v_{m-1,j} / 1 \leq j \leq n\}$$

$$S_m = \{v_{m,j} / 1 \leq j \leq n\}.$$

Case (i) If $n \equiv 0 \pmod{4k}$, where

k is the k -distance domination of G .

Then $|S_i| = n$, for all $1 \leq i \leq m$, then collect all the vertices from each S_i , $1 \leq i \leq m$ as follows:

- Let $t = \left\lceil \frac{n+1}{2} \right\rceil$. If m is odd then collect all vertices $\{v_{i,t} / i = 1, 3, 5, \dots, m\}$.
- ii) If m is even then collect all vertices $\{v_{i,t+2k} / i = 2, 4, 6, \dots, m\}$.
 $\Rightarrow D_k(G) = \{\{v_{m,k(4i-3)+1} / 1 \leq i \leq m, \text{ where } m \text{ is odd}\}$

$\cup \{v_{m,k(4i-1)+1} / 1 \leq i \leq m, \text{ where } m \text{ is even}\}$ is the required k -dominating set of G and $|D_k(G)| = \frac{m}{2} = m = \frac{n}{4k} = \left\lceil \frac{n}{4k} \right\rceil$ [since n is a multiple of $4k$].

Case (ii) If $n \not\equiv 0 \pmod{4k}$, where k is the k -distance domination of G . Then the last partition S_m containing less than n vertices, choose

$$p = \frac{2(m-1)rk+m+n}{2}$$

$$\Rightarrow D_k(G) = \{\{v_{m,k(4i-3)+1} / 1 \leq i \leq m, \text{ where } m \text{ is odd}\} \cup \{v_{m,k(4i-1)+1} / 1 \leq i \leq m, \text{ where } m \text{ is even}\} \cup \{v_p\}\}$$

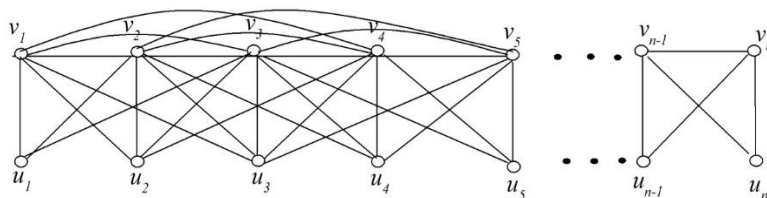
is the required k -dominating set of G and

$$|D_k(G)| = \frac{m-1}{2} + \frac{m-1}{2} + 1 = m = \left\lceil \frac{n}{4k} \right\rceil \Rightarrow \gamma_k(G) = \left\lceil \frac{n}{4k} \right\rceil.$$

Theorem: 1.1.5

Let C_n be the centipede graph with '2n' vertices. If $G = (C_n)^r$ then $\gamma_k(G) = \left\lceil \frac{n}{2rk-1} \right\rceil$, where $1 \leq r \leq \left\lfloor \frac{n}{2} \right\rfloor$.

Proof: Let $G = (C_n)^r$ be the r^{th} power of the centipede graph with '2n' vertices.



The vertex set of G is denoted by $V = V_1 \cup V_2$, where

$V_1 = \{v_i / 1 \leq i \leq n\}, V_2 = \{u_i / 1 \leq i \leq n\}$. Now divide the vertices of V_1 into m sets $S_i, 1 \leq i \leq m$ such that each set consists of $2rk-1$ vertices. Therefore,

$$S_1 = \{v_1, v_2, \dots, v_{2rk-1}\}$$

$$S_2 = \{v_{(2rk-1)+1}, v_{(2rk-1)+2}, \dots, v_{4rk-2}\}$$

$$S_3 = \{v_{(4rk-2)+1}, v_{(4rk-2)+2}, \dots, v_{6rk-3}\} \quad \cdot \quad \cdot \quad \cdot \quad \cdot$$

$$S_{m-1} = \{v_{(m-2)(2rk-1)+1}, v_{(m-2)(2rk-1)+2}, \dots, v_{(m-1)(2rk-1)}\}$$

$$S_m = \{v_{(m-1)(2rk-1)+1}, v_{(m-1)(2rk-1)+2}, \dots, v_n\}.$$

Case (i) If $n \equiv 0 \pmod{2rk-1}$, where $1 \leq r \leq \left\lfloor \frac{n}{2} \right\rfloor$ and k is the k -distance domination of G . Then

$|S_i| = 2rk-1$, for all $1 \leq i \leq m$, then collect all the middle vertices from each $S_i, 1 \leq i \leq m$

Therefore, $D_k(G) = \{v_{(2rk-1)i - (rk-1)} / 1 \leq i \leq m\}$ is the required k -dominating set of G and

$$\begin{aligned} |D_k(G)| &= m = \frac{n}{2rk-1} \\ &= \left\lceil \frac{n}{2rk-1} \right\rceil \quad [\text{since } n \text{ is a multiple of } 2rk-1]. \end{aligned}$$

Case (ii) If $n \not\equiv 0 \pmod{2rk-1}$, where $1 \leq r \leq \left\lfloor \frac{n}{2} \right\rfloor$ and k is the k -distance domination of G . Then the last partition S_m containing less than $2rk-1$ vertices,

Choose $p = \frac{(m-1)(2rk-1)+1+n}{2}$ Now, $D_k(G) = \{\{v_{(2rk-1)i - (rk-1)} / 1 \leq i \leq m-1\} \cup \{v_p\}\}$ is the required k -dominating set of G and

$$\begin{aligned} |D_k(G)| &= m-1+1 = m = \left\lceil \frac{n}{2rk-1} \right\rceil \\ \Rightarrow \gamma_k(G) &= \left\lceil \frac{n}{2rk-1} \right\rceil. \end{aligned}$$

Conclusion

The theory of domination parameter is used as models for communications and electrical networks, computer architectures and network optimization. We have found the k -domination numbers of some graphs such as path, cycle, $(C_n \times P_m)$ the product of cycle and path graphs and $(C_n)^r$ the power of cycle and the centipede graph with '2n' vertices.

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Sum Divisor Cordial labeling for Cartesian product of line Graphs

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ABSTRACT:

Graph labelling plays a vital role in identifying structural properties of graphs and has numerous applications in communications network, coding theory and scheduling. Among the various labelling schemes, mean labelling- introduced in the early 21st century assigns integer labels to the vertices of a graph so that edge corresponds to the mean of its endpoints. The paper focuses mainly on sum divisor cordial labeling for the Cartesian product of two line graph.

KEYWORDS:

Graph, Line graph, Cartesian product, Labeling, sum divisor cordial labeling

1. INTRODUCTION:

NOTATION

Let $e_{f(0)}$ denotes the number of edges labelled with 1 and $e_{f(0)}$ denotes the number of edges labelled with 0.

DEFINITION 1.1:

A **graph** G consists of a pair $(V(G), X(G))$ where $V(G)$ is a non-empty finite set whose elements are called **points** or **vertices** and $X(G)$ is a set of unordered pairs of distinct elements of $V(G)$. The elements of $X(G)$ are called the **lines** or **edges**.

DEFINITION 1.2:

A **line graph** $L(G)$ of a graph is another graph constructed as follows:

- Each vertices of $L(G)$ represents an edge of G .
- Two vertices of $L(G)$ are adjacent if and only if their corresponding edges in G share a common endpoint.

This transformation captures edge adjacency of the original graph.

DEFINITION 1.3:

The Cartesian Product of two graphs G_1 and G_2 can be written as $G_1 \square G_2$. Let $G_1 = (u_1, u_2)$ and $G_2 = (v_1, v_2)$, then the cartesian product of the graphs have the condition that u_1 and u_2 should be adjacent and have the same point v_1 or v_2 .

DEFINITION 1.4:

A Sum divisor cordial labeling of a graph G with vertex set V is a bijection $f : V \rightarrow \{1, 2, \dots, |V(G)|\}$ such that an edge uv is assigned the label 1 if 2 divides $f(u) + f(v)$ and 0 otherwise; and the number of edges labeled with 0 and the number of edges labeled with 1 differ by at most 1. The function f is called a sum divisor cordial labeling if $|e_f(0) - e_f(1)| \leq 1$. A graph which admits a sum divisor cordial labeling is called a sum divisor cordial graph.

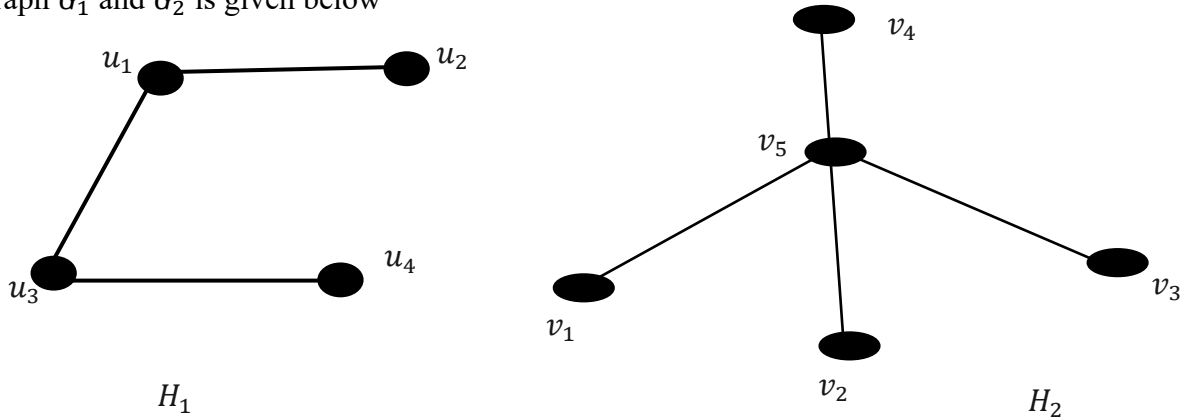
THEOREM

$L(G_1)$ and $L(G_2)$ are line graph of G_1 and G_2 . then Cartesian product of $L(G_1)$ and $L(G_2)$ is sum divisor cordial graph.

Proof

Let G_1 and G_2 be a two graph $|v(G_1)| = 4, |v(G_2)| = 5$

Graph G_1 and G_2 is given below



The line graph of $L(G_1)$ and $L(G_2)$ be two graph with $L(G_1) = 3, L(G_2) = 4$

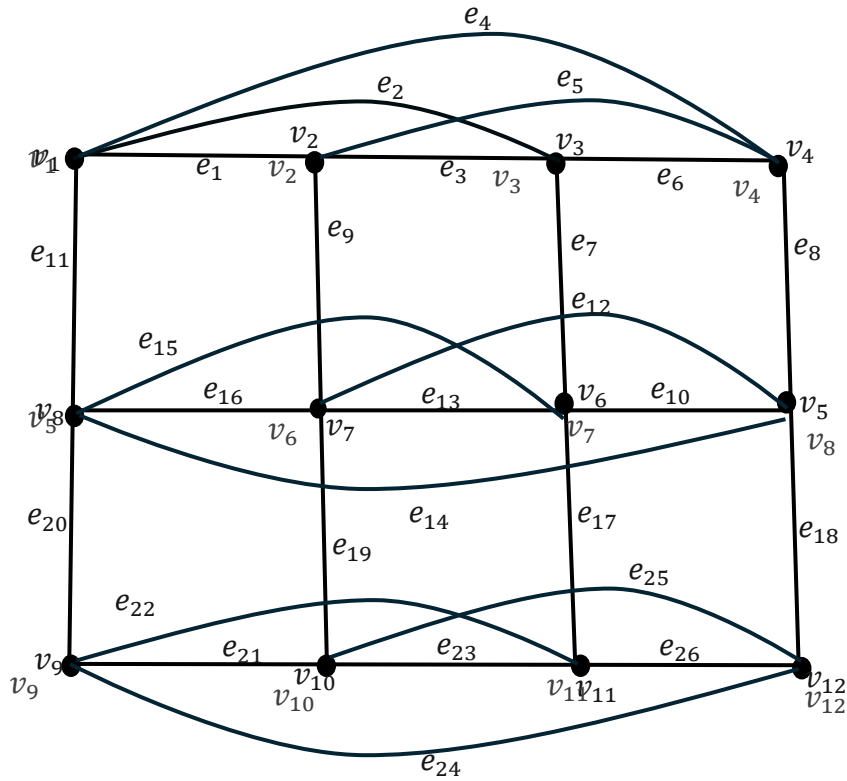
The Cartesian product of the graphs G_1 and G_2 is

$$\begin{aligned} G_1 \square G_2 &= \{x_1, x_2, x_3\} \square \{y_1, y_2, y_3, y_4\} \\ &= \{(x_1, y_1), (x_1, y_2), (x_1, y_3), (x_1, y_4), (x_2, y_1), (x_2, y_2), (x_2, y_3), (x_2, y_4), (x_3, y_1), \\ &\quad (x_3, y_2), (x_3, y_3), (x_3, y_4)\} \end{aligned}$$

The vertex and edges set of the graph is given below,

$$V(G) = \{v_i / 1 \leq i \leq 12\}$$

$$E(G) = \{e_i / 1 \leq i \leq 26\}$$



The vertex v_1 is adjacent to v_2 , v_3 , v_4 and v_5 . The vertex v_2 is adjacent to v_3 , v_4 , v_6 . The vertex v_3 is adjacent to v_1 , v_2 , v_4 , v_7 . The vertex v_4 is adjacent to v_1 , v_2 , v_3 and v_8 . The vertex v_5 is adjacent to v_1 , v_6 , v_7 , v_8 and v_9 . The vertex v_6 is adjacent to v_5 , v_7 , v_8 and v_{10} . The vertex v_7 is adjacent to v_5 , v_6 , v_8 , v_{11} . The vertex v_8 is adjacent to v_5 , v_6 , v_7 , v_{12} . The vertex v_9 is adjacent to v_5 , v_{10} , v_{11} , v_{12} . The vertex v_{10} is adjacent to v_6 , v_9 , v_{11} , v_{12} . The vertex v_{11} is adjacent to v_7 , v_9 , v_{10} , v_{12} . The vertex v_{12} is adjacent to v_8 , v_9 , v_{10} , v_{11} .

$$|e_f(0) - e_f(1)| \leq 1$$

Here $e_f(0) = 12$ and $e_f(1) = 14$

Thus

$$|e_f(0) - e_f(1)| = |12 - 14| \leq 1$$

Hence the graph admits sum divisor cardinal graph.

Theorem 2

Let G_1 and G_2 be two graphs and G be a Cartesian product of G_1 and G_2 . Then G admits sum divisor cardinal graph.

Proof:

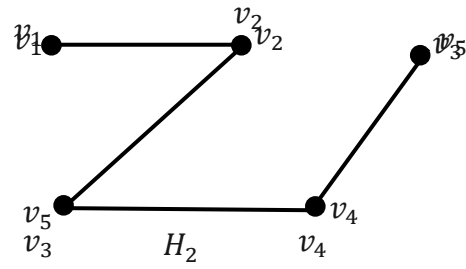
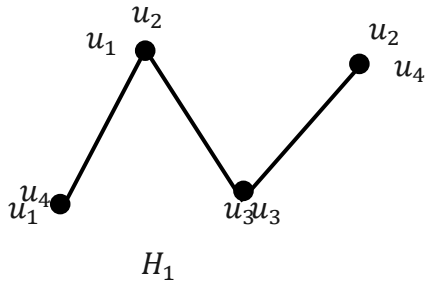
Let H_1 be a graph with vertex set, $|V(H_1)|=4$ and the edge set, $|E(H_1)|=3$

The line graph of H_1 has the vertex set, $|V(G_1)|=3$ and the edge set, $|E(G_2)|=2$

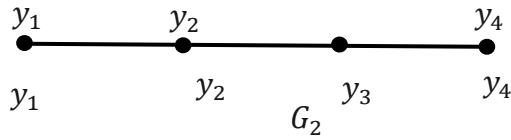
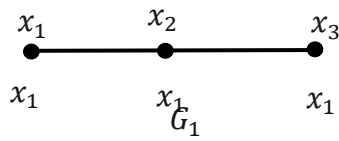
Let H_2 be a graph with the vertex set, $|V(H_2)|=5$ and the edge set, $|E(H_2)|=4$

The line graph of H_2 has the vertex set, $|V(G_2)|=4$ and the edge set, $|E(G_2)|=3$

Since G_1 and G_2 are the line graphs.



The line graphs are



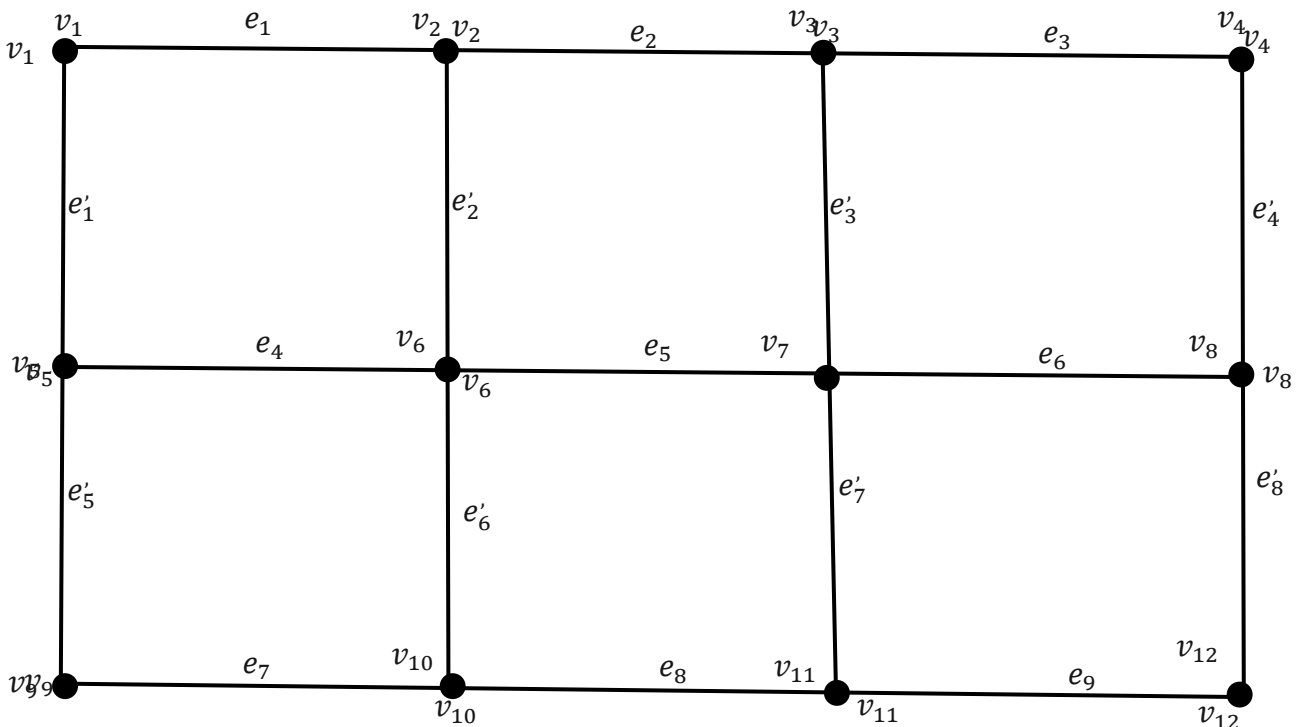
The cartesian product of the graphs G_1 and G_2 is

$$\begin{aligned} G_1 \square G_2 &= \{x_1, x_2, x_3\} \square \{y_1, y_2, y_3, y_4\} \\ &= \{(x_1, y_1), (x_1, y_2), (x_1, y_3), (x_1, y_4), (x_2, y_1), (x_2, y_2), (x_2, y_3), (x_2, y_4), (x_3, y_1), \\ &\quad (x_3, y_2), (x_3, y_3), (x_3, y_4)\} \end{aligned}$$

The vertex and edges set of the graph is given below,

$$V(G) = \{v_i / 1 \leq i \leq 12\}$$

$$E(G) = \{e_i / 1 \leq i \leq 17\}$$



The vertex v_1 is adjacent to v_2 and v_5 . The vertex v_2 is adjacent to v_3, v_1, v_6 . The vertex v_3 is adjacent to v_2, v_4, v_7 . The vertex v_4 is adjacent to v_3 and v_8 . The vertex v_5 is adjacent to v_1, v_6 , and v_9 . The vertex v_6 is adjacent to v_5, v_7, v_2 and v_{10} . The vertex v_7 is adjacent to v_3, v_6, v_8, v_{11} . The vertex v_8 is adjacent to v_6, v_7, v_{12} . The vertex v_9 is adjacent to v_8, v_{10} . The vertex v_{10} is adjacent to v_6, v_9, v_{11} . The vertex v_{11} is adjacent to v_7, v_{10}, v_{12} . The vertex v_{12} is adjacent to v_8, v_{11} .

$$|e_f(0) - e_f(1)| \leq 1$$

Here $e_f(0) = 9$ and $e_f(1) = 8$

Thus

$$|e_f(0) - e_f(1)| = |9 - 8| \leq 1$$

Hence the graph admits sum divisor cordial graph.

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Artificial Intelligence: Reshaping the World of E-Commerce – An Empirical Analysis

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Abstract

Artificial Intelligence (AI) has become a critical enabler in the evolution of the e-commerce sector, transforming traditional business models through data-driven decision-making, personalization, and automation. The present study examines consumer perception and the impact of AI applications in e-commerce platforms using primary data. A structured questionnaire was administered to 100 online consumers, and the collected data were analyzed using percentage analysis, mean score ranking, and chi-square testing. The results indicate that

AI-enabled features such as personalized recommendations and automated customer support significantly influence consumer purchasing behavior. However, concerns related to data privacy and trust persist. The study concludes that responsible and transparent implementation of AI is essential for sustained growth in the e-commerce industry.

Keywords: Artificial Intelligence, E-Commerce, Consumer Perception, Personalization, Statistical Analysis

1. Introduction

The rapid expansion of digital commerce has intensified competition among online retailers, necessitating the adoption of advanced technologies to enhance efficiency and customer engagement. Artificial Intelligence (AI), encompassing machine learning, natural language processing, and predictive analytics, has emerged as a transformative force in the e-commerce ecosystem. AI enables organizations to analyse large volumes of consumer data, predict purchasing patterns, and deliver personalized shopping experiences. Applications such as recommendation systems, chatbots, dynamic pricing, and fraud detection have fundamentally reshaped the online retail environment. This study seeks to examine the extent to which AI influences consumer behaviour and satisfaction in e-commerce platforms.

2. Review of Literature

Davenport and Ronanki (2018) emphasized that AI improves organizational performance by enhancing analytical capabilities and operational efficiency. McKinsey & Company (2021) reported that AI-driven recommendation systems contribute significantly to e-commerce revenues by increasing conversion rates. PwC (2022) observed that while consumers appreciate AI-enabled personalization, concerns regarding data privacy and transparency remain prominent. Accenture (2022) highlighted that ethical and responsible AI implementation strengthens customer trust and loyalty. Despite extensive research on AI applications, empirical studies focusing on consumer perception using primary data remain limited, thereby providing scope for the present study.

3. Core Concept: Artificial Intelligence as a Strategic Enabler in E-Commerce

The core concept underlying this study is that Artificial Intelligence functions not merely as a technological tool but as a **strategic enabler** within the e-commerce ecosystem. AI integrates data, algorithms, and automated decision-making to generate value across the entire customer journey. By leveraging consumer data, AI systems facilitate personalization, predictive insights, and real-time responsiveness, thereby enhancing customer experience and operational efficiency. In e-commerce, AI-driven mechanisms such as recommendation engines, chatbots, demand forecasting, and dynamic pricing convert raw data into actionable intelligence. This intelligence

supports informed managerial decision-making, improves customer satisfaction, and strengthens competitive advantage. However, the effectiveness of AI as a strategic enabler depends on responsible data governance, transparency, and ethical deployment, which directly influence consumer trust and long-term sustainability.

4. Objectives of the Study

The study is undertaken with the following objectives:

1. To assess consumer awareness of AI applications in e-commerce
2. To analyze the influence of AI features on online purchasing behavior
3. To measure consumer satisfaction with AI-enabled services
4. To identify challenges associated with AI adoption in e-commerce

5. Research Methodology

5.1 Research Design

A descriptive research design was adopted to analyze consumer perception of AI in e-commerce.

5.2 Data Collection

- **Nature of Data:** Primary data
- **Sample Size:** 100 respondents
- **Sampling Technique:** Convenience sampling
- **Research Instrument:** Structured questionnaire

5.3 Statistical Tools Applied

- Percentage analysis
- Mean score ranking
- Chi-square test

6. Data Analysis and Interpretation

Table 1: Awareness of AI Usage in E-Commerce

Awareness Level	Respondents	Percentage
Aware	82	82%
Not Aware	18	18%

Awareness Level	Respondents	Percentage
Total	100	100%

Interpretation:

The results indicate a high level of awareness regarding AI integration in e-commerce platforms.

Table 2: AI Features Used by Respondents

AI Feature	Respondents	Percentage
Product Recommendations	40	40%
Chatbots / Virtual Assistants	28	28%
Dynamic Pricing	20	20%
Voice Search	12	12%
Total	100	100%

Interpretation:

AI-based product recommendation systems are the most frequently used AI feature among consumers.

Table 3: Mean Score Analysis of Consumer Satisfaction

(5 = Strongly Agree, 1 = Strongly Disagree)

Statement	Mean Score	Rank
AI makes online shopping easier	4.32	I
AI recommendations match preferences	4.15	II
AI chatbots improve customer service	3.98	III
Trust in AI handling personal data	3.45	IV

Interpretation:

Respondents show the highest satisfaction with AI-enabled convenience and personalization, while trust in data handling remains comparatively lower.

Table 4: Chi-Square Test – Awareness and Purchase Influence

Particulars	Value
Calculated χ^2 Value	6.42
Table Value (df = 1, α = 0.05)	3.84

Particulars	Value
Result	Significant

Interpretation:

The analysis indicates a statistically significant relationship between AI awareness and consumer purchase decisions.

7. Findings and Suggestions

7.1 Findings

- A majority of consumers are aware of AI applications in e-commerce
- AI-enabled personalization significantly influences purchasing behavior
- AI improves convenience and overall shopping experience
- Data privacy and trust remain key concerns among consumers

7.2 Suggestions

- E-commerce firms should enhance transparency in AI-driven processes
- Strong data protection and privacy policies should be implemented
- Continuous improvement of AI-based customer support systems is recommended
- Consumer awareness regarding ethical AI usage should be promoted

8. Discussion of Findings

The empirical findings of the study reveal that Artificial Intelligence has a pronounced influence on consumer behaviour in the e-commerce environment. A high level of awareness regarding AI applications among respondents suggests increasing consumer familiarity with technology-driven shopping experiences. AI-based product recommendations and personalized services significantly enhance convenience and purchase intention, reinforcing AI's role in shaping buying behaviour. The chi-square analysis confirms a meaningful association between AI awareness and purchase influence, indicating that informed consumers are more receptive to AI-enabled features. Nevertheless, moderate concerns related to data privacy and trust persist, highlighting the need for ethical and transparent AI practices. Overall, the findings suggest that AI adoption offers substantial benefits for e-commerce platforms, provided consumer trust is adequately safeguarded.

9. Conclusion

The study demonstrates that Artificial Intelligence plays a significant role in reshaping the e-commerce industry by improving personalization, efficiency, and customer engagement.

The statistical evidence confirms that consumer awareness of AI positively influences purchasing decisions. While consumers largely perceive AI-enabled services as beneficial, issues related to data privacy and trust must be addressed to ensure long-term sustainability. Responsible, transparent, and ethical adoption of AI will be a key determinant of future success in the e-commerce sector.

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AI-Enabled English Language Teaching for the Next Generation

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Abstract

The rapid expansion of Artificial Intelligence (AI) in education has significantly reshaped pedagogical paradigms, particularly within English Language Teaching (ELT). This paper offers a conceptual and theoretical exploration of AI-enabled English language teaching for the next generation of learners. Drawing upon constructivist theory, Vygotsky's Zone of Proximal Development, and connectivist learning frameworks, the study examines how AI-driven technologies, such as natural language processing, intelligent tutoring systems, automated writing evaluation, and speech recognition tools, support personalized learning, formative assessment, and adaptive instruction. The paper argues that AI functions as an intelligence-amplifying scaffold rather than a pedagogical replacement, enhancing learner autonomy while preserving teacher agency. At the same time, it critically engages with concerns related to algorithmic bias, linguistic standardization, data privacy, digital inequity, and overdependence on automated feedback. By situating AI within human-centred pedagogical models, the study proposes a balanced framework for integrating AI into ELT classrooms through blended learning and critical digital literacy. Ultimately, the future of English language teaching lies in collaborative intelligence, where technological innovation and human interpretative expertise coexist to create equitable, ethically grounded, and intellectually rigorous learning environments.

Keywords

Artificial Intelligence, English Language Teaching, Natural Language Processing, Constructivism, Connectivism, Adaptive Learning, Digital Pedagogy and Collaborative Intelligence

AI-Enabled English Language Teaching for the Next Generation

Artificial Intelligence (AI) is rapidly transforming global education systems, compelling English Language Teaching (ELT) to re-evaluate its pedagogical foundations and instructional strategies. As English continues to function as a global lingua franca in academia, digital communication, and professional spaces, language classrooms must adapt to emerging technological realities. AI-enabled systems (driven by machine learning, natural language processing, and predictive analytics) offer new models of personalization, assessment, and learner engagement. Rather than replacing teachers, AI reconfigures educational ecosystems by enhancing instructional efficiency and supporting individualized learning pathways (Holmes et al. 12).

The theoretical justification for AI integration in ELT can be grounded in constructivist learning theory. Constructivism emphasizes that learners actively construct knowledge through interaction and reflection. AI-powered adaptive learning systems align with this principle by responding dynamically to learner input. These platforms analyse patterns of errors, adjust difficulty levels, and recommend targeted exercises, thus facilitating individualized knowledge construction. As Luckin argues, machine learning systems can support “intelligence amplification,” enabling learners to engage more deeply with content rather than passively receiving instruction (Luckin 45). In this sense, AI becomes a scaffold that strengthens cognitive engagement rather than diminishing it.

In addition to constructivism, Vygotsky’s concept of the Zone of Proximal Development (ZPD) offers a useful framework. AI-enabled tutoring systems can function as digital scaffolds, providing assistance precisely at the level where learners require support. Intelligent tutoring systems identify gaps between current competence and potential performance, offering incremental guidance. This mirrors the pedagogical role traditionally played by human instructors. Holmes et al. note that AI systems are particularly effective in offering timely feedback within this developmental zone, thereby accelerating skill acquisition (23).

Connectivism further strengthens the theoretical foundation for AI-enabled ELT. According to connectivist theory, learning in the digital age occurs across networks of information sources. AI tools connect learners to extensive linguistic corpora, authentic multimedia materials, and interactive conversational platforms. Through NLP-driven applications, students interact with vast databases of real-world language usage. Godwin-Jones observes that AI technologies extend language exposure beyond classroom boundaries, creating immersive and context-rich learning environments (3). Thus, AI expands the ecology of learning rather than confining it.

Natural language processing (NLP) has significantly altered writing instruction. Automated Writing Evaluation (AWE) tools analyse syntactic complexity, lexical density, and discourse organization. While critics argue that automated scoring cannot fully capture rhetorical nuance, such systems provide valuable formative feedback. Zawacki-Richter et al. highlight that AI-driven feedback mechanisms increase revision frequency and learner self-awareness (9). By offering immediate correction, AI supports iterative drafting processes, reinforcing writing as a recursive activity.

Similarly, speech recognition technologies enhance pronunciation and oral fluency. Traditional classrooms often lack sufficient time for individualized speaking practice. AI-based pronunciation software evaluates stress patterns, intonation, and phonetic accuracy. This technological intervention democratizes speaking practice, especially for learners who experience communication anxiety. As Warschauer and Healey suggest, technology-mediated environments reduce affective barriers and encourage experimentation with language (63).

Beyond skill development, AI invites a rethinking of literary studies within ELT contexts. AI-driven text analysis tools enable corpus-based approaches to literature. Students can examine frequency patterns, stylistic markers, and thematic clusters in literary texts. Such computational approaches foster data-informed literary interpretation. However, critical engagement remains essential. Human interpretative insight must guide analytical conclusions, ensuring that quantitative analysis does not replace hermeneutic depth. Thus, AI functions as an analytical aid rather than a substitute for literary sensibility.

Ethical considerations, however, complicate AI integration. Algorithmic bias presents significant risks. Language models trained on dominant linguistic datasets may privilege standardized varieties of English while marginalizing global dialects. Such bias can reinforce linguistic hierarchies. Holmes et al. caution that ethical governance frameworks are necessary to prevent discriminatory outcomes (76). Institutions must ensure inclusive datasets and culturally responsive implementation strategies.

Data privacy is another pressing issue. AI systems depend on extensive data collection to personalize instruction. Transparent data policies and institutional accountability are therefore essential. Luckin emphasizes that responsible AI in education requires regulatory oversight and ethical literacy among educators (102). Without such safeguards, technological advancement may compromise learner autonomy.

Furthermore, overdependence on AI may diminish critical thinking skills. If students rely excessively on automated grammar corrections or AI-generated suggestions, independent linguistic judgment may weaken. Zawacki-Richter et al. note that technological assistance must be balanced with pedagogical intentionality (15). Teachers must design activities that encourage reflective engagement rather than passive acceptance of machine output.

The teacher's role consequently evolves rather than diminishes. In AI-enabled classrooms, educators become facilitators of digital literacy, ethical reasoning, and critical interpretation. They contextualize AI-generated feedback and ensure meaningful learning experiences. Human empathy, cultural understanding, and interpretative nuance remain beyond algorithmic reach. As Holmes et al. affirm, AI augments rather than replaces professional expertise (131).

A balanced pedagogical model therefore requires blended learning strategies. AI tools should complement face-to-face interaction, collaborative discussion, and interpretative dialogue. Professional development programmes must equip teachers with technological competence and ethical awareness. Students must also develop critical AI literacy (the ability to question, evaluate, and responsibly use intelligent systems).

In conclusion, AI-enabled English language teaching represents a transformative evolution in educational practice. Grounded in constructivist, sociocultural, and connectivist theories, AI enhances personalization, scaffolding, and formative assessment. Nevertheless, ethical responsibility, inclusivity, and human-centred pedagogy must guide implementation. The future of ELT lies in collaborative intelligence, where artificial systems support, but do not supplant, human creativity and critical inquiry. When thoughtfully integrated, AI prepares the next generation of learners for a technologically sophisticated yet ethically grounded global society.

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Exposure to Celebrity Advertisements and Consumer Perception toward Sports Brands

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Abstract

Celebrity endorsements have become a dominant marketing strategy in the sports industry, influencing brand recognition, consumer trust, and purchase intention. The present study examines the exposure of respondents to celebrity advertisements across different media platforms and evaluates their perception of athletes, brands, and advertising elements. Using primary data collected from 588 respondents, the research identifies television and YouTube as the most influential media channels, while prominent athletes such as Virat Kohli and M. S. Dhoni emerge as highly impactful figures in sports advertising. The findings also reveal that celebrity appearance, product performance, and visual appeal are major attraction factors in advertisements. Although a significant proportion of respondents believe that celebrity endorsements help in discovering new sports brands, opinions remain divided regarding the authenticity of celebrity product usage. The study contributes to understanding how celebrity-driven marketing strategies shape consumer awareness and trust in the competitive sports brand market.

Keywords

Celebrity Advertising, Sports Marketing, Consumer Exposure, Brand Trust, Athlete Endorsements, Media Influence, Sports Brands.

Introduction

In the contemporary marketing environment, celebrity endorsements have become a powerful promotional tool, particularly within the sports industry. Brands frequently collaborate with well-known athletes to create emotional connections, enhance brand visibility, and strengthen consumer engagement. The rapid growth of digital media platforms has further transformed the way audiences encounter celebrity advertisements, shifting exposure from traditional media such as television and newspapers toward online channels like YouTube and Instagram.

Sports celebrities possess strong symbolic value, often representing performance, discipline, and success. Their influence extends beyond athletic achievements, shaping consumer perceptions and brand preferences. As audiences become more digitally connected, marketers must understand which media channels generate maximum exposure and which celebrity figures resonate most effectively with consumers. The success of celebrity-driven campaigns largely depends on audience trust, perceived authenticity, and the attractiveness of the advertisement content.

Despite the widespread use of celebrity endorsements, consumer attitudes toward these advertisements remain complex. While some consumers perceive endorsements as credible and inspiring, others question whether celebrities genuinely use the products they promote.

Understanding these perceptions is essential for marketers aiming to design impactful campaigns that balance promotional appeal with authenticity.

This study focuses on analyzing exposure patterns to celebrity advertisements and evaluating how various factors—including media platforms, athlete popularity, and advertisement features—affect consumer responses toward sports brands. By examining distribution-wise data from respondents, the research provides insights into evolving advertising trends and consumer expectations in the sports marketing domain.

Objectives of the Study

The main objectives of this study are:

1. To analyze the media platforms through which respondents are most frequently exposed to celebrity sports advertisements.
2. To identify the athletes perceived as most influential in sports brand promotions.

Review of Literature

Schaefer, Parker, and Kent (2010) conducted a comparative study examining the attitudes of American and Chinese consumers toward athlete celebrity endorsers. Their findings revealed that cultural context plays a significant role in shaping consumer perceptions, with Chinese audiences placing greater emphasis on credibility and reputation, while American consumers were more influenced by entertainment value and popularity. The study highlights the importance of understanding regional differences when designing celebrity-based marketing strategies.

Somdutta, Biswas, Mahmood, Hussain, and Kathleen O'Donnell (2009) explored celebrity endorsements from a cross-cultural perspective, focusing on how advertisements influence consumer perceptions and brand evaluation. The research demonstrated that celebrity credibility, attractiveness, and expertise significantly affect audience attitudes toward advertisements. The authors emphasized that the effectiveness of celebrity endorsements depends on alignment between the celebrity's image and the brand message, reinforcing the concept of congruence in advertising.

Thomaselli (2008) analyzed the branding challenges faced in transforming basketball star LeBron James into a global marketing icon. The study discussed how endorsement success relies not only on athletic performance but also on media narratives and strategic positioning. It provided insights into the evolving dynamics of sports marketing, where advertisers aim to build long-term brand equity through celebrity partnerships rather than short-term promotional gains.

Thwaites, Lowe, Monkhouse, and Barnes (2012) investigated the impact of negative publicity on celebrity endorsements and found that unfavorable media coverage can significantly damage brand perception and consumer trust. Their research highlighted that while celebrity endorsements can enhance brand awareness, they also carry reputational risks, emphasizing the need for companies to carefully evaluate the public image and behavior of endorsers.

Venkatesakumar, Sathyanarayanan, and Sathish (2012) examined regional differences in celebrity endorsement effectiveness within the Indian context, focusing on the North–South divide. The study revealed variations in consumer preferences influenced by language, cultural values, and regional media exposure. It suggested that localized marketing strategies are crucial for maximizing the effectiveness of celebrity advertising campaigns across diverse audiences.

Wei and Lu (2013) analyzed the combined impact of celebrity endorsements and online customer reviews on female consumers' shopping behavior. Their findings indicated that digital word-of-mouth and social media interactions significantly enhance the persuasive power of celebrity advertising. The research demonstrated that consumers often rely on both celebrity influence and peer feedback when making purchase decisions in online environments.

Zipporah and Mberia (2014) examined the overall effects of celebrity endorsements in advertisements and concluded that celebrity presence increases brand recall, attention, and purchase intention. However, the authors also noted that excessive reliance on celebrity appeal without product relevance may reduce long-term effectiveness. Their work underscores the importance of balancing celebrity attractiveness with meaningful product communication.

Haque (2016) focused on the impact of celebrity endorsement on students' buying behavior toward personal care products. The study found that young consumers are highly responsive to celebrity-based promotions, particularly when the endorsers are perceived as relatable and trustworthy. It highlighted the role of youth-oriented marketing strategies and the growing influence of media personalities among student populations.

Sheth (2018) compared the effectiveness of celebrity-endorsed advertisements with animated character endorsements. The research revealed that while celebrity endorsements often generate higher attention and credibility, animated characters may appeal more to specific audience segments depending on product category and target demographics. The study suggests that marketers should adopt a strategic approach when selecting endorsement styles to match consumer expectations.

Research Methodology

The present study adopts a descriptive research design to analyze respondents' exposure to celebrity advertisements and their perceptions toward sports brand endorsements. Primary data were collected from 588 respondents using a structured questionnaire consisting of multiple-choice questions related to media exposure, influential athletes, advertisement attraction factors, brand trust, and recognition of sports products. A convenience sampling technique was applied to select participants who were familiar with sports advertisements, ensuring relevance to the research objectives. The data collection process maintained confidentiality and voluntary participation, and responses were systematically compiled for analysis. The study mainly relied on percentage analysis and frequency distribution to interpret the data and identify patterns in consumer attitudes toward celebrity endorsements. Secondary sources such as journals and previous research studies were also reviewed to support the theoretical background and strengthen the analytical framework of the research.

EXPOSURE TO CELEBRITY ADVERTISEMENTS

Table1: Exposure to Celebrity Advertisements(Distribution wise of the Respondents)

S.No	Question	Options	Frequency (f)	Percentage (%)
1	Which media do you most often see celebrity sports ads on?	Television	220	37.4
		YouTube	145	24.7
		Instagram	100	17
		Facebook	50	8.5
		X (Twitter)	25	4.3

		Newspapers	30	5.1
		Billboards	15	2.6
		Others	3	0.5
		Total	588	100
2	Which of the following athletes do you find most influential in sports advertisements?	Virat Kohli	180	30.6
		M. S. Dhoni	120	20.4
		Rohit Sharma	60	10.2
		Hardik Pandya	30	5.1
		P. V. Sindhu	40	6.8
		Neeraj Chopra	25	4.3
		Lionel Messi	35	6
		Cristiano Ronaldo	35	6
		Roger Federer	25	4.3
		Serena Williams	20	3.4
		Others	18	3.1
		Total	588	100
3	Which sports company's advertisement featuring a celebrity caught your attention recently?	Nike	110	18.7
		Adidas	85	14.5
		Puma	50	8.5
		Reebok	30	5.1
		Under Armour	20	3.4
		Asics	10	1.7
		Yonex	25	4.3
		New Balance	20	3.4
		Nivia	15	2.6
		Cosco	10	1.7
		SG	10	1.7
		SS	8	1.4
		Decathlon	15	2.6
		Li-Ning	8	1.4
		HRX	12	2
		Bose / JBL	5	0.9
		Red Bull	5	0.9
		GoPro	3	0.5
		Skechers	2	0.3
		Fila	2	0.3
		Lotto	2	0.3
		Spalding	3	0.5

		Speedo	3	0.5
		Wilson	2	0.3
		Head	2	0.3
		Others	8	1.4
		Total	588	100
4	Which factor in the celebrity ad attracted you most?	Celebrity Appearance	160	27.2
		Product Performance	140	23.8
		Visual Appeal	100	17
		Emotional Message	80	13.6
		Music / Slogan	50	8.5
		Brand Prestige	40	6.8
		Social Media Presence	18	3.1
		Total	588	100
5	Do you believe celebrity advertisements create trust in the brand?	Yes	260	44.2
		No	80	13.6
		Sometimes	248	42.2
		Total	588	100
6	Do celebrity endorsements help you recognize new sports brands or products?	Yes	400	68
		No	188	32
		Total	588	100
7	Do you think celebrities actually use the products they endorse?	Always	50	8.5
		Often	120	20.4
		Sometimes	220	37.4
		Rarely	120	20.4
		Never	78	13.3
		Total	588	100

Source: Primary data

Exposure to Celebrity Advertisements – Analysis and Interpretation

The distribution-wise analysis of respondents indicates that television (37.4%) remains the most dominant platform for viewing celebrity sports advertisements, followed by YouTube (24.7%) and Instagram (17%). This highlights the continued relevance of traditional media alongside digital platforms. Social media channels such as Facebook and X (Twitter) recorded comparatively lower exposure, suggesting that visually engaging platforms are more effective for sports-related promotional content.

Regarding athlete influence, Virat Kohli (30.6%) emerged as the most impactful celebrity in sports advertising, followed by M. S. Dhoni (20.4%) and Rohit Sharma (10.2%). International icons like Lionel Messi and Cristiano Ronaldo also attracted attention, demonstrating the global appeal of sports celebrities. The dominance of cricket personalities reflects the cultural significance of cricket within the Indian context.

In terms of brand recall, Nike (18.7%) and Adidas (14.5%) advertisements featuring celebrities captured the highest attention among respondents. Other brands such as Puma, Reebok, and Yonex also gained moderate visibility. The wide distribution of brands indicates a competitive advertising landscape where celebrity association plays a key role in differentiating sports products.

The study further reveals that celebrity appearance (27.2%) and product performance (23.8%) are the most attractive elements in advertisements, followed by visual appeal and emotional messaging. This suggests that consumers value both the charisma of the celebrity and the functional benefits of the product. Interestingly, only a smaller proportion of respondents considered social media presence as a primary attraction factor.

Trust perception toward celebrity advertisements shows mixed responses, with 44.2% believing that endorsements create brand trust, while 42.2% reported that trust is developed only sometimes. A significant majority (68%) agreed that celebrity endorsements help them recognize new sports brands or products, highlighting the awareness-building power of such campaigns. However, skepticism remains regarding authenticity, as many respondents indicated that celebrities “sometimes” or “rarely” use the products they endorse.

Conclusion

The findings of the study reveal that celebrity advertisements play a significant role in shaping consumer awareness, brand recognition, and perceptions toward sports products. Traditional media such as television continues to be a dominant platform for exposure, while digital channels like YouTube and Instagram are rapidly gaining importance among audiences. Popular athletes, particularly cricket celebrities, were found to have strong influence on viewers, highlighting the cultural relevance of sports icons in marketing communication. The results also indicate that factors such as celebrity appearance, product performance, and visual appeal are key elements that attract consumer attention. Although a large proportion of respondents believe that celebrity endorsements help them recognize new sports brands, opinions remain divided regarding the authenticity of celebrity product usage, suggesting a cautious level of consumer trust.

Based on these insights, it is recommended that sports companies adopt a balanced celebrity endorsement strategy that emphasizes authenticity, transparency, and product relevance. Marketers should focus on selecting celebrities whose image aligns with brand values and who can communicate genuine product benefits rather than relying solely on popularity. Increasing the use of interactive digital platforms and engaging storytelling techniques may further strengthen audience connection and trust. Companies should also consider regional preferences and audience expectations when designing advertising campaigns, ensuring that celebrity presence complements product performance and brand credibility. Future advertising strategies can benefit from combining celebrity influence with customer reviews and experiential marketing to enhance long-term consumer loyalty and brand effectiveness.

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AI Science To Real-World Power: Skills For The Next Generation

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ABSTRACT

Artificial Intelligence (AI) has rapidly evolved from a scientific concept into a practical force shaping education, business, governance, healthcare, and everyday life. While AI-driven systems are transforming industries, the true advantage lies not merely in access to technology but in the ability to use it effectively. This article explores how AI science is converted into real-world power through skill development, particularly for the next generation of learners and workforce entrants. The paper explains AI in simple terms, highlights real-world applications across sectors, and identifies the essential skill clusters required for students to remain relevant in an AI-integrated world. It further discusses how educational institutions can redesign learning ecosystems through AI literacy, project-based learning, interdisciplinary training, and ethical awareness. The study concludes that AI will not replace human potential; instead, individuals who master AI tools, critical thinking, creativity, and responsible decision-making will lead the

future. The article recommends a structured approach to integrating AI readiness into curricula to ensure employability, entrepreneurship, and inclusive development.

Keywords: Artificial Intelligence, Future Skills, AI Literacy, Employability, Education Transformation, Innovation, Digital Competence

1. INTRODUCTION

Artificial Intelligence (AI) is no longer a futuristic term limited to laboratories or advanced research institutions. It has become a daily reality, influencing the way people learn, work, communicate, shop, travel, and even make decisions. The global workforce is undergoing a rapid transition where traditional job roles are being reshaped by automation, predictive analytics, and generative AI systems. In this context, the next generation—students, early-career professionals, and aspiring entrepreneurs—must be equipped with relevant skills to convert AI from a technological concept into real-world advantage.

Education systems historically focused on knowledge acquisition, examinations, and memorization. However, the AI era demands a shift toward skills, application, adaptability, and innovation. The relevance of AI is not confined to engineering or computer science disciplines alone. Commerce, management, humanities, law, healthcare, education, and social sciences are equally impacted by AI-driven transformation. Therefore, AI readiness must become a foundational component of modern education and career preparation.

This article presents AI as a bridge between science and practical power. It highlights the skills that enable individuals to effectively use AI tools, interpret AI outputs, make ethical decisions, and innovate within their chosen fields. The article also explores how institutions and educators can create structured pathways to prepare learners for AI-integrated careers and responsible citizenship.

2. UNDERSTANDING AI SCIENCE: FROM CONCEPT TO CAPABILITY

AI refers to systems or machines designed to perform tasks that typically require human intelligence. These tasks include understanding language, recognizing patterns, making predictions, solving problems, generating content, and assisting in decision-making. The power of AI arises from its ability to process large volumes of data, identify patterns faster than humans, and produce outputs that can guide or automate actions.

AI SCIENCE INCLUDES MULTIPLE LAYERS.

2.1 Machine Learning (ML)

Machine Learning is a subset of AI where systems learn from data. Instead of being explicitly programmed for every situation, ML models detect patterns and improve performance over time. For example, ML systems can predict student performance based on attendance, assignments, and test scores.

2.2 Deep Learning (DL)

Deep Learning is a more advanced subset of ML inspired by the human brain. It uses neural networks to process complex data such as images, voice, and video. Facial recognition, voice assistants, and medical image analysis often rely on deep learning.

2.3 Generative AI

Generative AI refers to AI systems that create new content such as text, images, audio, video, and code. Tools like ChatGPT, Gemini, and Copilot are examples. Generative AI is particularly relevant for education, business communication, digital marketing, design, and entrepreneurship. The understanding of AI science is important not to become an AI engineer necessarily, but to become an informed user, capable decision-maker, and responsible contributor in an AI-driven environment.

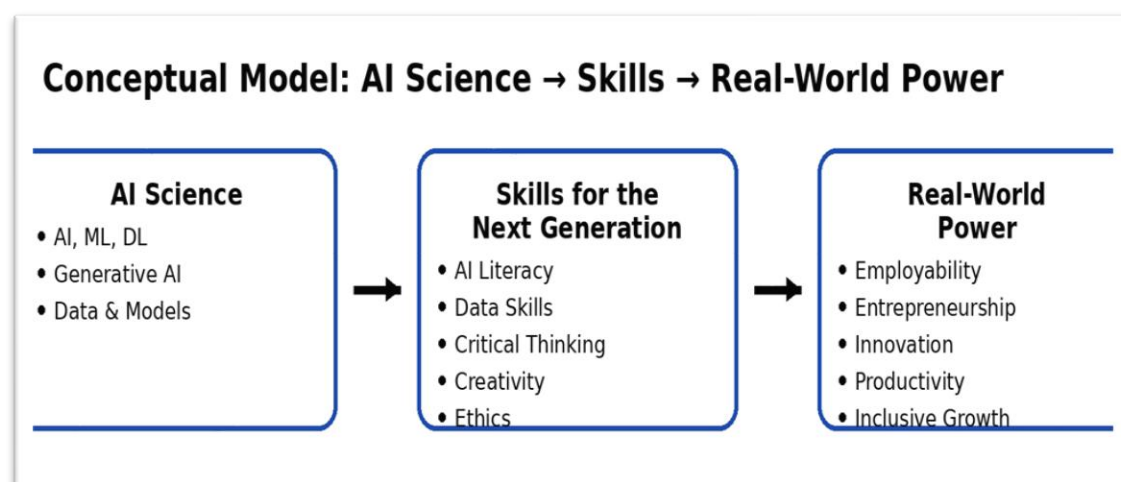


Figure 1. Translating AI science into practical outcomes through skill development.

3. AI IN THE REAL WORLD: APPLICATIONS ACROSS SECTORS

The transformation created by AI can be observed across multiple sectors. AI is becoming a universal productivity layer, improving efficiency, reducing time consumption, and enabling data-driven decision-making.

3.1 AI in Education

AI supports education through personalized learning, intelligent tutoring systems, automated evaluation, content creation, and learning analytics. AI can assist teachers in designing lesson plans, generating quizzes, and tracking student progress. Students can use AI tools to improve writing, language learning, research, and project development.

3.2 AI in Business and Commerce

AI is widely used in customer service (chatbots), marketing automation, sales forecasting, inventory management, fraud detection, and consumer behavior analysis. In commerce, AI helps businesses understand trends, predict demand, and create personalized customer experiences.

3.3 AI in Healthcare

AI supports diagnosis, medical imaging, drug discovery, hospital management, and patient monitoring. AI-driven tools can help doctors identify diseases at early stages and assist in personalized treatment planning.

3.4 AI in Agriculture and Rural Development

AI is used for crop monitoring, soil analysis, weather prediction, pest detection, and supply chain optimization. AI applications can contribute to sustainable agriculture and improved rural livelihoods.

3.5 AI in Governance and Society

Governments use AI for smart city management, public service delivery, security surveillance, and digital administration. AI can improve citizen services but also raises concerns related to privacy and misuse.

The above examples demonstrate that AI is not limited to one domain. It has become an essential tool for modern development, productivity, and innovation.

4. SKILLS FOR THE NEXT GENERATION: CONVERTING AI INTO POWER

The next generation must develop skill clusters that enable them to effectively interact with AI tools and apply them to real-world challenges. AI itself is not the power; the skills to use AI strategically are the power.

4.1 AI Literacy and Tool Competence

AI literacy includes understanding what AI is, how it works, and how it should be used. Students must learn how to use AI tools for research, writing, summarization, and content creation; how

to write effective prompts; how to verify AI-generated outputs; and how to understand limitations such as hallucinations and bias. AI literacy is becoming as important as basic computer literacy.

4.2 Data Literacy and Analytical Thinking

AI depends on data. Therefore, the next generation must be able to interpret graphs, dashboards, and datasets; use tools like Excel, Google Sheets, and Power BI basics; understand data privacy and data ethics; and make decisions based on evidence rather than assumptions. Data literacy strengthens employability, especially in commerce and management sectors.

4.3 Critical Thinking and Problem-Solving Skills

AI tools can provide answers, but humans must evaluate them. Critical thinking helps learners identify misinformation and manipulation, question the accuracy of AI outputs, apply logic and reasoning, and solve real-world problems creatively. This skill becomes essential in an era of deepfakes and AI-generated misinformation.

4.4 Creativity, Innovation, and Entrepreneurship

AI tools support creativity through design assistance, content generation, and ideation. Students can use AI to develop business ideas, create marketing content, design posters, brochures, and presentations, and build prototypes and pitch decks. AI enhances entrepreneurial ability by reducing entry barriers and increasing productivity.

4.5 Communication, Collaboration, and Emotional Intelligence

Despite AI growth, human-centered skills remain irreplaceable. The next generation must strengthen communication skills, teamwork and leadership, emotional intelligence, and negotiation and persuasion. AI cannot replace empathy, cultural understanding, and human relationships.

4.6 Ethical Awareness and Responsible AI Use

AI introduces risks such as bias, privacy violations, plagiarism, and misuse. Students must learn academic integrity in AI use, responsible content creation, data privacy principles, and avoiding discrimination and stereotyping. Ethics is a core skill, not an optional topic.

5. CAREER READINESS: AI AS A SKILL LAYER FOR EVERY PROFESSION

A key misconception is that AI careers are limited to programmers or engineers. In reality, AI acts as a skill layer across professions.

5.1 AI in Commerce and Management Careers

Students in commerce can use AI in business analytics, digital marketing, financial forecasting, customer relationship management, and e-commerce strategy. Roles include Business Analyst, Digital Marketing Strategist, Financial Data Associate, and AI-assisted Entrepreneur.

5.2 AI in Education Careers

Teachers and education professionals can use AI for content development, lesson planning, student engagement tools, and learning analytics. New roles include Instructional Designer, EdTech Consultant, and Digital Learning Coordinator.

5.3 AI in Media and Communication Careers

AI supports video editing, content strategy, social media automation, and branding and storytelling. Roles include AI Content Creator, Digital Brand Strategist, and Media Planner.

5.4 AI in Social and Community Development

AI can support NGOs and community initiatives through data-driven planning, impact measurement, public awareness campaigns, and rural skill development models. This is especially relevant for women empowerment and inclusive development initiatives.

6. ROLE OF EDUCATIONAL INSTITUTIONS AND TEACHERS

Institutions play a central role in ensuring AI readiness. AI should not be taught only as a technical subject; it must be integrated into learning and career preparation.

6.1 Curriculum Integration

Institutions should include AI literacy modules in all departments, data literacy and digital tools training, and interdisciplinary projects involving AI use.

6.2 Project-Based and Skill-Based Learning

Students must be evaluated through projects and presentations, case studies and real-world tasks, and internships and field-based learning. This approach ensures skill development beyond theoretical knowledge.

6.3 Industry Partnerships and Skill Ecosystems

Collaboration with industries, startups, and professional bodies can provide expert sessions, certification programs, internship opportunities, and AI-enabled entrepreneurship training. Such partnerships strengthen employability and innovation.

6.4 Faculty Development and Institutional Readiness

Teachers require AI tool training, updated teaching methodologies, ethical AI guidance, and digital pedagogy support. Faculty development ensures sustainability of AI integration.

7. CHALLENGES AND ETHICAL CONSIDERATIONS

While AI offers significant benefits, it also raises challenges.

7.1 Digital Divide

Not all students have equal access to devices, internet, and AI tools. Institutions must work toward inclusive digital access.

7.2 Bias and Fairness

AI systems may reflect biases present in data. Students must learn fairness principles and avoid blindly trusting AI outputs.

7.3 Academic Integrity and Plagiarism

Generative AI can lead to unethical copying and misuse. Institutions must define clear AI usage guidelines and teach responsible learning practices.

7.4 Privacy and Data Security

Students must understand that AI tools often store or process user data. Responsible usage and awareness of privacy risks are essential.

8. CONCLUSION AND RECOMMENDATIONS

AI is transforming every sector, and its influence will continue to grow. However, the real-world power of AI depends on human capability, not only technology availability. The next generation must develop AI literacy, data skills, critical thinking, creativity, communication, and ethical responsibility. These skills ensure that learners become innovators and leaders rather than passive users of technology.

Educational institutions must redesign learning ecosystems through interdisciplinary curricula, project-based evaluation, faculty training, and industry partnerships. AI should be positioned not as a threat but as an opportunity for inclusive growth, entrepreneurship, and employability. With the right skills, the next generation can convert AI science into real-world power and contribute to national and global development.

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Applications of Artificial Intelligence in the Pure Mathematical Concept of Nano Topology

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Abstract

Nano topology is a generalized topological framework derived from rough set theory and approximation spaces. It provides a structured way to study uncertainty using lower and upper approximations defined through equivalence relations. Artificial Intelligence (AI), particularly symbolic computation and machine learning, offers computational tools that simplify nano topological calculations and enable exploration of large approximation spaces. This paper presents the mathematical foundation of nano topology, illustrates approximation computations through examples, and discusses how AI techniques support research in pure mathematics. The study highlights the role of computational intelligence in extending approximation-based topology toward practical and theoretical applications.

Keywords

Artificial Intelligence, Nano topology, Rough Sets, Approximation Space, Mathematical Computing, Symbolic Computation

1. Introduction

Topology is one of the central branches of pure mathematics dealing with the structure of space and continuity. Over time, several generalized forms of topology have been developed to model uncertainty and discrete structures. One such development is **nano topology**, which originates from rough set theory.

Rough set theory, introduced by **Zdzisław Pawlak**, provides tools for dealing with incomplete or uncertain information. Using approximation operators, it becomes possible to construct new topological structures known as nano topological spaces.

At the same time, Artificial Intelligence has become an important tool in mathematical research. AI systems assist in symbolic reasoning, automated computation, and pattern recognition. These capabilities make AI particularly useful for handling approximation-based structures such as nano topology.

This paper explains the mathematical structure of nano topology and discusses how AI contributes to its computation and analysis.

2. Approximation Spaces and Nano topology

Let U be a non-empty finite set called the **universe**. Let R be an equivalence relation on U . The pair (U, R) is called an **approximation space**. The equivalence relation partitions U into equivalence classes:

$$U/R = \{E_1, E_2, \dots, E_n\}.$$

For a subset $X \subseteq U$, nanotopology is constructed using approximation operators.

3. Nano Approximation Operators

Lower Approximation

The nano lower approximation contains elements that certainly belong to X .

$$L_R(X) = \cup\{E: E \subseteq X\}$$

Upper Approximation

The nano upper approximation contains elements that possibly belong to X .

$$U_R(X) = \cup\{E: E \cap X \neq \emptyset\}$$

Boundary Region

$$B_R(X) = U_R(X) - L_R(X)$$

Nano Topology

$$\tau_R(X) = \{U, \emptyset, L_R(X), U_R(X), B_R(X)\}.$$

These sets form a nano topological space.

4. Computational View of Nano topology

Nano topology is naturally computational because:

- it uses finite sets,
- it depends on classification,
- it involves repeated approximation calculations.

This makes it suitable for **algorithmic** implementation, where Artificial Intelligence tools can assist in computation and pattern discovery.

4.1. Examples: Computation in Nano topology

Example 4.1.1:

Let

$$U = \{1,2,3,4\}$$

Equivalence classes:

$$\{\{1,2\}, \{3\}, \{4\}\}$$

Let

$$X = \{1,3\}.$$

Lower approximation:

$$L_R(X) = \{3\}$$

Upper approximation:

$$U_R(X) = \{1,2,3\}$$

Boundary:

$$B_R(X) = \{1,2\}$$

Nano topology:

$$\tau_R(X) = \{\emptyset, U, \{3\}, \{1,2,3\}, \{1,2\}\}.$$

Example 4.2: Algorithmic Computation

Steps:

1. Input U, R, X
2. Generate equivalence classes
3. Test subset condition
4. Test intersection condition
5. Construct nano topology

This process resembles classification algorithms in AI.

Example 4.3: Nano Interior and Closure

$$Nint(X) = L_R(X)$$

$$Ncl(X) = U_R(X)$$

From Example 4.1.1:

$$Nint(X) = \{3\}, Ncl(X) = \{1,2,3\}.$$

Example 4.3.1:

Let

$$U = \{a, b, c, d, e\}.$$

Equivalence classes:

$$\{\{a, b\}, \{c, d\}, \{e\}\}.$$

Let

$$X = \{a, c, e\}.$$

$$L_R(X) = \{e\}$$

$$U_R(X) = \{a, b, c, d, e\}$$

$$B_R(X) = \{a, b, c, d\}.$$

5. Artificial Intelligence and Mathematical Computation

AI contributes to pure mathematics mainly through computational assistance. In nano topology, AI systems can:

- compute equivalence partitions,

- evaluate approximation operators,
- classify nano sets,
- verify structural properties.

Symbolic computation systems already perform many of these tasks automatically.

6. Algorithmic Representation of Nano Approximation

A simple computational model:

Input:

- Universe U
- Equivalence relation R
- Subset X

Output:

- $L_R(X), U_R(X), B_R(X)$

Procedure:

- generate partition of U
- check subset inclusion
- check intersection condition
- compute unions

This structure resembles rule-based AI systems.

7. Real-Life and Computational Applications

7.1 Data Classification

Nano approximation concepts are closely related to classification problems in AI.

7.1.1 Decision Support Systems

Approximation-based reasoning is used in:

- medical diagnosis
- recommendation systems
- uncertainty modeling.

7.1.2 Knowledge Representation

Nano topology connects with:

- granular computing
- database approximation
- information systems.

7.1.3 Mathematical Software Systems

AI-supported mathematical tools help:

- visualize discrete structures,
- test conjectures,
- generate examples.

8. Discussion

Nano topology represents a bridge between topology, discrete mathematics, and computational reasoning. Because its construction depends on approximation spaces, it aligns naturally with AI-based classification methods.

AI does not replace mathematical proof but supports exploration, computation, and experimentation in approximation-based topology.

The relationship between nano topology and AI demonstrates how theoretical mathematics benefits from computational intelligence.

9. Conclusion

Nano topology provides a structured method for studying approximation-based topological spaces derived from rough set theory. Artificial Intelligence enhances this field by automating computations, supporting symbolic reasoning, and enabling exploration of complex approximation spaces. The integration of AI with pure mathematics highlights the growing importance of computational tools in theoretical research.

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